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Experimental and Numerical Investigation on Surface Roughness in Additively Manufactured Rocket Engines

Tiziano Santese

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Chair:
Prof. Dr.-Ing. Mirko Hornung
Examiners:
1. Prof. Dr.-Ing. Chiara Manfletti
2. Prof. Francesco Nasuti, Ph.D.

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TIZIANO SANTESE

EXPERIMENTAL AND NUMERICAL INVESTIGATION ON SURFACE ROUGHNESS IN ADDITIVELY MANUFACTURED ROCKET ENGINES

Dedicato alla mia famiglia

Abstract

Additive Manufacturing (AM) is reshaping the design and production of aerospace components by enabling geometric flexibility and optimising flow and weight. However, this advancement introduces new engineering challenges, such as the inherently high surface roughness produced by AM processes, which can critically affect the performance of propulsion systems operating under extreme thermal and fluid dynamic conditions.
This work focuses on understanding and quantifying the impact of AM-induced surface roughness on the aerodynamic and thermal performance of rocket engine components. The central motivation is the need for accurate prediction tools and design strategies that account for roughness effects, particularly in high-stakes applications such as space propulsion, where small deviations can lead to system-level failures.

A multi-scale methodology is adopted to investigate these effects. High-fidelity Direct Numerical Simulations (DNS) are used to calculate fundamental interactions between the flow and the rough surfaces, revealing significant increases in skin friction and alterations in thermal boundary layers. These numerical results are confirmed by experimental tests on components manufactured via Laser Beam Melting (LBM), which showed friction factors up to 15 times higher than smooth theoretical values, and heat transfer enhancements up to 35%. Significant azimuthal temperature variations, up to 200 K, linked to process-induced surface roughness, highlight the need for detailed surface characterisation. Industrial Reynolds-Averaged Navier-Stokes (RANS) models are also evaluated. While sand-grain roughness analogies can reproduce pressure losses when properly tuned, predicting wall heat transfer remains challenging. A modified turbulent Prandtl number model shows improvement, suggesting a path forward for more accurate simulation tools.

These findings provide a clearer picture of when and how roughness matters, and they offer practical guidelines for model calibration, quality control, and component design. The outcomes are directly applicable to industry and extend to other high-performance systems such as gas turbines and heat exchangers.
Despite these advances, several key questions remain open. Future research must aim to generalise roughness modelling across different AM processes and geometric configurations. Data-driven approaches, such as machine learning, still require further development to build accurate models informed by high-resolution surface measurements and experimental data. Addressing these challenges will be essential to fully harness the potential of Additive Manufacturing in aerospace propulsion and other high-performance engineering applications.

Zusammenfassung

Additive Fertigung (AM) verändert die Konstruktion und Herstellung von Luft- und Raumfahrtkomponenten grundlegend, da sie geometrische Flexibilität sowie Optimierung von Strömung und Gewicht ermöglicht. Diese Fortschritte bringen jedoch neue ingenieurtechnische Herausforderungen mit sich, da die durch AM-Prozesse inhärent erzeugte hohe Oberflächenrauheit die Leistung von Antriebssystemen, die unter extremen thermischen und strömungsdynamischen Bedingungen arbeiten, kritisch beeinflussen kann.

Diese Arbeit konzentriert sich auf das Verständnis und die Quantifizierung der Auswirkungen von durch AM induzierter Oberflächenrauheit auf die aerodynamische und thermische Leistung von Raketentriebwerkskomponenten. Die zentrale Motivation ist der Bedarf an präzisen Vorhersagewerkzeugen und Entwurfsstrategien, die Rauheitseffekte berücksichtigen, insbesondere in sicherheitskritischen Anwendungen wie der Raumfahrtantriebe, bei denen kleine Abweichungen zu systemweiten Ausfällen führen können.

Eine multiskalige Methodik wird angewendet, um diese Effekte zu untersuchen. Hochaufgelöste Direkte Numerische Simulationen (DNS) werden genutzt, um grundlegende Wechselwirkungen zwischen Strömung und rauen Oberflächen zu berechnen. Diese zeigen signifikante Zunahmen des Reibungswiderstands sowie Veränderungen in den thermischen Grenzschichten. Die numerischen Ergebnisse werden durch experimentelle Tests an mittels Laserstrahlschmelzen (LBM) gefertigten Komponenten bestätigt, bei denen Reibungsbeiwerte bis zu dem 15-fachen der glatten theoretischen Werte und Wärmeübertragungssteigerungen bis zu 35% beobachtet wurden. Signifikante azimutale Temperaturunterschiede von bis zu $200\,K$, die mit prozessbedingter Oberflächenrauheit zusammenhängen, unterstreichen die Notwendigkeit einer detaillierten Oberflächencharakterisierung.

Auch industrielle Reynolds-Averaged Navier-Stokes (RANS)-Modelle werden bewertet. Während Analogieansätze mit Sandkornrauheit den Druckverlust bei entsprechender Kalibrierung reproduzieren können, bleibt die Wandwärmeübertragung schwer vorherzusagen. Ein modifiziertes Modell der turbulenten Prandtl-Zahl zeigt Verbesserungen und deutet auf einen möglichen Weg zu genaueren Simulationswerkzeugen hin.

Diese Ergebnisse liefern ein klareres Bild davon, wann und wie Rauheit eine Rolle spielt, und bieten praktische Richtlinien für Modellkalibrierung, Qualitätskontrolle und

Bauteilauslegung. Die Erkenntnisse sind direkt auf die Industrie anwendbar und lassen sich auf andere Hochleistungssysteme wie Gasturbinen und Wärmetauscher übertragen.

Trotz dieser Fortschritte bleiben mehrere zentrale Fragen offen. Zukünftige Forschung muss darauf abzielen, die Modellierung der Rauheit über verschiedene AM-Prozesse und geometrische Konfigurationen hinweg zu verallgemeinern. Datengetriebene Ansätze, wie etwa maschinelles Lernen, müssen noch entwickelt werden, um präzise Modelle zu erstellen, die durch hochauflösende Oberflächenmessungen und experimentelle Daten gestützt werden. Die Bewältigung dieser Herausforderungen wird entscheidend sein, um das volle Potenzial der Additiven Fertigung in der Raumfahrtantriebstechnik und anderen hochleistungsfähigen Ingenieuranwendungen auszuschöpfen.

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List of Symbols

Acronyms

CH_4	Methane
EoS	Equation of state
GO_x	Gaseous Oxygen
$N2$	Gaseous Nitrogen
AM	Additive Manufacturing
BC	Boundary Conditions
CEA	Chemical Equilibrium with Applications
CFD	Computational Fluid-Dynamics
CT	Computed Tomography
DNS	Direct Numerical Simulation
EDM	Electron-Discharge Machining
FVM	Finite Volume Method
IC	Initial conditions
ISS	International Space Station
LBM	Laser Beam Melting
LRE	Liquid Rocket Engine
NASA	National aeronautics and space administration
RANS	Reynolds-Averaged Numerical Simulation
SA	Spalart-Allmaras
SSME	Space Shuttle Main Engine

General

$\dot{m}$	Mass flow
$\dot{q}$	Heat flux
η	Kolmogorov scale
Γ	Momentum source
γ	Adiabatic index
κ	Von Karman constant
μ	Dynamic viscosity
ν	Kinematic viscosity
Φ	Energy source
ρ	Density
τ	Viscous tensor
θ	Temperature deviation
ε	Convergence indicator
A	Area
c^*	Characteristic speed
C_f	Skin friction coefficient
C_t	Convective Time unit
d	Diameter
d_H	Hydraulic diameter
F	Thrust
f	Darcy–Weisbach coefficient
g_0	Gravital acceleration
h	Half-length of duct side

h_g	Hot gas convection coefficient
I_{sp}	Specific Impulse
k	Thermal conductivity or turbulent kinetic energy
k	Turbulent kinetic energy
k_s	Sand-grain parameter
k_s^+	Roughness Reynolds number
L_c	Characteristic length
M	Mach number
OF	Oxidiser to Fuel ratio
p	Pressure
Pr	Prandtl number
Q	Heat
r	Radius, or radial distance
Re	Reynolds number
St	Stanton number
T	Temperature
t	Time
u, v, w	Velocity components
u_τ	Viscous velocity
y	Vertical wall distance

Superscript and subscript

$(-)'$	Instantaneous fluctuation
$(-)^+$	Wall unit
$(-)_b$	Bulk quantity
$(-)_c$	Chamber or Characteristic
$(-)_i$	Index notation
$(-)_t$	Throat
$(-)_{avg}$	Average
$(-)_{in}$	Inlet
$(-)_{out}$	Outlet
$(-)_{wc}$	Wall - coolant side
$(-)_{wg}$	Wall - hot gas side

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1 Introduction and Objectives

In the last three decades, space activities have focused on a collaborative international approach, with programs like the International Space Station (ISS), where multiple nations have collaborated. This led to increased attention on scientific research, satellite communications, and Earth observation while laying the ground for future commercial activities. However, space remained a niche field, with high entry costs, limited technological infrastructure, and a no-risk attitude due to the extreme conditions of the space environment.

In the last years, the space sector has grown significantly and become a leading global industry. Not only national agencies but also new private companies are spreading worldwide, from the most famous American companies, like SpaceX and Blue Origin, to emerging European start-ups, such as Isar Aerospace, The Exploration Company, and Maia Space. According to the Space Foundation report [1], the global space economy was valued at more than 570 billion in 2023, representing a 7% increase from the previous year, which demonstrates the rapid expansion of this sector despite global economic challenges. Projections from the same reports indicate that the space industry could reach the milestone of $1 trillion value by 2040, driven by advancements in satellite technologies, space tourism, and space exploration.
Commercial services, such as Earth observation, satellite communications, and the global navigation satellite system (GNSS), now drive the sector's growth, offering services to industries like telecommunications, agriculture, environmental monitoring, and defence. The demand for space services has increased significantly with the worldwide reliance on digital infrastructure in everyday life, including phones, the internet, and televisions.

The economic potential of this market is evident: government expenses remain significant, led by the United States, and are steadily rising globally, from the European Union to India, China, and Russia. However, private investments have also grown significantly. In 2023, space companies, mostly startups, have raised $17.9 billion in private capital, a record, according to the Space Capital report, [2], which reflects the growing attention to space-related technologies in the last years.

Alongside this economic growth of the space sector, the technology has significantly benefited from the larger interest in space activities and has reached new levels of maturity, enabling faster development cycles, enhanced mission capabilities, and more cost-effective solutions across launch, satellite, and in-orbit services.
However, this rapid advancement brings with it a new set of challenges. The increasing

frequency of launches and the growing number of satellites in orbit raise serious concerns about space traffic management and collision risks. Additionally, the demand for higher-performance systems, such as more efficient propulsion, long-duration life support, and advanced materials, pushes the limits of current engineering capabilities.
Traditional production methods often fall short when it comes to manufacturing lightweight and customised components. In response to these challenges, additive manufacturing (AM) is increasingly being adopted as a transformative approach within the space industry.

AM has changed the design and production phases, allowing for free (or nearly) design choices and a reduction in production cost and time.
These techniques are based on the superposition, or addition, of layers of material until the final shape is obtained, contrary to traditional milling and turning processes, which are based on the removal of material.
This new concept of production has been adopted in recent decades, especially in the aerospace sector, where the optimisation of every component needs to be pushed to the technological limit. The number of connections between components needs to be minimised to reduce the number of failure modes, and even a few grams of material can have a significant cost and impact on the global design. However, it is essential to consider that the new process also presents new challenges. In their book, [3], Gradl et al. present all the pros and cons of the innovative technique, bringing attention to new steps and procedures that must be introduced in the production chain. In the introduction, they say, [3]:

"*As with any new process, it is tempting to become captivated by the excitement of AM and its application and to give little forethought to the potential economic and technical challenges.*"

This sentence perfectly summarises the dual nature of AM. While it presents groundbreaking opportunities for innovation, it also introduces complex challenges that must not be overlooked. Indeed, new challenges and failure modes are introduced, and they must be taken into account and properly addressed. One example is the failure that occurred during the testing of the LLAMA engine at the NASA facilities, presented in Fig. 1.1.
During operation, after only 9 seconds from the ignition, the additively manufactured chamber fails in its cylindrical part, opening up in two parts with a clear and neat cut. Three identical chambers were printed with the same process, parameters, and machine. One of the chambers was tested for thousands of seconds under hot operations without failing, showing that the failure was not due to a design issue.
The failure was instead due to a process interruption during the printing of the combustion chamber, which resulted in internal defects in the material. This example is useful to understand that every engineering rule and methodology has to be challenged when using a new technique, such as AM. As presented in the chapters of this work,

the traditional understanding of flow behaviour over smooth surfaces and the common methods to evaluate heat transfer are no longer valid in the presence of AM components.

Figure 1.1: LLAMA failure, picture adapted from [4].

In general, many aspects of the 3D printing process are not fully understood or controlled, which lays the groundwork for ongoing research worldwide. Universities and industries are striving to master new processes, fine-tune the parameters involved, and control the features of the final product. Additionally, when dealing with space applications, the margin of error is very limited or non-existent, as the operating conditions are extremely harsh. Thus, the implementation of a new process is possible only after thorough investigations, testing, and inspections to ensure the overall quality of the product in terms of structural strength, hydraulic operations, and thermal performance.

This work does not aim to solve all the new challenges emerging from the use of this technology, but rather focuses on one specific issue: the impact of surface roughness on the performance of propulsion components manufactured through additive processes. The objective of this study is to quantify the performance alterations induced by the intrinsic roughness of AM surfaces, with particular attention to their aerodynamic and thermal implications under representative operating conditions.
Given the importance of reliability in space applications, where even minor deviations in component behaviour can have critical consequences, this research begins with the assumption that AM-induced surface roughness is not only a geometric imperfection but a functional feature that must be properly characterised and accounted for. The work is based on the hypothesis that surface roughness significantly alters flow behaviour and heat transfer in internal cooling channels and combustion chamber surfaces, leading to increased pressure losses, higher wall temperatures, and enhanced coolant heat pickup capabilities. Additionally, it is assumed that the operating conditions of the targeted application significantly influence the impact of roughness on performance.

The central research questions addressed in this study are:

1. To what extent does surface roughness induced by AM influence flow dynamics and heat transfer within propulsion components?
2. Does the effect of surface roughness remain consistent across varying operating conditions, particularly with changes in Reynolds number?
3. What is the impact of small-scale, process-induced surface non-uniformities on the overall performance of propulsion components?
4. How accurately do current industrial roughness models represent the physical behaviour observed in the experiments?
5. In what ways can simulation models be enhanced or calibrated to more accurately predict the influence of surface roughness?

By addressing these questions, this study helps bridge the gap between fundamental fluid dynamics and practical engineering applications in the context of AM.
To answer these questions, the study employs a multi-scale approach, combining Direct Numerical Simulations (DNS), experimental testing of AM-manufactured components, and Reynolds-Averaged Navier-Stokes (RANS) simulations, which are commonly used in industrial design. This methodology allows for a comprehensive understanding of roughness-induced phenomena and enables the evaluation of current modelling strategies through a direct comparison with test data.

In the next chapter, a few basic concepts of space propulsion are used to present the application field. Afterwards, a brief introduction to the AM techniques is presented, highlighting the pros, cons, and challenges of this technique, with a focus on the most commonly used process in space applications.
In Chapter 3, the most important concepts of turbulent flows are revisited, with attention on their simulation in numerical environments. Here, the state of the art of surface roughness modelling is presented and compared to the well-known literature on the smooth-wall theory.
In Chapter 4, the experimental facility is introduced, which is available at the Chair of Space Mobility and Propulsion at TUM and has been used for all the experiments presented in this work. The new test articles and the thrust chamber are described along with the instrumentation used to record the test data.
The presentation of the results begins in Chapter 5, where the DNS simulations are introduced and analysed in detail. This chapter aims to investigate, on a small-scale level, the features in the flow that are induced by the presence of a rough element. Here, several simulations are performed under different flow conditions and in three different geometrical domains to compare the results between a smooth duct, a ribbed duct, and a randomly rough duct.
In Chapter 6, the results from the experiments are presented. In this chapter, the topic of investigation shifts to a more applied framework, focusing on classical rocket

operating conditions, while maintaining the dualism between traditionally manufactured and additively manufactured combustion chambers. The difference in performance is presented using the temperature and pressure measurements available from the setup.
Lastly, Chapter 7 returns to the numerical frame. RANS simulations are performed, and the most recent roughness models are implemented to account for the AM process. The parameters are adjusted to better match the experimental results. The objective of this chapter is to provide a quantitative analysis of the accuracy of currently available models for surface roughness.

2 Fundamentals

2.1 Basics About Rocketry

A chemical propulsion system often features a combustion chamber where the chemical reaction occurs. The hot gases are then accelerated through the nozzle, where their expansion generates the thrust desired by the system. A simplified representation of a thrust chamber is given in Fig. 2.1.

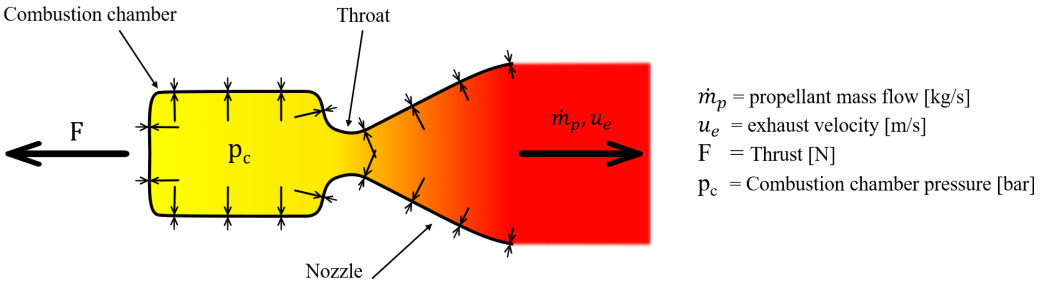

Figure 2.1: Basic rocket schematic.

The thrust of a rocket is directly proportional to the amount of gases that are expelled per unit of time, the hot gases' velocity, and the pressure contribution given by the nozzle, [5, 6].

Several kinds of propulsion systems exist. Thrust can be created with one propellant only, *monopropellant*, or two, *bipropellant*. The bipropellant systems can be further subdivided into liquids, solids or hybrids. In the frame of this work, attention will be paid to chemical propulsive systems featuring two liquid propellants, called liquid rocket engines (LRE).

2.1.1 Cooling Technologies

Liquid rocket engines require a significant effort in thermal management. The combustion taking place in the chamber reaches temperatures in the order of $3000 - 3500\ K$, and no material would be capable of both withstanding these temperatures and facing the high-pressure loads without being cooled down.
The evaluation of the heat flux that the chamber wall must deal with is not easy and depends on many characteristic quantities of the combustion, such as pressure, propellants, chamber geometry, and others. However, an easy and relatively accurate formulation

for the hot gas convection coefficient was proposed by Bartz in [7], and its formulation is reported in Eq. (2.1).

$$h_g = \frac{0.026}{d_t^{0.2}} \left(\frac{\mu_g^{0.2} c_p}{P_r^{0.6}} \right) \left(\frac{p_c}{c^*} \right)^{0.8} \left(\frac{d_t}{r_t} \right)^{0.1} \left(\frac{A_t}{A} \right)^{0.9} \sigma \tag{2.1}$$

$$\text{with} \quad \sigma = \left[\frac{T_{wg}}{2T_c} \left(1 + \frac{\gamma - 1}{2} M^2 \right) + \frac{1}{2} \right]^{-0.68} \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-0.12} \tag{2.2}$$

The symbols are detailed in the nomenclature section. This formulation considers the hot-gas properties, the pressure and efficiency of the combustion, the chamber's geometry, and the flow's velocity.
From this equation, an estimate of the heat flux can be obtained, which can vary from 1 to $150 - 200\ MW/m^2$. Additionally, as it is visible from Eq. (2.1), the heat flux also varies as a function of the area of the chamber, reaching its maximum at the throat, where the flow is sonic.
Different cooling technologies have been developed to maintain the engine's wall temperature within a reasonable range. Fig. 2.2 presents the usual application range of the most common cooling technologies. As it is visible, simple techniques such as ablative and radiative cooling are efficient solutions with relatively low heat flux levels. As the combustion chamber pressure increases, the heat flux also rises proportionally, necessitating more complex thermal management systems. Regenerative cooling, either alone or in conjunction with film cooling, encompasses the widest range of applications and is indeed the most commonly used in LRE applications.

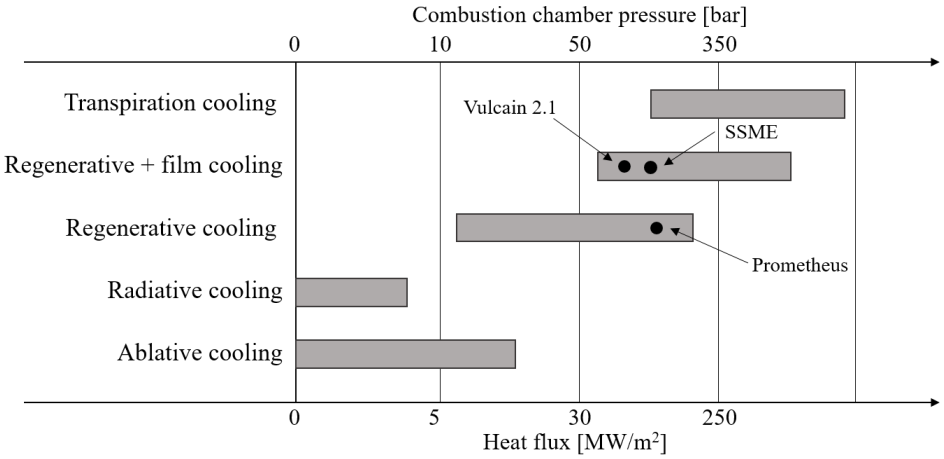

Figure 2.2: Common range of application of several cooling technologies. Image readapted from [8].

Typically, the cooling channels are milled in the turned liner, which is usually made of high-resistant copper, like CuAgZr, used for the Vulcain, and the SSME or CuCrZr, used for the RD-120. The liner is then closed with a high-strength jacket that resists

the high pressures in the chamber and the cooling channels. The introduction of additive manufacturing completely changes the traditional manufacturing process of cooling channels, making it possible to design new cooling paths and avoid complex and expensive procedures. Fig. 2.3 shows a schematic of the final structure and introduces some important design parameters.

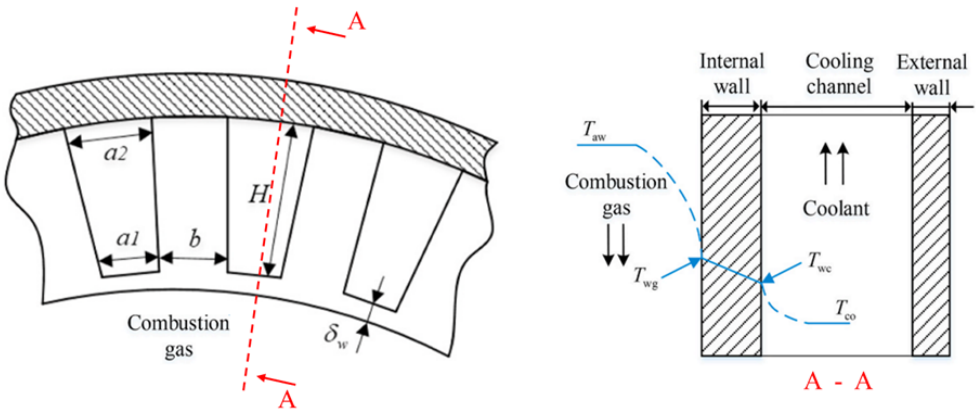

Figure 2.3: Schematic of a regenerative cooling system. Image readapted from [9].

The δ_w is usually referred to as "hot-gas wall thickness", while b is the "fin thickness", and a and H are the dimensions of the cooling channel. Besides, on the right of Fig. 2.3, a simplification of the heat transfer process occurring in the presence of a regenerative circuit is presented. The hot gases at temperature T_{aw} exchange by convection with the hot wall surface at temperature T_{wg}. The same happens on the "cold" side, where the coolant at T_{co} exchanges with the wall at T_{wc}. Inside the wall, the heat is transferred by conduction. In this simplification, the heat flux can be expressed as:

$$\dot{q} = h_g(T_{aw} - T_{wg}) \tag{2.3}$$

where h_g has been defined in Eq. (2.1), while T_{aw} needs a more precise definition: it is called "adiabatic wall temperature" and is the temperature that an insulating wall would assume when exposed to the hot gases. This value corresponds to the total temperature of the hot gas flow if stagnating on the wall, [10].

Every cooling methodology has its drawbacks, resulting in a more or less pronounced loss of engine performance or an increase in system weight. When dealing with regenerative circuits, it is of great interest to optimise the coolant path to reduce the pressure losses (directly related to the weight of the system). Furthermore, accurate thermal and hydraulic performance predictions enable accurate estimations of structural temperatures and chamber life. For first-stage applications, an error of 40 K in the prediction of the hot-gas wall temperature can lead to a life of the component that is half of the expected one, [11].

2.2 Additive Manufacturing

Additive manufacturing technologies are transforming industries such as aerospace, automotive, energy, and medical, enabling faster and more cost-effective production, greater design freedom, and reduced lead times. AM, especially metal-based, experienced significant growth in the 2010-2020 period. Aerospace benefits from AM, given the complexity of the components required to work in extreme environments. Particular aspects of AM that influence the business case for propulsion applications are:

1 High complexity application: part complexity that was previously not possible or too expensive can now be obtained and used to increase structural efficiency, reduce system mass, or it may allow for flow features that are not possible with traditional methods.

2 Low-production-volume application: aerospace propulsion components are typically produced in medium or small production volumes and often have long lead times. AM requires no custom tooling and can manufacture complex designs that, with traditional techniques, would require multiple assembly operations.

3 Time-critical applications: lead time is often an economic consideration. Metal AM compresses lead time through the use of digital designs and manufacturing.

Talking more generally about cost, schedule, and performance improvements, the characteristics of the AM process that make it attractive for propulsion component applications are many. However, despite the numerous advantages, AM is not always the best technique to use for every component and must be considered as another available manufacturing process, which must be weighed against traditional techniques. A detailed overview of the available processes is given in [3]. In this section, only a few aspects are mentioned to help the understanding of the following results.

2.2.1 AM Techniques

The term "Additive Manufacturing" is quite general and indicates a generic process that requires adding material to create an object, in contrast to the more common material removal. In the frame of AM processes, several techniques exist with different features, characteristics, and fields of application. The most commonly used are: *Laser powder bed fusion* or L-PBF, *Laser powder direct energy deposition* or LP-DED, *Electron beam powder bed fusion*, or EB-PBD, *Arc-wire directed energy deposition*, or AW-DED, and *Cold Gas Spray*, or CGS. A comparison of these procedures is reported in appendix A.1 and in [3].

Among these, L-PBF is the most mature and widely used metal AM technology because it offers high precision, excellent material properties, and supports a wide range of metals. Its long development history, industrial adoption (especially in aerospace and medical), and well-established standards make it the best choice for critical, high-performance parts. As L-PBF was also used for the production of the components tested in this

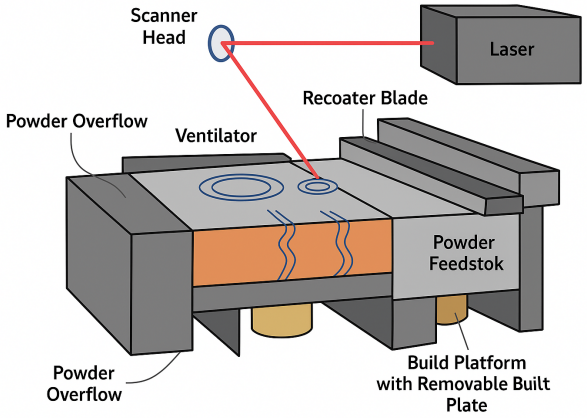

Figure 2.4: Schematic of an L-PBF machine.

work, only this process is detailed.
L-PBF uses a laser to precisely melt thin layers of metal powder to form a solid cross-section. This is done layer by layer until a full part has been built.
To achieve a solid layer, several parameters, including laser power, scan speed, and layer thickness, must be aligned to produce stable melt pools. The process is presented in Fig. 2.4.
At a macroscopic level, nearly all L-PBF processes begin with the deposition of an extremely thin layer of metal powder, typically just tens of micrometres thick, onto a flat build plate. One common method for distributing this powder involves a feedstock tank, which is incrementally raised by a precisely controlled piston. This piston elevates the powder in the vertical direction, allowing a defined amount to be delivered to the build area. Simultaneously, the build platform is lowered by an equivalent distance using a second piston, ensuring consistent layer thickness. A recoater blade or scraper then moves across the surface, spreading the powder evenly from the raised feedstock platform onto the build plate.

Recoating can be performed in one direction or both directions, and any surplus powder is collected in a catch pan located just beyond the build area.
After each new layer of powder is spread, the laser system, mounted above the build platform, activates. The laser follows a predefined toolpath derived from a CAD model, selectively melting the powder layer as it moves. The laser beam is directed and focused by mirrors mounted on galvanometers, which control its position and direction.
To prevent oxidation during melting, the entire process takes place within a controlled environment, typically filled with an inert gas, such as argon.

Each one of these parameters affects the final product in several ways. The thickness of each powder layer influences both the total printing time and the surface quality of the component: a thicker powder layer leads to increased surface roughness but faster printing. The laser calibration and the prescribed toolpath are of great importance

for heat management during the process, which can lead to thermal deformations of the printed part. The design of the component itself must also be compatible with the process. For example, highly inclined surfaces cannot be printed without support structures or are printed with increased rugosity and a lower material density caused by the numerous imperfections.
All these aspects, and many others, alter the final product, which will deviate more or less from the nominal design file. More details can be found in [3].

2.2.2 Performance

The geometrical imperfections introduced in the previous section induce performance deviations which are difficult to predict and can be non-negligible in the overall functional behaviour of the part.
For example, the performance of the component is highly dependent on its material properties and surface quality, and both aspects differ significantly between traditionally manufactured parts and additively manufactured parts.
AM parts present variability in microstructure and porosity, which can introduce uncertainties in mechanical properties. The variability in grain morphology and the presence of defects (such as lack-of-fusion pores, keyhole defects, or gas porosity, [3]) contribute to lowering the mechanical strength, ductility, and fracture toughness, [12, 13]. Additionally, since AM materials often do not have standardised processing parameters across different machines and manufacturers, the exact properties of a given component remain unknown unless extensive material characterisation is performed, [14].

The surface quality of the ALM components often results in surface roughness, which is particularly challenging, as it affects mechanical performance, friction loss, and thermal behaviour. The stair-stepping effect caused by layer deposition, along with partially sintered powder particles, results in significantly rougher surfaces compared to those of traditionally machined components. This high roughness not only diminishes the fatigue life of AM parts but also affects their functional performance in applications involving contact mechanics, fluid dynamics, or heat transfer.
Surface roughness has a direct influence on fluid friction losses and energy dissipation. When fluids flow over rough surfaces, turbulence is induced, increasing drag and resistance to motion. This effect is particularly significant in hydraulic systems, injectors, and cooling channels, where the Reynolds number is high. The irregularities on the surface create near-wall turbulence, leading to higher pressure drops and requiring greater input energy to maintain desired flow rates, [15]. In high-performance applications, excessive roughness can reduce the system's efficiency and result in significant energy losses. While post-processing methods, such as laser polishing, electropolishing, and mechanical grinding, can improve surface quality, they introduce additional costs and may not always be feasible for complex geometries.
In addition to frictional effects, surface roughness plays a critical role in heat transfer efficiency. In applications such as heat exchangers and cooling channels, roughness

affects the flow and the thermal performance by promoting turbulence and enhancing convective heat transfer, [16]. The trade-off between increased pressure losses and enhanced heat transfer capabilities is currently an area of research.

In summary, the performance of AM components is linked to uncertain material properties and rough surfaces, leading to unknown mechanical behaviour and altered hydraulic performance, respectively.
In this work, the different mechanical behaviour is left outside of the investigation, while the process-induced surface roughness is analysed and its effects on the performance of the part are quantified. Once the printing parameters are set, several components are printed, tested and compared against the traditionally manufactured counterparts. Additionally, a smoothing technique, Electro-Discharge Machining (EDM), is applied to some components to decrease the roughness of the surfaces and analyse the impact of this post-processing technique on the part's performance.

3 Numerical Methods

3.1 Introduction to the Numerical Analysis

A multi-disciplinary approach is followed to investigate the effect of surface roughness on the flow. Experimental testing is used to replicate real engine operating conditions, and is presented in Chapter 6, while numerical simulations explore small-scale phenomena and evaluate the predictive accuracy of computational tools.
Both DNS (Direct Numerical Simulation) and RANS (Reynolds Averaged Navier Stokes) are employed in this work. DNS reveals the physical interactions between rough walls and flow, whereas RANS enables comparison between test data and predictions from a faster and more practical model.
This chapter reviews the fundamentals of turbulent flow theory as a basis for the numerical models presented. It introduces the Navier–Stokes equations and explains their application in the DNS model. The RANS equations and the turbulence models examined are then described, along with the necessary adjustments to represent a rough wall. This chapter is not intended to be fully exhaustive, and a more detailed explanation of these concepts can be found in Batchelor's books [17], Pope, [18] and Panton, [19].

3.2 Governing Equations of Turbulent Flows

3.2.1 Navier-Stokes Equations

The law of conservation of mass states that for any system closed to all transfers of matter, the mass of the system must remain constant over time. For a generic fluid, this concept is summarised in Eq. (3.1).

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u_i) = 0 \tag{3.1}$$

where ρ is the density of the fluid and u_i are the velocity components.
A complete set of equations capable of solving the motion of the fluid is obtained if the momentum equation is also considered, as shown in Eq. (3.2). Based on Newton's second law, this equation relates the fluid acceleration to the body forces and external forces experienced by the fluid itself.

$$\frac{\partial \rho u_j}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_i} = -\frac{\partial p \delta_{ij}}{\partial x_i} - \frac{\partial \tau_{ij}}{\partial x_i} \qquad \text{for } j = [1, 3] \tag{3.2}$$

where u_i is the i-th component of the velocity vector, x_i the i-th component of the coordinate vector, δ_{ij} is the Kronecker-Delta and τ_{ij} is the viscous tensor. Furthermore,

the energy equation is derived to account for the transport of energy. In this work, the energy equation is defined in terms of the total energy E, as in Eq. (3.3).

$$\frac{\partial \rho E}{\partial t} + \frac{\partial \rho u_i E}{\partial x_i} = -\frac{\partial u_j p \delta_{ij}}{\partial x_i} - \frac{\partial u_j \tau_{ij}}{\partial x_i} - \frac{\partial q_i}{\partial x_i} \tag{3.3}$$

where q_i is the i-th component of the energy flux, that in the case of single-specie flows is equal to the conduction component only: $q_i = -k\frac{\partial T}{\partial x_i}$. To complete the system of equations, a relation between pressure, temperature, and density is required, known as the Equation of State (EoS). Several equations have been proposed in the literature, each one with a specific range of application. A good overview is given in [20].
In this work, the equation of state from Span et al. [21] is used in Chapter 7 for the calculation of the thermodynamic properties of nitrogen.

3.2.2 Turbulent Channel Flow

This work focuses on wall-bounded flows, which are common engineering applications, such as the flow in cooling channels or around aircraft and ships. In these cases, the velocity vector can be considered parallel to the wall, and the near-wall behaviours are found to respect universal behaviours. As sketched in Fig. 3.1, a long duct ($L >> h$) is considered, which has a high aspect ratio, $b >> h$.

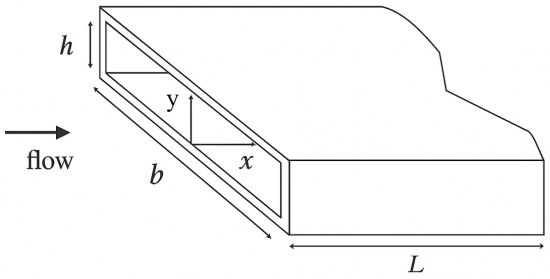

Figure 3.1: Duct flow reference system.

The mean flow is predominantly in the x direction, while the mean velocity varies in the vertical direction, y. The extent of the spanwise direction z is large enough to consider the flow independent of that direction. The Reynolds number, defined as $Re = L_c \rho u / \mu$, is used as an indicator to define the flow as turbulent if the Re is higher than 3000.

At the wall, being the velocity equal to zero, the wall shear stress is entirely due to the viscous contribution, [18]:

$$\tau_w = \rho \nu \left(\frac{du}{dy} \right)_{y=0} \tag{3.4}$$

where ν is the kinematic viscosity. In this case, it is possible to define viscous scales that are appropriate for velocity and length scales in the near-wall region. These are the friction velocity and the viscous length scale:

$$u_\tau = \sqrt{\tau_w / \rho} \tag{3.5}$$

$$l^+ = \nu/u_\tau \tag{3.6}$$

These parameters allow us to measure the distance from the wall in wall units, denoted by:

$$y^+ = \frac{y}{l^+} = \frac{u_\tau y}{\nu} \tag{3.7}$$

It can be noticed that y^+ has the same formulation as the Reynolds number, and can be intended as a local Reynolds number, whose magnitude defines the importance of the viscous and turbulent phenomena.
In 1925, Prandtl postulated that in flows with a high Reynolds number, there is an inner layer close to the wall in which the velocity profile is determined by the viscous scales only, [22]. More generally, he defined a dimensionless velocity as:

$$u^+ = u/u_\tau \tag{3.8}$$

such that this velocity is a function of the wall unit, $u^+ = f_w(y^+)$, and that this function is universal. It was found that for every flow, $u^+ = y^+$ for $y^+ < 5$. This layer is the *viscous layer*.
The viscous layer, as defined by Prandtl, has been demonstrated in numerous studies that successfully replicate the same velocity profile in the vicinity of the wall, regardless of the flow conditions. However, this applies only to smooth walls. In Chapter 5, the velocity profile close to the wall is calculated and presented for a smooth and a rough duct, highlighting important differences.

Von Karman then extended this theory, [23]. He formulated the so-called log-law, as in Eq. (3.9).

$$u^+ = \frac{1}{\kappa} \ln y^+ + A \tag{3.9}$$

where $\kappa = 0.41$ is the von Karman constant and $A = 5.2$. This formulation has found excellent agreement with DNS data for $y^+ > 30$ and $y/\delta < 0.25$. This region is referred to as the *log-law* region. In between the viscous sublayer and the logarithmic region, a "buffer layer" is found.
Contrary to the viscous layer, the logarithmic layer is found both on smooth and rough walls, as shown in Chapter 5.
Lastly, no universal law is found for the outer flow, namely $y/\delta > 0.2$. The resulting velocity profile, scaled with respect to wall units and plotted against the y^+ is called *law of the wall*.

3.3 DNS

3.3.1 Cost

The brute-force approach that solves the Navier-Stokes equations directly as presented in section 3.2.1, is called Direct Numerical Simulation and resolves *all* the scales of the motion, from the smallest to the largest vortex. This approach was not feasible until

some decades ago, when computing power did not allow such simulations.

Since all the scales of the motion are resolved, it is trivial to understand that the resolution requirements significantly determine the computational cost. Every cell must be small enough to resolve the smallest vortex, while the simulation domain has to be large enough to ensure that the vortices can develop freely and be representative of a real domain. In this section, only a few major constraints are reported, while for a more complete overview of the topic, the reference book [18] is suggested.
The resolution of the smallest dissipative motion is determined by the dimension of the smallest eddy (or vortex), which is defined by the *Kolmogorov length scale*:

$$\eta = \left(\frac{\nu^3}{\varepsilon}\right)^{1/4} \tag{3.10}$$

where ε is the dissipation rate.
It can be found that, to satisfactorily resolve all the scales of motions, the ratio between the grid size Δx and the Kolmogorov length scale has to respect Eq. (3.11), [18].

$$\frac{\Delta x}{\eta} < 2.1 \tag{3.11}$$

The stringent requirement mainly drives the maximum Reynolds number that can be simulated with a given grid. A dependence of the number of grid elements can be extrapolated as a function of the Reynolds number, which leads to $N^3 \propto Re^{9/4}$.
Additionally, to advance the solution in time, it is necessary that a fluid particle moves only in a fraction of the grid spacing Δx in the timestep Δt. In practice, the Courant number is imposed as follows:

$$Co = \frac{u\Delta t}{\Delta x} \leq C_{max} \tag{3.12}$$

It is concluded that the number of floating point operations required to complete the simulation is proportional to the product of the number of grid cells, N^3, and the timesteps to be performed, M. The previous trend with the Reynolds number is thus updated in $N^3 M \propto Re^3$. It is then clear that to perform a simulation with a Reynolds number of 10^6, which is very common in the aerospace industry, a total of 10^{18} floating-point operations would be needed, which would require hundreds of years to be performed.
The cost and computational effort of DNS are not suitable for the daily activities of industrial engineers. However, DNS provides an extremely powerful research tool. With the growing computational power of the last few decades, DNS has seen a steep growth, leading to an increasingly larger number of works. The primary reason is that a detailed study of the flow in a small portion of the domain can lead to significant discoveries and enhance the understanding of flow patterns. These discoveries can later be modelled and implemented in cheaper simulations adopted in industry.

3.3.2 Implementation

Since the cost of each simulation is very high, the preparation time for a simulation is typically lengthy, and many parameters must be correctly set to avoid wasting computing resources. A preliminary simulation is run first to assess the correct implementation of the mesh, the solver, the boundary conditions (BC), the initial conditions (IC), and the timestep.

In the case of duct flows, good practice requires the length of the simulated duct to be at least 20 times longer than the hydraulic diameter, [24], such that a completely developed flow is obtained in terms of velocity profile and boundary layer thickness. This requirement is bypassed with the use of cyclic boundary conditions, [25]. Cyclic, or periodic, boundary conditions are used in computational fluid dynamics simulations to define a repeating domain. The two boundary faces are then treated as physically connected. This results in the flow entering/exiting one surface and exiting/entering the other linked surface. With this procedure, infinitely long domains can be simulated using a much shorter one.
However, when using periodic BC between the inlet and outlet, fixed inlet velocity or mass flow boundary conditions cannot be set, while only an initial velocity field can be decided. Iteration after iteration, the average bulk velocity of the fluid decreases due to friction losses, causing parameters such as the mass flow or, more generally, the bulk Reynolds number to decrease over time. This effect is highly undesired. Thus, to counteract it, a source term is added to the momentum equation in Eq. (3.2), which becomes:

$$\rho\frac{\partial u_j}{\partial t} + \rho\frac{\partial u_i u_j}{\partial x_i} = -\frac{\partial p\delta_{ij}}{\partial x_i} - \frac{\partial \tau_{ij}}{\partial x_i} + \Gamma \tag{3.13}$$

where Γ is an additional pressure gradient required to balance the friction losses and maintain a constant mass flow rate along the streamwise direction during the simulation. In the simulations presented in Chapter 5, a constant Reynolds number is desired throughout the duration of the simulation. Thus, a uniform in-space value of Γ is used.

As for the momentum equation, the energy equation is also affected by the use of periodic boundary conditions. By setting a fixed heat transfer boundary condition on one wall, the fluid absorbs a certain amount of energy every time it flows through the duct. Thus, its bulk temperature increases with the simulation time. Again, this is an undesired effect, and a source term is added to the energy equation in Eq. (3.3), resulting in:

$$\rho\frac{\partial E}{\partial t} + \rho\frac{\partial u_i E}{\partial x_i} = -\frac{\partial u_j p\delta_{ij}}{\partial x_i} - \frac{\partial u_j \tau_{ij}}{\partial x_i} - \frac{\partial q_i}{\partial x_i} + \Phi \tag{3.14}$$

where Φ is a heating term, which is adjusted in such a way that the bulk temperature T_b is kept constant in time. This approach is taken from the work of Patankar et al. in [26], where the periodicity characteristics of heated flows in close ducts are investigated. Note that a thermally developed flow, which allows periodic boundary conditions, only occurs in the case of incompressible flow. Therefore, the density in Eq. (3.13) and

Eq. (3.14) has been taken out of the derivatives as it will be considered as a constant in the frame of this work.

As mentioned, a fixed inlet velocity cannot be set when cyclic BCs are used. However, an initial velocity field $u_i(x)$ can be defined in the domain, and then a momentum source term Γ can be used to control the desired bulk velocity. If a uniform velocity field is assigned, a very long simulation time is required before the flow turns turbulent. This transition time is very expensive and not useful for the simulation output, as the time average of the quantities needs to be performed over a time interval in which the flow is fully turbulent.
To save computational time and facilitate the transition from laminar to turbulent flow as quickly as possible, artificial turbulence is computed and applied as an initial velocity field to the simulation domain. This artificial turbulence model has been developed in our research group and presented in detail in [27]. It elaborates the strategy proposed by Shur et al. in [28], with a superposition of randomised harmonics following a reference spectrum as initially proposed by Kraichnan et al. in [29]. In this way, the synthetically generated velocity field respects the statistics of a generic turbulent flow. More details on the statistical description of the harmonics and their validation can be found in [27], which is further utilised in [30–33].
This artificial turbulence is imposed throughout the entire domain only at the first time step of the simulation, allowing for a faster transition to a fully turbulent flow. The flow is considered turbulent after 200 convective times (defined as $C_t = h/u_b$, with h being the half-height of the duct, or the channel radius) as suggested in [34].

3.4 RANS

3.4.1 Reynolds-Average

Turbulence, characterised by its complex and unpredictable dynamics across various scales, poses significant challenges for accurate numerical simulation. The Reynolds-Averaged Navier-Stokes (RANS) equations offer a practical solution. By averaging flow variables over time, RANS equations provide a computationally efficient approach for modelling turbulent flows, thus facilitating their application in engineering contexts.
These equations have their foundations in the Reynolds Decomposition, introduced in [35], where every quantity is split into its time-averaged part and its fluctuating part. Additionally, a Favre average is needed for a non-constant density fluid, which corresponds to a density-weighted average, [36].
By splitting every field and substituting this definition into the continuity, momentum, and energy equations presented respectively in Eq. (3.1), Eq. (3.2), and Eq. (3.3), the final form of the Reynolds-Averaged Navier-Stokes equations is obtained as follows:

$$\frac{\partial \overline{\rho}}{\partial t} + \frac{\partial \overline{\rho u_i}}{\partial x_i} = 0 \tag{3.15}$$

$$\frac{\partial \overline{\rho u_j}}{\partial t} + \frac{\partial \overline{\rho u_i u_j}}{\partial x_i} + \frac{\partial \overline{p}\overline{\delta_{ij}}}{\partial x_i} = \frac{\partial}{\partial x_i}(\overline{\tau_{ij}} - \overline{\rho u_i' u_j'}) \tag{3.16}$$

$$\frac{\partial \overline{\rho}\overline{E}}{\partial t} + \frac{\partial \overline{\rho u_i}\overline{E}}{\partial x_i} + \frac{\partial \overline{u_i}\overline{p}\overline{\delta_{ij}}}{\partial x_i} = -\frac{\partial}{\partial x_i}(\overline{q_i} - \overline{\rho u_i' E'}) + \overline{\tau_{ij}\frac{\partial u_i}{\partial x_j}} \tag{3.17}$$

The momentum Reynolds-averaged equation in Eq. (3.16) appears to be similar to the Navier-Stokes equation in Eq. (3.2), apart from the presence of the term $\overline{\rho u_i' u_j'}$. The term $\overline{u_i' u_j'}$ indicates the so-called "Reynolds stresses", which play a crucial role with respect to the non-averaged case.
Indeed, if this term were zero, the two equations would match.
For a general three-dimensional flow, there are then four independent equations governing the mean velocity field, namely the mean continuity equation, Eq. (3.15), and three mean momentum equations, Eq. (3.16). However, these four equations contain more than four unknowns because the Reynolds stresses must be determined in addition to the three velocity components and the pressure. This is called "the closure problem".
This problem can only be solved with models and approximations which try to replicate the real behaviour of these stresses. The most famous approach was advanced by Boussinesq in 1877, [37], whose hypothesis states that the deviatoric part of the Reynolds stresses is proportional to the mean rate of strain, as follows:

$$\begin{aligned} -\overline{u_i' u_j'} + \frac{2}{3} k \delta_{ij} =& \nu_T \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) \\ =& \, 2\nu_T \overline{S_{ij}} \end{aligned} \tag{3.18}$$

where k is the kinetic energy, defined as $k = 0.5\overline{u_i u_i}$, while ν_T is a positive scalar coefficient called *turbulent viscosity* or *eddy viscosity*. The turbulent viscosity hypothesis assumes that the Reynolds stresses anisotropy, $a_{ij} \equiv \overline{u_i' u_j'} - 2/3 k \delta_{ij}$, is determined by the mean velocity gradients $\partial \overline{u_i}/\partial x_j$, and that this relation is the one shown in Eq. (3.18), where S_{ij} is the mean rate-of-strain tensor.
In the following sections, this work presents only two models to evaluate this turbulent viscosity and close the set of equations, which are the two models used for the simulations presented in Chapter 7. For more information on the derivation of these or other models, refer to other textbooks as [18, 38, 39].

3.4.2 Spalart-Allmaras

The Spalart-Allmaras (SA) turbulence model is a RANS approach that requires the solution of a single governing equation, which describes the eddy turbulent viscosity. It was first presented in 1992 by the authors who gave the model its name, [40].
The model was specifically developed for use in aerodynamic applications involving wall-bounded systems as well as in turbomachinery applications. The SA model includes multiple parameters and coefficients, and it is computationally efficient because there is only one differential equation to solve. In its original form, this model is effective at low-Reynolds numbers if the boundary layer region (near-wall region) is properly

resolved, meaning it has a very fine mesh. Afterwards, the model was extended to allow for independent application, blending the viscous formulation into a logarithmic formulation.
The transport equation is presented in Eq. (3.19).

$$\begin{aligned}\frac{\partial\overline{\rho\nu}_{sa}}{\partial t}+\frac{\partial\overline{\rho\nu}_{sa}\overline{u_i}}{\partial x_i} =& C_{b1}\overline{\rho}\overline{S}\overline{\nu}_{sa}-C_{w1}\overline{\rho}f_w\left(\frac{\overline{\nu}_{sa}}{d}\right)^2+\\ &\frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left((\overline{\mu}+\overline{\rho\nu}_{sa})\frac{\partial\overline{\nu}_{sa}}{\partial x_j}\right)+C_{b2}\overline{\rho}\left(\frac{\partial\overline{\nu}_{sa}}{\partial x_j}\right)^2\right]\end{aligned} \tag{3.19}$$

where $\overline{\nu}_{sa}$ is the modified turbulent viscosity and is linked to the eddy turbulent viscosity as $\nu_T = f_{v1}\overline{\nu}_{sa}$. All the coefficients and functions C_{b1}, $\overline{S}$, C_{w1}, f_w, d, σ, C_{b2} and f_{v1} are listed in [40].

3.4.3 $k-\omega$ SST

The $k-\omega$ SST turbulence model is a two-equation eddy-viscosity model that has become very popular in aerospace industries thanks to its flexibility and wide range of applications. The model was proposed by Menter in 1993, [41].
The shear stress transport (SST) formulation combines the best of two previously published models, $k-\varepsilon$ and the $k-\omega$, where k is the turbulent kinetic energy, ε is the dissipation rate, and ω is the specific dissipation. Using a $k-\omega$ formulation close to the boundaries allows the model to be directly usable down to the walls without requiring any additional damping functions. The SST formulation also switches to a $k-\varepsilon$ behaviour in the free-stream, thereby avoiding the problem of the $k-\omega$ being too sensitive to the inlet free-stream turbulence properties. To achieve this, the two models are multiplied by a blending function and then added together. The blending function is designed to be one in the near-wall region, activating the $k-\omega$ model, and zero in the free-stream region, switching to the $k-\varepsilon$. The two governing equations are presented in Eq. (3.20) and Eq. (3.21).

$$\frac{\partial\overline{\rho k}}{\partial t}+\frac{\partial\overline{\rho k u_i}}{\partial x_i}=\frac{\partial}{\partial x_j}\left(\Gamma_k\frac{\partial\overline{k}}{\partial x_j}\right)+\overline{\mu_T}S^2-\overline{\rho}\beta^*\overline{k\omega}+G_b \tag{3.20}$$

$$\frac{\partial\overline{\rho\omega}}{\partial t}+\frac{\partial\overline{\rho\omega u_i}}{\partial x_i}=\frac{\partial}{\partial x_j}\left(\Gamma_\omega\frac{\partial\overline{\omega}}{\partial x_j}\right)+\frac{\alpha\alpha^*}{\overline{\nu_T}}\overline{\mu_T}S_{ij}^2-\overline{\rho}\beta\overline{\omega^2}+G_{\omega b} \tag{3.21}$$

From these two equations, it is possible to compute the turbulent eddy viscosity as follows:

$$\nu_T=\frac{a_1 k}{max(a_1\omega;\Omega F_2)} \tag{3.22}$$

All the coefficients and the functions Γ_k, Γ_ω, β^*, β, α, α^*, a_1, Ω and F_2 are given and explained in the original work, [41].

3.5 Roughness Characterisation and Modelling

3.5.1 Wall-Similarity Hypothesis

The equations of the flow presented in the previous paragraph are only true under the assumption of smooth walls. In reality, highly irregular surfaces are often encountered. Surface roughness can be the result of the normal use of the product (wear and tear) or from phenomena like erosion, corrosion, and deposition of materials such as in gas turbines, deposits in combustion chambers, ice accretion on aircraft wings, or biofouling on ship hulls, [42–45]. Atmospheric boundary layer flows are also a field where the flow over terrain, plants, and urban buildings presents the same features as a flow over rough surfaces [46]. In these cases, the flow is influenced by rough elements, and the flow patterns are different from what is expected from the smooth wall theory.

Several methods exist to capture and quantify the roughness or topography of the surface, the most common of which are optical or tactile profilometry, X-ray Computed Tomography (CT), or scanning electron microscopy. The best approach must be selected based on the resolution and accessibility of the desired measurement point. Once a measurement is obtained, profile-based roughness parameters are used to describe the surface. The most common parameters are presented in Tab. 3.1, and explained in Fig. 3.2.

Parameter	Description	Mathematical description
R_a	Arithmetic mean deviation	$\frac{1}{L}\int \lvert y\rvert dx$
R_q	Root-mean-square roughness	$\sqrt{\frac{1}{L}\int y^2 dx}$
R_z	Maximum peak-to-valley height	$max(y) - min(y)$

Table 3.1: Profile-based roughness parameters: y is the vertical dimension, and L is the length of the measurement.

Darcy pioneered the study of roughness, [47], and showed that roughness influences the pressure drop by increasing the drag force and the blockage effects. The roughness influences the boundary layer and, thus, the heat and momentum transfer over the interface between the surface and the fluid, increasing the turbulence, destroying the viscous sublayer, and forming a roughness sublayer.

To create new engineering models for the boundary layer over rough surfaces, the structure of these types of flows must be understood. Currently, the greatest part of engineering models considers the roughness as a small perturbation to the smooth wall boundary layer theory, thus extending or correcting this model.
These models are based on the *wall-similarity hypothesis*, which is an extension of Townsend's Reynolds number similarity, [48]. In his book, [48], Townsend states that flows with similar Reynolds numbers, large enough to be fully turbulent, are characterised by nearly similar turbulent structures. This hypothesis is then extended to rough

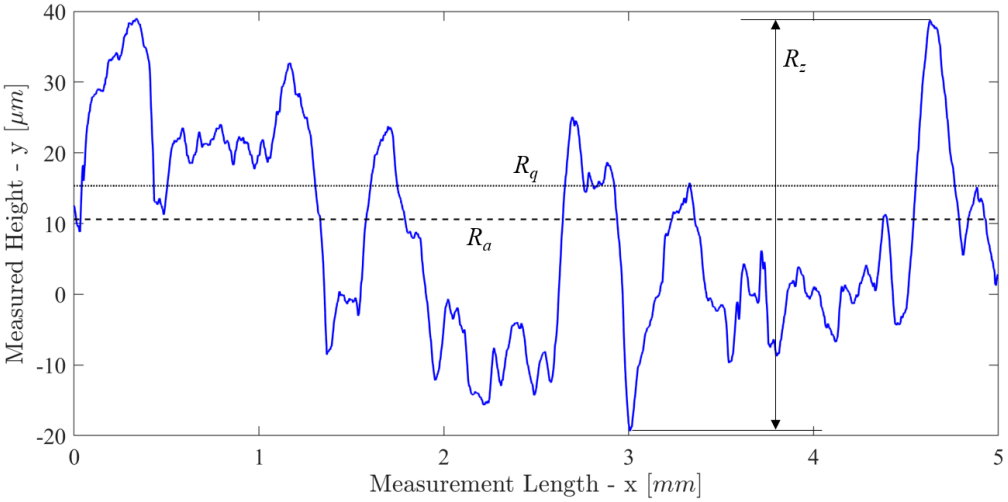

Figure 3.2: Representative roughness measurement, along with statistical roughness dimensions from Tab. 3.1.

surfaces. According to [48], if the Reynolds number is sufficiently high outside the viscous sublayer, the turbulent structures are independent of the surface conditions and the viscosity, except for wall shear stress. This means that the logarithmic law presented in section 3.2.2 is not affected by the roughness but is shifted due to the wall displacement. This hypothesis, later labelled as "wall similarity", was supported by several numerical [49–51] and experimental [52–54] works.

3.5.2 Sand-grain Roughness

When deriving the log-law in section 3.2.2, it is assumed that the near-wall flow is governed by viscosity, thus the name viscous sublayer. However, this is only true for hydrodynamically smooth walls, characterised by an average roughness height smaller than the thickness of the viscous sublayer. When this assumption does not apply, the effects of roughness on the flow must be considered by defining a roughness scale that reflects these effects. In literature, the most used parameter is the *equivalent sand-grain roughness height*. The equivalent sand-grain roughness approach was presented in the works from Nikuradse, [55, 56] and Schlichting [57].
This approach is derived from the experiments of Nikuradse, which were conducted with circular pipes internally coated with uniform spheres packed as densely as possible. The diameter of those spheres is called sand-grain roughness height, as in Fig. 3.3a, for the sake of similarity with the sandpaper. This concept is based on the definition of a single length scale that would characterise the roughness. However, when the roughness deviates from the uniform distribution observed in Nikuradse's experiments, a single parameter cannot be used anymore, and geometrical parameters may be employed, as shown in Tab. 3.1. Therefore, an "equivalent" sand grain roughness, k_s was introduced by Colebrook and White, [58], as in Fig. 3.3b. The equivalent sand grain height is an

equivalent roughness height for irregular and non-uniform surfaces that produces the same results as the uniform sand grain of Nikuradse's experiments.

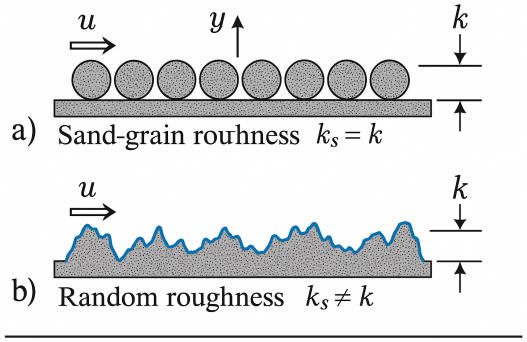

Figure 3.3: Comparison between a) roughness made of spherical elements and b) random roughness profile. Geometrical roughness height k coincides with the equivalent sand-grain roughness k_s only in the first case.

As stated by the similarity law, if the Reynolds number is high enough, the roughness modifies the law of the wall by shifting the velocity profile. In this context, the "roughness Reynolds number" is defined as:

$$k_s^+ = \frac{k_s u_\tau}{\nu} \tag{3.23}$$

Based on this number, the flow regimes are classified into three main categories, [55]:

- Hydraulically smooth regime: if $k_s^+ < k_{smooth}$, the roughness protrusions are entirely embedded in the viscous sublayer. Thus, skin friction and drag are not affected.
- Transitionally smooth: if $k_{smooth} < k_s^+ < k_{rough}$, both the Reynolds number and the roughness size affect the skin friction and the drag.
- Fully rough regime: if $k_s^+ > k_{rough}$, the rough elements protrude outside the viscous sublayer, which is destroyed, and they shift the logarithmic profile downwards. The friction drag increases significantly and is independent of the Reynolds number.

where k_{smooth} and k_{rough} are thresholds for which different values have been suggested in the literature, [15]. In this work, attention is focused on the fully rough regime, which is characteristic of high roughness or high Reynolds number flows.

As mentioned, the equivalent sand grain roughness is not directly correlated to the geometrical dimensions of the roughness. However, it is related to the effects of the roughness on the flow, thus, to the increase in friction and drag that are caused.
Up to now, no precise and universal correlation exists for determining the exact value of k_s across different geometries. Numerous studies and empirical correlations, some

summarised in Tab. 3.2, have been proposed for specific flows, applications, and roughness characteristics, but their validity is typically limited to small application ranges. Other investigations, [54, 59, 60], have highlighted that a single roughness parameter cannot fully capture the effects of roughness. Despite these extensive efforts, predicting k_s remains challenging and demands case-specific parameter tuning for each process or phenomenon generating the roughness.
Consequently, experiments remain essential for quantifying the pressure drop induced by surface roughness or variations in the flow's velocity profile, and for extrapolating an equivalent sand-grain parameter. This methodology has been adopted in this work, as will be presented in Chapter 6 and Chapter 7.

3.5.3 Mean Flow over Rough Walls

The major impact of surface roughness on the flow is the perturbation in the boundary layer, which leads to an increase in wall shear stress and, consequently, to increased heat and momentum transfer close to the wall. Depending on the flow regime, the viscous sublayer may be partly or completely destroyed and replaced by a roughness sublayer. However, according to the wall-similarity hypothesis, the outer layer remains unaffected. Based on these considerations, researchers have attempted to modify the existing smooth-wall models presented in section 3.2.2 to incorporate roughness effects. In 1954, Clauser and Hama introduced a parameter called the *roughness function*, denoted as ΔU^+, to emulate the effects of a rough surface through a velocity shift in the log-law [66, 67]. Indeed, they observed that the surface roughness shifts the log-law downwards, indicating a reduced momentum with respect to the smooth case.
Clauser readapted the logarithmic velocity profile for turbulent flow presented in Eq. (3.9) as follows:

$$u^+ = \frac{1}{\kappa} \ln y^+ + A - \Delta U^+ \tag{3.24}$$

where κ and A have been introduced in Eq. (3.9), while the roughness function ΔU^+ is the new addition. Hama ran an extensive experimental study to better define the value of this parameter and proposed, for fully rough flows, the relation in Eq. (3.25).

$$\Delta U^+ = \frac{1}{\kappa} \ln k_s^+ + A - B \tag{3.25}$$

Schlichting proved that B is a constant with value 8.5, [57], and a significant amount of data agreed with it up to relatively low values of $k_s^+ \approx 100$. However, many other relations estimate the value of the roughness function, for example, the Grigson relation, [68], the White relation, [69] and the more recent De Marchis correlation, [70]. They mostly differ for high roughness Reynolds number, as discussed in Chapter 7.

3.5.4 Heat Transfer on Rough Surfaces

As for the velocity profile, the Reynolds analogy can also be used to estimate the wall's temperature profile. However, since the analogy is not maintained on rough walls, several

Researcher(s)	Roughness characteristics or application	Correlation
Botros [60]	Colebrook roughness	$k_s = 1.306R_q + 0.078R_q^2$
Sigal and Danberg [61]	A 2D dense arrangement of transverse roughness (bars, rods and ribs)	$k_s/k = 0.0164\Lambda_s^3.78$ if $\Lambda_s \leq 4.93$ $k_s/k = 139\Lambda_s$ if $\Lambda_s > 4.93$
van Rij et al. [59]	3D staggered patterns of spheres, spherical segments and cones	$k_s/k = 0.00321\Lambda_s^{4.925}$ if $1.4 \leq \Lambda_s < 4.89$ $k_s/k = 8$ if $4.89 \leq \Lambda_s \leq 13.25$ $k_s/k = 151.71\Lambda_s^{-1.1379}$ if $13.25 < \Lambda_s < 100$
Bons [62]	Real turbine blades roughness	$k_s/R_{ref} = 0.019\alpha_f^2 + 0.0736\alpha_f$ if $\alpha_f < 11°$
Flack and Schultz [63]	Sand-paper, commercial pipes, pyramids and packed spheres	$k_s = 4.43R_q(1 + s_k)^{1.37}$
Flack et al. [64]	Roughness with negative skewness	$k_s = 2.91R_q(2 + s_k)^{-0.284}$
Stimpson et al. [65]	3D random roughness generated from additive manufacturing processes	$k_s/D = 18\frac{R_a}{D} - 0.05$ if $\frac{R_a}{D} > 0.028$

Table 3.2: Correlations for the evaluation of k_s. Note: Λ_s is the roughness density parameter; α_f is the mean forward-facing roughness slope; s_k is the skewness; R_{ref} is the average of several maximum heights taken from a piece of surface.

studies were performed to predict the mean temperature profile over rough surfaces.
Kays and Crawford, [71], suggested a simplified relation for the thermal law of the wall, which has been used for decades. The equation is reported in Eq. (3.26).

$$T^+ = \Delta T_0^+ + \frac{Pr_t}{\kappa} \ln\left(\frac{32.6y^+}{k_s^+}\right) \tag{3.26}$$

where Pr_t stands for the turbulent Prandtl number. Setting $\Delta T_0^+ = 0$ would replicate the Reynolds analogy assumption, otherwise, its value must be determined empirically. This parameter accounts for the conductive heat transfer occurring in semi-stagnant fluid in the roughness cavities, [71].
Dipprey and Sabersky suggested in [72] an empirical correlation for the thermal correction ΔT_0^+, as:

$$\Delta T_0^+ = \frac{1}{0.8} k_s^{+m} Pr^n \tag{3.27}$$

with $m = 0.2$ and $n = 0.44$.
It is clear that in addition to the velocity shift ΔU^+, a thermal shift near the wall occurs, ΔT_0^+.
Surface roughness is also responsible for larger-scale motions that affect the core flow. Secondary motions of turbulent flow pipes and ducts were first categorised by Prandtl, [22], as primary and secondary, whether they occur due to the curvature of the channel or to the presence of corners in the cross-section of the duct. Despite their weak intensities, these motions can significantly influence the flow, alter the mean velocity profile, and enhance mixing, leading to improved heat transfer performance.
Surface roughness generates secondary flows, as shown in [73–75]. New recirculation areas have been found, both in the streamwise direction, caused by the flow impacting on the rough element and on the cross-section plane, [76]. When dealing with smooth ducts, the magnitude of these motions is in the magnitude of $1-2\%$ of the bulk velocity of the flow. Instead, for rough surfaces, their magnitude is increased, enhancing mixing and also affecting the core flow.
The effects of surface roughness on the heat transfer process are not yet universally quantified, however, it is known that heat transfer enhancement is usually lower than friction augmentation, thus, many researchers have tried to correlate the two increases with respect to the smooth case, [62, 77, 78], as shown in Chapter 7.

3.5.5 Roughness in CFD

Once the physical effects of surface roughness on fluid patterns and heat exchange are described, models are created and implemented to replicate those effects in CFD simulations.
Two turbulence models are considered in this work. For the SA model, the roughness correction developed by Boeing is used, [79, 80], while for the $k-\omega$ SST this work uses the correction from Aupoix, [81].
For both models, the correction changes the wall value of the turbulent parameters,

namely the modified eddy viscosity $\overline{\nu}_{sa}$ for the Spalart-Allmaras and the turbulent kinetic energy k, and specific dissipation ω, for the $k-\omega$ SST model. The wall boundary condition implemented in the roughness correction of the Spalart-Allmaras model is reported in Eq. (3.28)

$$\frac{\partial \overline{\nu}}{\partial n} = \frac{\overline{\nu_w}}{d_w + d_0(k_s)} \qquad \text{with} \qquad d_0 = 0.03 k_s \tag{3.28}$$

while the $k-\omega$ SST boundary conditions are shown in Eq. (3.29). For more details on the implementation of these wall boundary conditions and the turbulence models themselves, see [40, 41, 80, 81].

$$\begin{aligned} k_w^+ &= \max(0; k_0^+) \quad \text{with} \quad k_0^+ = \frac{1}{\sqrt{\beta^*}} \tanh\left[\left(\frac{\frac{\ln k_s^+}{30}}{\ln 10} + 1 - \tanh\frac{k_s^+}{125}\right)\tanh\frac{k_s^+}{125}\right] \\ \omega_w^+ &= \frac{300}{k_s^{+2}}\left(\frac{15}{4k_s^+}\right)^{-1} + \frac{191}{k_s^+}\left(1 - \exp\left(-\frac{k_s^+}{250}\right)\right) \end{aligned} \tag{3.29}$$

In both models, the only variable set by the user is the sand grain roughness k_s.
These corrections are needed to account for the velocity shift that models the presence of roughness. Nevertheless, they also affect the heat transfer performance of the fluid in the near-wall region.

Both the roughness corrections were found to simulate too high heat transfer performance, and the same author who developed the correction proposed fixing the heat flux by adjusting the turbulent conductivity. To do so, the turbulent Prandtl number, Pr_t, which is usually kept constant at a value of around $0.8-0.9$ is adjusted close to the wall, as described in [82] and indicated in Eq. (3.30).

$$Pr_t = Pr_{smooth} + Pr_{rough} \qquad \text{where} \qquad Pr_{rough} = (A\Delta U^{+2} + B\Delta U^+)\exp(-y/k_s) \tag{3.30}$$

In Eq. (3.30), the y stands for the distance from the wall, while the ΔU^+ can be estimated with empirical relations [68–70]. Lastly, A and B are usually polynomials depending on some geometrical features, [82], or fluid properties, [83].
These corrections have received considerable attention recently due to their ease of implementation. However, they have been developed and tested for open air flows, and they still need to be verified for internal flows with high Reynolds numbers, as discussed in Chapter 7.

4 Experimental Setup

4.1 TUM Space Propulsion Lab

All the experimental tests presented in this work have been run at the Space Propulsion Laboratory at TUM.
The test-bench allows experiments with gaseous oxygen (GOx) and gaseous methane (CH4), which are fed to the combustion chamber from an external gas storage, as explained in appendix A.2.
The setup, detailed in appendix A.2 has already been used in the past years for experimental research in heat transfer analysis, [84–86], combustion stability, [87], life-fatigue prediction, [88, 89], injector behavior, [90–92], film cooling, [93–95], and new ignition techniques, [96, 97]. An extension for cryogenic liquid oxygen (LOx) has also been set up and tested, [98], but will not be presented in the current work.

4.2 7-element Combustor

4.2.1 Traditional Setup

A schematic section is shown in Fig. 4.2. The combustor is fed with gaseous oxygen and methane, which enter the injector head tangentially and radially, respectively. The two propellants have separate manifolds to avoid pre-mixed mixtures. A sinter plate is present in each manifold to make the flow as uniform as possible.
The injector features seven shear coaxial tubes with no recess and no tapering angle. The distance from the wall and the distance between injectors are kept constant and equal to half the diameter of the injectors, resulting in a pattern of six elements displaced circumferentially and one in the centre. A schematic view is provided in Fig. 4.1. The injector dimensions are summarised in Tab. 4.1. For more information on the injector and its behaviour, see [87, 90].
The combustion chamber is made of CuCrZr and features four segments and a nozzle, denominated with letters "A-B-C-D-N" as visible in Fig. 4.2. The segments are held together by eight rods and hydraulic nuts, which limit the tension when the copper segments heat up and expand longitudinally.

GOx diameter	[mm]	4	GOx post wall thickness	[mm]	0.5
GOx post recess	[mm]	0	GCH4 diameter	[mm]	6
Inj. Area ratio A_{CH4}/A_{GOx}	[-]	0.7	$(L/D)_{GOx}$	[-]	19.6

Table 4.1: Injector parameters.

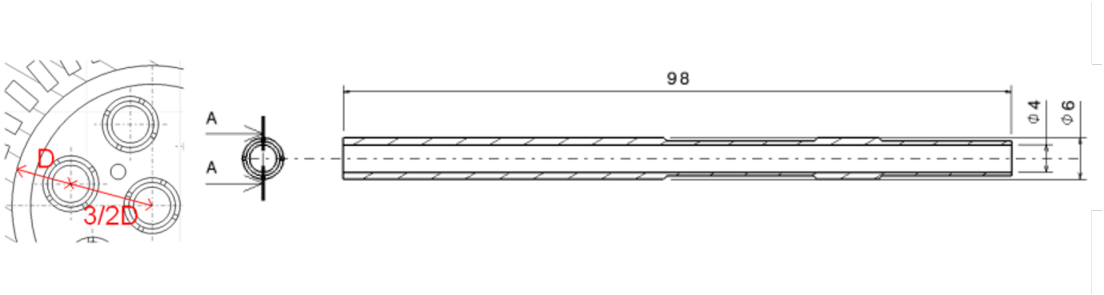

Figure 4.1: Injector element pattern (left) and cut of one element (right). Dimensions are in $[mm]$.

Every segment is equipped with cooling channels where flowing water absorbs heat from the combustion, maintaining the structural temperature within an acceptable range. The water passes through transition manifolds between the segments, where the coolant's pressure and temperature are measured. The water circuit that cools the chamber consists of three transition manifolds, one inlet manifold, and one outlet manifold. These manifolds enable the measurement of coolant temperature and pressure at multiple axial positions of the chamber. The water flows in the opposite direction with respect to the hot gases. The nozzle is cooled down by a separate circuit that has only one inlet and one outlet manifold. Some of the main geometrical features of the combustion chamber are summed up in Tab. 4.2.

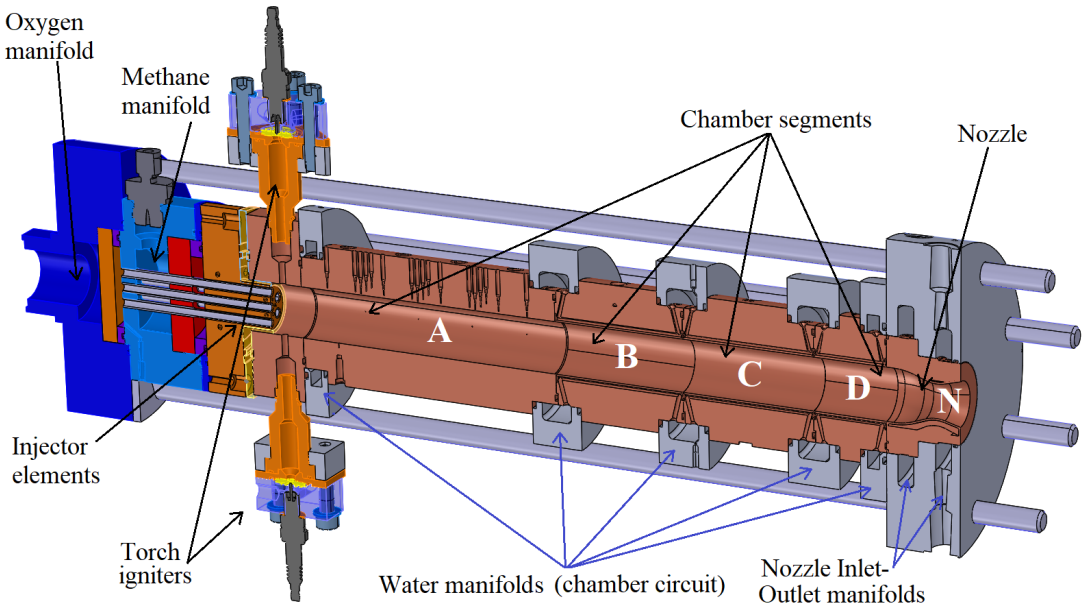

Figure 4.2: 7-element combustor: traditional setup.

The experimental setup is equipped with all the necessary instrumentation required to fully characterise the performance of the combustion and of the cooling circuit. Pres-

Chamber diameter	$[mm]$	30	Chamber length	$[mm]$	375
Throat diameter	$[mm]$	19	Expansion ratio	$[-]$	1.73
Contraction ratio	$[-]$	2.5			

Table 4.2: Combustion chamber parameters.

sure transducers are distributed along the axial direction of the chamber and are used to measure the static pressure in the combustion chamber along this direction, giving information on the acceleration of the hot gases. 90 thermocouples type T are spring-mounted in the combustion chamber wall, along the axial and azimuthal direction as shown in Fig. 4.3a, and at different radial depths, as visible in Fig. 4.3b. The spring system, visible in Fig. 4.3b, grants continuous contact between the thermocouple's tip and the wall. This leads to a precise measurement despite the vibrations of the chamber during operations.

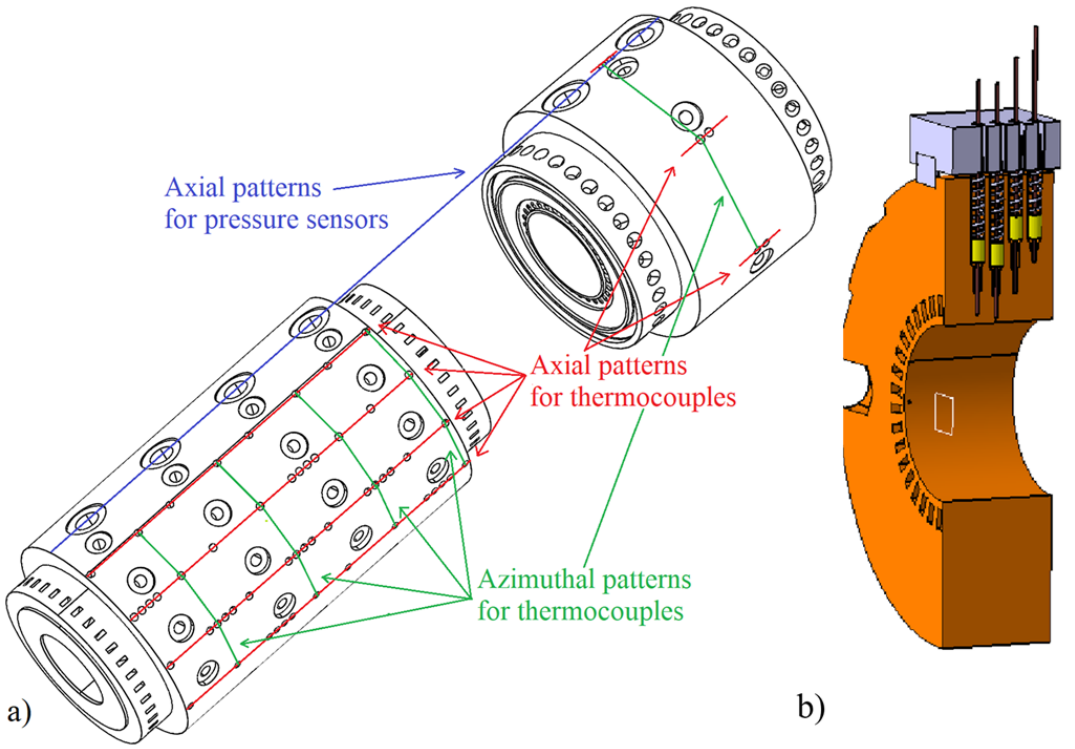

Figure 4.3: a) Exemplary patterns of thermocouples and pressure sensors port on segments A and B; b) mounting system for thermocouples placed in the combustion chamber wall.

Six thermocouples are placed in every water manifold, see Fig. 4.2, to measure the temperature of the flowing water after every segment and in six azimuthal positions (every 60°).
A complete overview of the equipped measurement instrumentation is given in Tab. 4.3.

Instrument	Location	Amount	Type
Pressure sensor	Chamber	13	WIKA A-10
Pressure sensor	Injector-Head	2	WIKA A-10
Pressure sensor	Cooling circuit	9	WIKA A-10
Dynamic pressure sensor	Injector-Head/Chamber	3	Kistler 6053BB60
Thermocouple	Hot-wall	90	T
Thermocouple	Injector-Head	2	T
Thermocouple	Cooling circuit	25	T
Mass-flow meter	GOx/CH4 Feed-lines	2	Rheonik RHM 12
Differential pressure	Cooling circuit	2	WIKA TRONIC 892

Table 4.3: Overview of the equipped instrumentation on the feedlines and on the engine.

4.2.2 New Setup

The aim of the experimental work is to test new additively manufactured segments, whose performance must be compared with that of traditionally manufactured ones. The new setup is shown in Fig. 4.4.

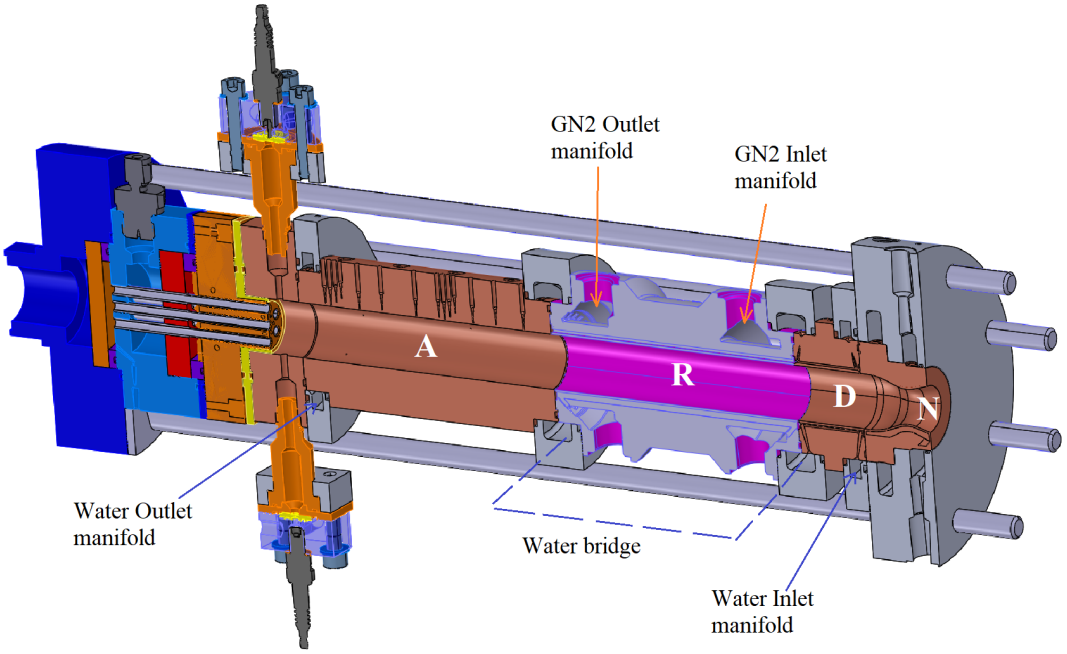

Figure 4.4: 7-element combustor: new setup.

The position of the new segment will be identified as “R” (substituting the positions “B” and “C”). The test article is as long as the sum of the two segments that have been substituted. By doing so, perfect comparability is ensured between the performance of

the traditional setup and that of the one with the AM segment.

The water circuit is bridged with flexible pipes from segment D to segment A, as shown in Fig. 4.4 with a dashed line.
Fig. 4.5 provides a clear view of the position and configuration of the test article. In the picture, it is possible to see the segment installed right after the first segment, "A", while the smaller segment, "D", and the nozzle are still missing.

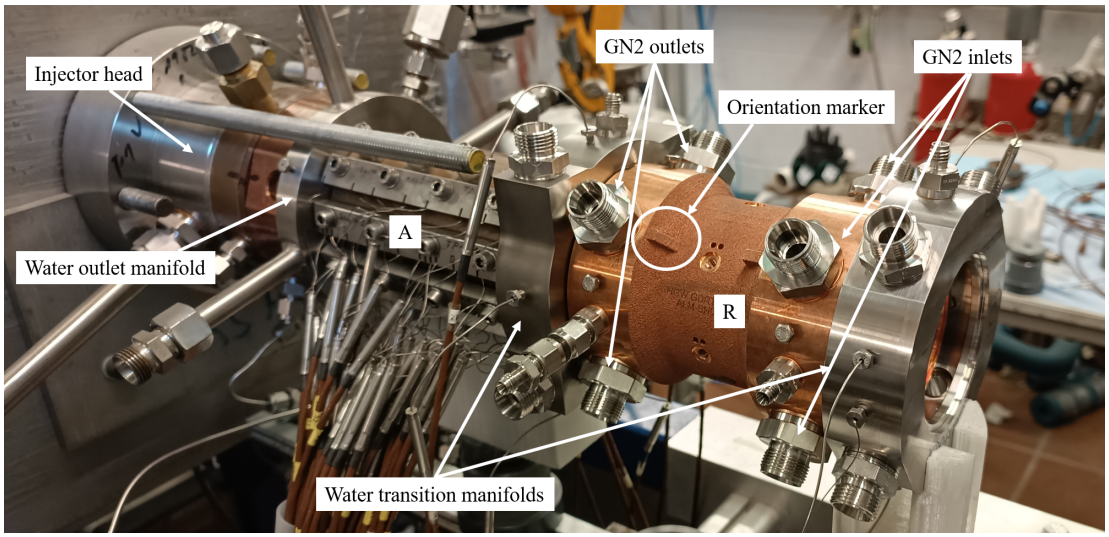

Figure 4.5: Test article during assembly procedure of the combustion chamber.

Fig. 4.5 also provides a good insight into the modularity of the chamber and the number of sensors with which the chamber is equipped.
Four AM segments have been printed and tested to compare the different effects that the post-processing techniques may have.
Additionally, a fifth segment, featuring cooling channels with increased height, is tested to investigate the effect of rough surfaces on a larger section. All the test articles have rectangular cooling channels.
The printed and tested specimens are summarised in Tab. 4.4.

Segment Name	Production Technique	Channel Shape	Combustion Chamber	Cooling Channels
1SR	LBM	Nominal	Machined	As-built
1SS	LBM	Nominal	Machined	EDM
1RS	LBM	Nominal	As-built	EDM
1RR	LBM	Nominal	As-built	As-built
2RR	LBM	Increased	As-built	As-built

Table 4.4: Printed and tested specimens characteristics.

The nomenclature used for each article is made of one number and two letters: the first number indicates the shape of the cooling channels. It is easy to remember that all segments have cooling channels with the same rectangular shape (identified with the number "1") except for the fifth article, whose channels have a larger aspect ratio and are identified with the number "2".
The second letter identifies the surface treatment of the combustion chamber, which can be smooth or rough, thus "S" or "R", respectively. The same logic applies to the third letter, which refers to the cooling channels.
The letter "S" is used to indicate smooth surfaces, both for the chamber and the cooling channels. However, the cooling channels are small and tortuous cavities, making the machining process unfeasible. Therefore, various smoothing processes are employed. The combustion chamber is machined while the cooling channels are treated with EDM (Electron-Discharge Machining). Both techniques smoothen the surface, however, the machining process delivers very smooth surfaces (average roughness in the order of 2 μm) while the electron discharge machining process decreases the roughness but to a lower extent (average roughness still in the order of $5 - 20$ μm), as reviewed in [99].
All the specimens are printed through the laser powder-bed fusion process. The difference between a machined and an as-built combustion chamber surface is visible in Fig. 4.6, where the segments 1SR and 1RR are compared.

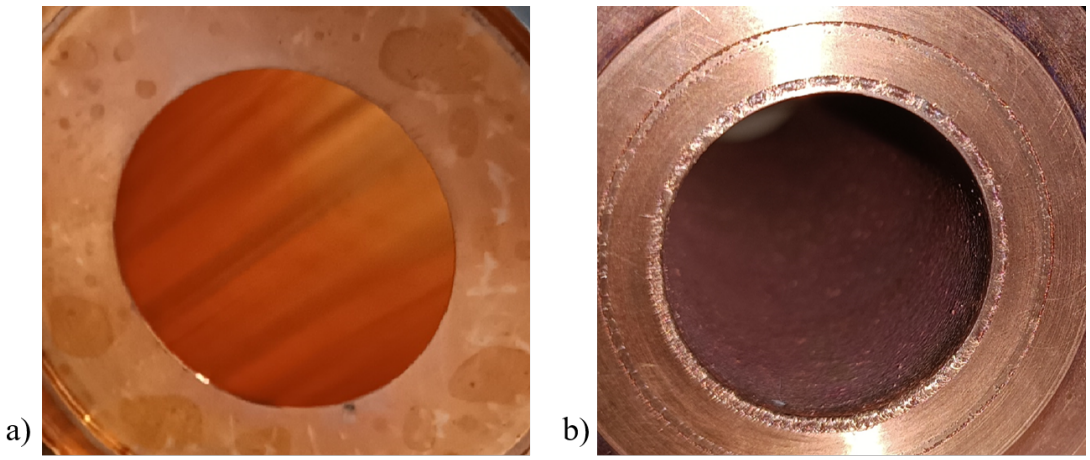

Figure 4.6: Comparison of combustion chamber surface quality between a) machined and b) as-built.

4.2.3 Ignition Procedure

The ignition of the flow occurs in a fraction of a second, and it is a crucial process for ensuring the stability of combustion and repeatable performance.
An igniter ring is designed and placed upstream of the first segment, "A", as can be seen in Fig. 4.4. A cut-view of the injector-head and the igniter ring is provided in Fig. 4.7a. This ring features two symmetric ports, which are placed symmetrically to each other,

such that the hot gases from the igniters are flowing in front of at least three injector elements.

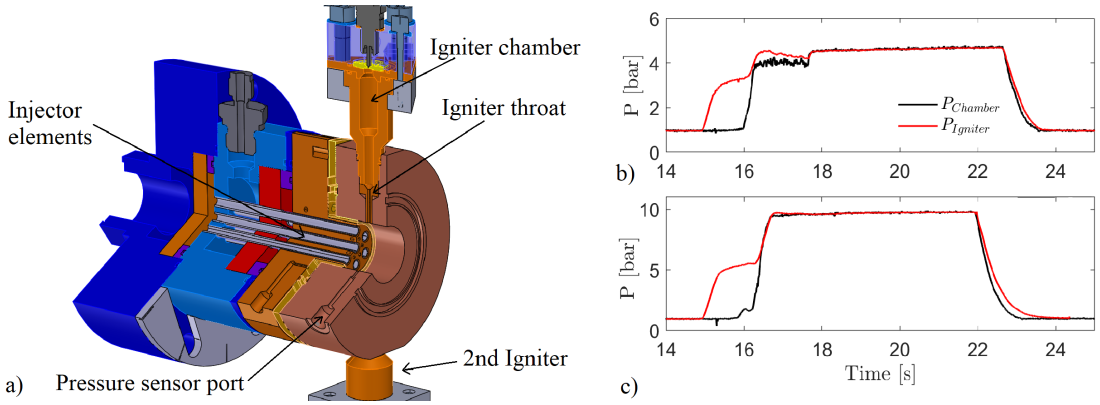

Figure 4.7: a) Cut section of the igniter ring design; b) behaviour of the combustion chamber pressure (black) and igniter chamber pressure (red) in two different ignition sequences.

Test activities have shown that one igniter is enough to start the main combustion in the chamber. However, the second igniter is also used for the uniformity and stability of the flame, as it has been proven to lead to more stable combustion.
In addition, the flow from the igniters can impact the main flow radially, disturbing it and causing instabilities. This can be seen in Fig. 4.7b, where the igniter's pressure is close to the desired chamber pressure, and the two flows interact.
A solution is found, as visible in Fig. 4.7c: the ignition of the main flow occurs at a higher target combustion chamber pressure than the igniter chamber pressure. In this way, the higher pressure in the main chamber is not influenced by the igniter behaviour.

5 Phenomenological Study through DNS

5.1 Motivation and Objectives

Due to the small characteristic dimension of the surface roughness, experiments are only capable of measuring global behaviours on a large scale because the dimension of the measurement probes largely overcomes the dimension of the phenomena. In this case, numerical simulations are used to simulate the flow patterns and vortical structures that form around the rough element. Despite their computational cost, DNS is the most suitable tool for recreating the interaction between a rough surface and the flow, allowing the evaluation of flow properties at every point within the domain.

In the past, numerous experiments and simulations have been conducted on ducts with square or rectangular cross-sections. Square ducts are first studied in Brundrett and Burroughs, [100], in which bulk Reynolds numbers $Re_b = d_H u_b/\nu$ (with d_H being the hydraulic diameter, u_b the bulk velocity and ν the kinematic viscosity) of 33.000 and 67.000 are considered.
It is known that the presence of corners creates secondary motions, as presented by [22] and [55], who explained that secondary motions are indicators of non-zero streamwise vorticity, which in turn is caused by a Reynolds stress gradient near the corner. In square ducts, the secondary motions are known to come in the form of eight counter-rotating vortices, which bring high-momentum flow from the core region to the corners and low-velocity flow from the near wall region to the core flow and are characterised by an intensity in the order of 1–2% of the bulk velocity, [101].

Numerically, secondary flows can only be caught if all the walls are simulated through no-slip boundary conditions. The first DNS of a square duct is that of Gavrilakis [102], which utilised a bulk Reynolds number of 4410. Higher Reynolds numbers are studied by Vinuesa et al., [103], Zhu et al., [104], and more recently by Pirozzoli et al., [24], Modesti et al., [105] and Zhang et al., [106].
In particular, in [24] and [107], Pirozzoli and Modesti analyse the relation between secondary flows and the bulk Reynolds number by creating a wide database of square duct flows, ranging from a Reynolds number of 4.400 to 84.000. They simulate a square duct using DNS and suggest a differentiation between a “core recirculation” and a “corner recirculation”. The former is characterised by a velocity scale associated with the bulk velocity of the flow and a length scale associated with the half-channel height. The latter has a velocity scale that is still similar to the bulk velocity, but a length scale that matches the inner-wall scaling length. At low bulk Reynolds numbers, the two circulations coincide, whereas at higher Reynolds numbers, the corner circulation moves

nearer and nearer to the duct corners, while the core circulation emerges in the bulk flow.

The geometry of a rough surface is difficult to simulate due to the high frequencies in the roughness profile, which are introduced by the scanning process, as explained in [108]. Thus, discrete roughness elements are often employed in numerical simulations to recreate the roughness protrusions within the boundary layer. For this purpose, riblets, cubes, or cones are used. Many efforts have been made to characterise the effects of such discrete elements in terms of heat transfer enhancement and momentum loss. However, most of these works did not simulate the side walls and only applied periodic boundary conditions on the side of the duct, thereby reducing the simulation's complexity to a 2D case. A few examples are the simulations performed by Leonardi et al. [109], Nagano et al. [110], Orlandi et al. [111], and Miyake et al. [112, 113]. The results of these works highlight new recirculation zones between rough elements, which enhance the diffusion mechanism from the wall to the core fluid region.
The only work in which a ribbed wall is simulated in a duct with four walls is the one from Mahmoodi-Jezeh and Wang, [114], where the same square duct is simulated three times, with different riblet dimensions, showing how the statistics of the flow vary with varying blockage ratio (BR) and how this influences the recirculation between riblets.
However, the understanding of the effects of the interaction between the riblets (or, more generally, the rough elements) and the side walls is missing, and although the influence of the Reynolds number has been thoroughly investigated for turbulent flows in smooth ducts, its influence on these newly generated vortices has never been discussed before. Thus, major phenomena related to the transport of momentum and energy must be quantified again, taking into account more complex wall geometries.

The objective of this chapter is to investigate the new patterns and effects that are found in this setup, which aims to resemble real cooling channel applications.
In this work, a similar setup to the one implemented in [114] is used. Firstly, an incompressible fluid is simulated through DNS in a square duct with transversal riblets on one wall. This setup aims to validate our simulations against similar domains found in the open literature and understand the influence of the Reynolds number on the flow.
Secondly, real roughness profiles are used to create a duct with rough walls. The same Reynolds numbers are simulated to facilitate a reasonable comparison of the two different cases. Heat flux is added to this second case to calculate the enhancement in heat transfer compared to the smooth case. The objective of the second test case is to simulate a scenario as closely as possible to real applications within the computational frame.

5.2 Flow over Riblets in a Duct

In this section, the results from simulations of a ribbed duct are presented. This work has already been published in [76] and is partially reported in this thesis to illustrate how the flow interacts with a wall protrusion.

5.2.1 Methodology

Each test case has been solved with the EBI (Engler-Bunte-Institut)-DNS code, [115–117], based on the open-source code OpenFOAM, [118], which solves the conservation equations for mass and momentum using a finite volume method (FVM). The same solver has already been widely used in previous works of our research group to simulate turbulent combustion flows [30–33]. The continuity and the momentum equations are solved as reported in Eq. (3.1) and Eq. (3.13).
The equations are solved in OpenFOAM using the PIMPLE algorithm with two correctors and two outer-corrector loops. The numerical algorithm has a second-order accuracy in both spatial and temporal discretisations. The discretised equations are iteratively solved until the residuals drop below 10^{-6} and the Courant number never exceeds 0.9, automatically adjusting the timestep at every iteration. For more information on the solver, see [117].
To facilitate understanding of the turbulence pattern and allow comparison with previous studies, the fluid is considered incompressible and has uniform properties.
Regarding the simulation setup, periodicity has been imposed between the inlet and outlet, and source terms have been added to the momentum and energy equations, as explained in section 3.3.2. The bulk velocity, $u_b = 1/A \int u \, dA$ is used to set the Reynolds number of the simulation. All the sides of the square duct are simulated through no-slip boundary conditions. The velocity field is initialised with a uniform streamwise velocity component, while artificial turbulence has been used to define the wall-normal and spanwise velocity components, as previously explained in section 3.3.2 .
The time average of each quantity is evaluated using Reynolds averaging, which is indicated with the overline, $\overline{(\cdot)}$.
The simulated domain is shown in Fig. 5.1.

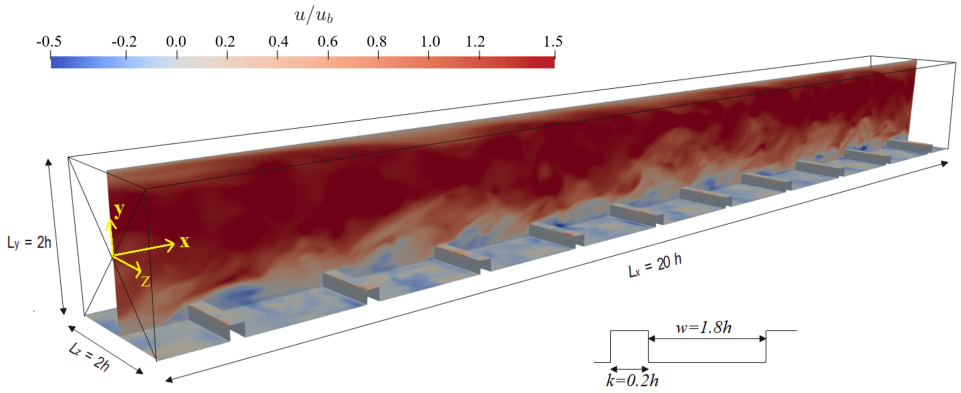

Figure 5.1: The contour of a generic instantaneous velocity field is presented along with the simulated domain and the relevant dimensions

A square channel is considered with the sides of length $2h$, and channel length of $20h$, where h is the half-duct height. The riblets are placed only on one wall, from now on

referred to as the "bottom wall". The riblets are squared with side k equal to $0.2h$ and spacing w equal to $1.8h$ between each other, repeating the same geometry with a period of $2h$. The channel length is chosen so that typical large-scale turbulent structures of the flow are properly captured. According to previous studies, a length of $4\pi h$ is considered enough, as used in [114, 119, 120].
To confirm this choice, the streamwise two-point correlation R_{uu} is computed. Due to the complexity of the flow, the two-point auto-correlation coefficient is defined as in Eq. (5.1), [114] [121, 122].

$$R_{uu}(x_{ref}, \Delta x) = \frac{\overline{u'(x_{ref})u'(x_{ref} + \Delta x)}}{\sqrt{\overline{u'^2(x_{ref})}\ \overline{u'^2(x_{ref} + \Delta x)}}} \tag{5.1}$$

where the notation $(\cdot)'$ stands for the instantaneous fluctuation of a certain quantity with respect to its time-averaged value. The correlation is evaluated considering the vertical centreline of the section at the centre of the domain, namely $x = 10h$ and $z = 0$. Different points along this line are used to evaluate the streamwise two-point autocorrelation. The results are shown in Fig. 5.2a. The value of R_{uu} drops to values close to 0 in a portion of the domain, which is less than a quarter of the total length of the domain. It can be stated that the domain is sufficiently long to allow the turbulent structures to develop completely.

Concerning the dimension of the mesh cells, precursor simulations have been run to assess the required dimensions, as presented in [123]. Due to the complex geometry and the strong ejection of fluid resulting when hitting the riblets, a uniform mesh has been preferred. As investigated in [124], the cell size must respect the requirement imposed by the Kolmogorov length, Eq. (3.11). The ratio between the resolution and the Kolmogorov scale is shown in Fig. 5.2b.

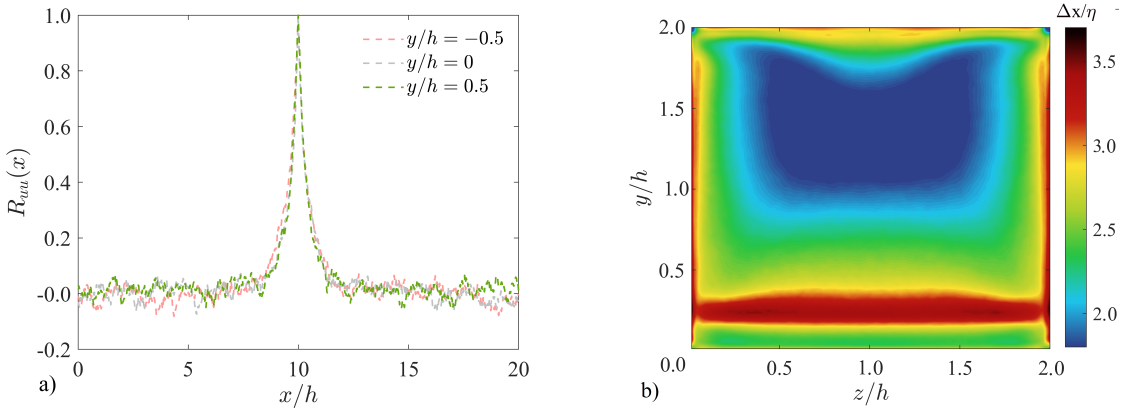

Figure 5.2: a) Streamwise two-point autocorrelation for the smooth (solid-line) and the ribbed (dashed-line) duct simulation; b) contour of the ratio between resolution and Kolmogorov scale, at $x/h = 0.6$.

The ratio never exceeds a value of 4 and assumes its highest values near the crest of the riblet, where the flow is shed in small vortices. Even if not strictly respecting the DNS requirements of Eq. (3.11), it is proved in [124] that this resolution is enough to correctly resolve the turbulence features of the simulation, alleviating its cost.
In total, four simulations are performed within the same square duct. One simulation has four smooth walls and has been used for validation and comparison purposes, see appendix A.3. The other three are performed with a fixed geometry having a ribbed wall, while the bulk Reynolds number is varied. Details on the implemented mesh, the Reynolds number, the convergence indicators, and the Kolmogorov length are provided in Tab. 5.1 for each simulation.

Simulation	Re_b	N_x	N_y	N_z	ε_p x 10^3	$(\Delta x/\eta)_{avg}$	$t_s/(h/u_b)$
S	3300	80	80	800	6.6	2.84	1920
R1	4400	120	120	1200	1.7	2.19	1918
R2	8800	140	140	1400	1.5	2.08	2044
R3	13200	160	160	1600	2.0	2.36	1890

Table 5.1: Simulations data: N_x, N_y, N_z are the number of cells in each direction, ε_p is a convergence indicator defined in [34], η is the Kolmogorov length; t_s is the simulation time.

Due to the geometrical domain, the friction velocity is evaluated following the approach used by Leonardi et al. in [125], defining $u_\tau = (\overline{P_d} + \overline{C_f})^{0.5}$, where $\overline{P_d}$ stands for the pressure drag while $\overline{C_f}$ is the skin friction drag. For a duct with riblets on one wall, the skin friction drag can be evaluated as follows,[125]:

$$\overline{C_f} = 1/w \int_w 1/Re_b^{-1} (\partial \overline{u}/\partial y)_{y=0} dx \tag{5.2}$$

The pressure drag instead, in case of periodicity in x and constant mass flow, can be derived as, [125]:

$$\overline{P_d} = 1/w \int_w 1/Re_b^{-1} (\partial \overline{u}/\partial y)_{y=k} dx - 1/w \int_w (\overline{uv})_{y=k} dx - 1/w \int_w 1/Re_b^{-1} (\partial \overline{u}/\partial y)_{y=0} dx \tag{5.3}$$

5.2.2 Velocity Field and Drag Coefficient

The main objective of this chapter is to highlight the effects of the side walls, the influence of the bulk Reynolds number, and the comparison between a smooth wall and a wall with transverse riblets.
Fig. 5.3 presents the flow pattern in the region between two riblets. The section is taken at $z = 0$ and shows the contour of the velocity magnitude underneath the streamlines of the flow. Three major recirculation regions are distinguished: a bigger bubble is evident after the riblet, and two smaller vortices are created in the corners right after the first

riblet and right before the following one. Their centres of rotation are respectively called "A", "B", and "C". After the recirculation zone "A", the flow re-attaches on the bottom of the channel, as it happens for *k-type* roughness, [109], and then starts rotating in a smaller vortex before the next riblet.

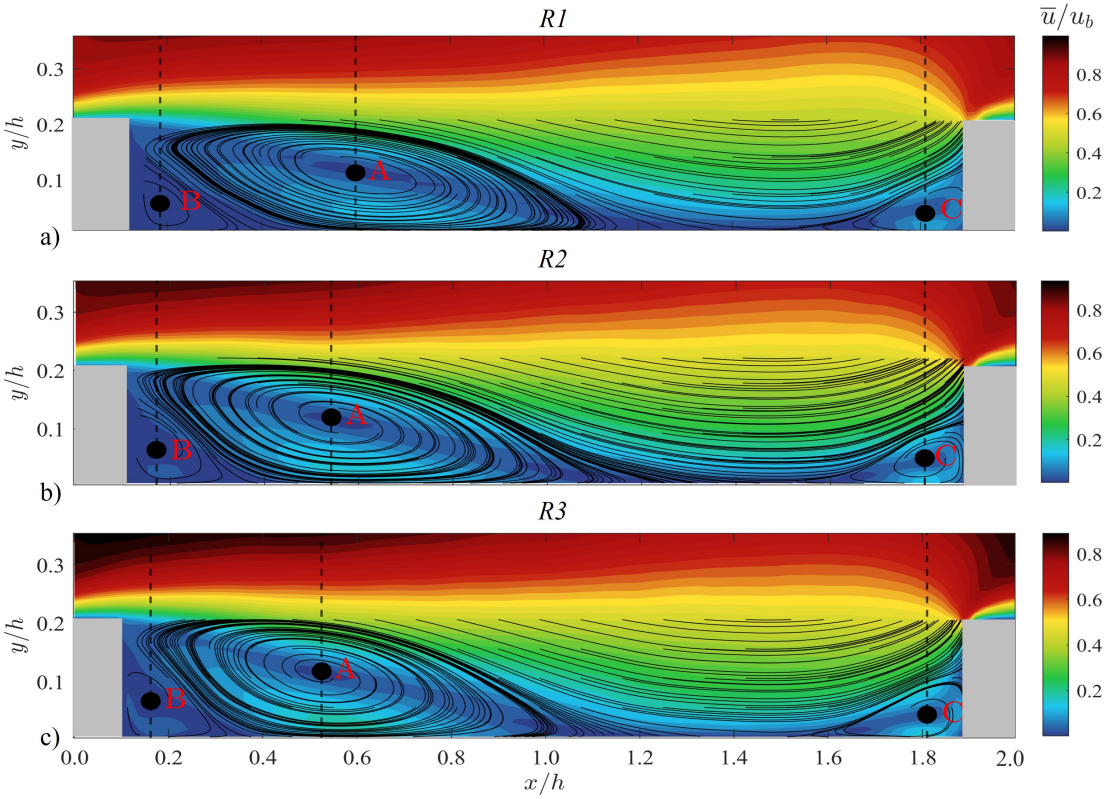

Figure 5.3: Recirculation region between two riblets: contour of the ratio between the velocity magnitude and the bulk velocity, and superposition of streamlines for simulations R1, R2 and R3. Points "A-B-C" highlight the centre of the recirculations.

It is found that the position of the smaller vortices "B" and "C" does not change significantly with respect to the Reynolds number. On the other hand, the bigger vortex, rotating around "A", gets smaller with increasing Re and the point "A" moves to the left, towards the riblet that generates it. Overall, the behaviour of the flow does not show a remarkable change with increasing Re, as it depends mostly on the geometry of the roughness elements, as investigated in [114].

As well explained in [126], the blockage ratio k/δ, where k is the height of the rough element and δ the diameter of the channel, differentiates the flow over rough surfaces and the flow over obstacles since it measures the influence of the roughness on the logarithmic layer. It is suggested that ratios of δ/k higher than 40 should be considered to retrieve any similarity of the near-wall flow with canonical flows. However, the same riblet dimensions used in this work are also employed in [109, 125, 127], and the authors found

that the similarity law of the dimensionless velocity profile is respected. To assess if the simulations performed in this work can be addressed as "flow over rough surfaces", the velocity profile is presented in Fig. 5.4a. In the figure, the quantities are made dimensionless with respect to wall units and are presented alongside other works with similar w/k ratios, where w represents the riblet spacing. Fig. 5.4a shows that the logarithmic region can still be found, and the results agree well with previous works.

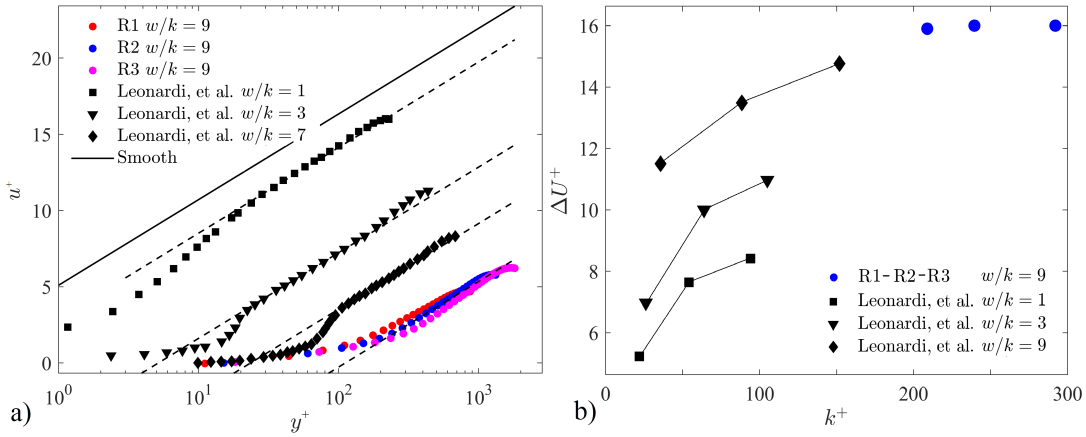

Figure 5.4: a) Non-dimensional velocity profile for simulations R1, R2 and R3 and results from Leonardi et al. [127], at different w/k ratios. Dashed lines are $u^+ = 1/0.41 \ln y^+ + C$, with C being 2.9, -6.2, -7.7, -11. b) Roughness function ΔU^+ against the roughness Reynolds number k_s^+, and compared with results from Leonardi et al. [109].

Additionally, the roughness function ΔU^+ is presented in Fig. 5.4b, again compared with the results from [109]. The results are presented as a function of the roughness Reynolds number $k_s^+ = k u_\tau / \nu$, and are close to the results from the simulations from Leonardi et al., [109], especially the ones having the same w/k ratio. However, the side walls simulated in our work increase the calculated friction velocity, resulting in higher roughness Reynolds numbers compared to those in the work by Leonardi et al. [109].

Moving back to the recirculation zones presented in Fig. 5.3, although the dimension and position of the streamwise vortices do not change with the Re, a strong impact is detected on the secondary flows visible on the plane perpendicular to the direction of the flow.
Fig. 5.5a presents the contour of the cross-flow velocity magnitude divided by the bulk velocity, $VW = \sqrt{\overline{v}^2 + \overline{w}^2}/u_b$, for the simulations R1 and R3, namely the lowest and highest Re of this work. The same representation is given in Fig. 5.5b for two smooth wall simulations taken from the work of Pirozzoli et al. in [24], with bulk Reynolds numbers equal to 4410 and 17800.
The presence of the riblets causes the flow to start recirculating, and the vortex occupies the entire channel, as indicated by the arrows, which represent the direction of the in-plane velocity vectors. When increasing the bulk Reynolds number, the cross-flow

velocity near the walls increases from 8% to 15% of the bulk velocity. This is particularly evident close to the sidewall and the bottom corners.

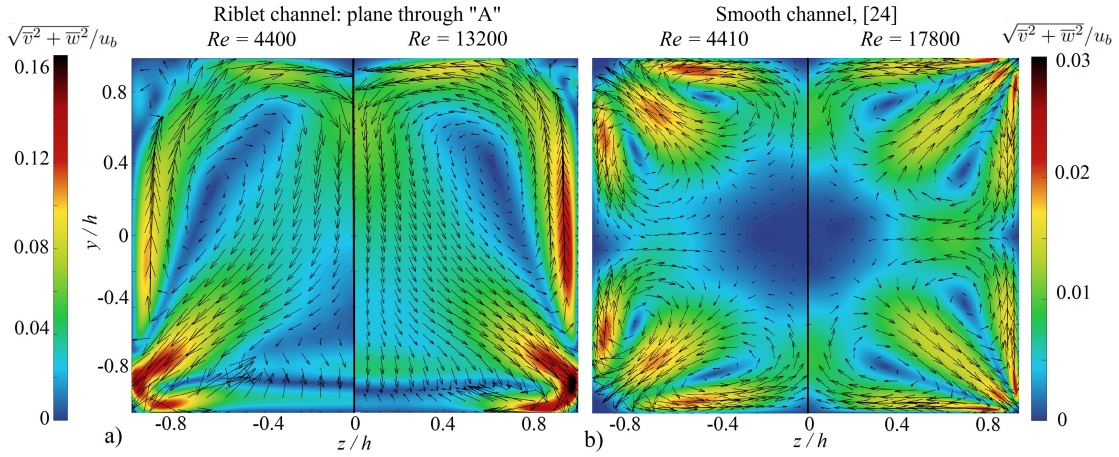

Figure 5.5: Contour of the dimensionless cross-velocity magnitude, with superimposed arrows presenting the in-plane velocity vector on a plane perpendicular to the flow. a) plane passing through the vortex-centre "A" for simulations R1 and R3; b) smooth square duct flow from Pirozzoli et al. [24], at similar Re.

Additionally, the typical secondary flows observed in smooth duct flows, which are characterised by two counter-rotating vortices near the corners, are illustrated in the Fig. 5.5b at a comparable Reynolds number for reference.
The symmetric rotating structures visible in the smooth wall simulations are no longer visible when riblets are simulated on the bottom wall. This absence is attributed to their average intensity, which remains between 1-3% of the bulk velocity, as depicted for two Reynolds numbers in Fig. 5.5b. In contrast, the vortices generated by the presence of riblets exhibit greater strength, up to 15% of the bulk velocity, namely an order of magnitude higher than those observed in the smooth case. Consequently, the conventional corner-generated secondary flows are replaced by larger vortices occupying the entire cross-section.

This result highlights the importance of studying rough surfaces. Traditionally, corner vortices have been recognised for their role in transferring momentum from the core flow to the corners, [101]. Therefore, the emergence of these new flow structures driven by rough elements necessitates a thorough investigation. Additionally, the dependence on the Reynolds number suggests that these roughness-induced secondary flows may be critical in applications with high Reynolds numbers. Fig. 5.6 displays the vertical component of the velocity vector, scaled with the bulk velocity, at two spanwise positions: $z/h = -0.95$, to capture the behaviour near the side wall, and $z/h = -0.5$, to illustrate the transition from the wall towards a region dominated by the core flow. For comparison, the same quantities from the smooth duct simulations in [24] are reported in Fig. 5.6.

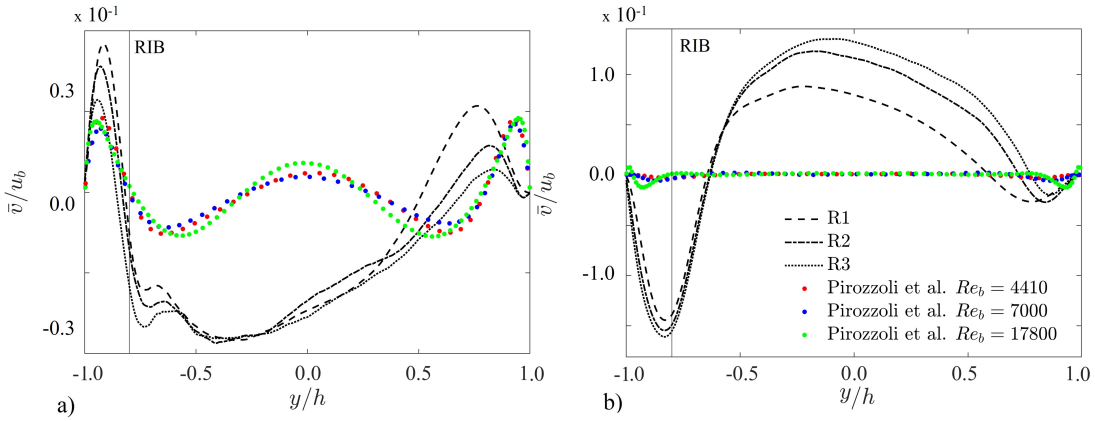

Figure 5.6: Vertical velocity component along the vertical direction, on the plane passing through "A" and for spanwise position a) $z/h = -0.5$ and b) $z/h = -0.95$. The smooth case results from Pirozzoli et al. [24] are overimposed for comparison.

These spanwise positions will be consistently used throughout this chapter to demonstrate the interactions with the rib and side walls.

In Fig. 5.6a, the influence of the side wall is minimal, and the profiles converge in the central region of the duct, particularly for $-0.5 < y/h < 0.5$, where the flow is unaffected by the Reynolds number. Closer to the top and bottom walls, the profiles diverge and exhibit peaks at both ends, decreasing their magnitudes as Re increases. This behaviour is due to the core flow strengthening and expanding with higher Re, which pushes side effects closer to the walls and reduces their influence. Comparing with the smooth simulations, it becomes clear that the riblet enhances the peak near the ribbed wall, while the peak near the top wall shows similar behaviour to the smooth case, though with a more pronounced dependence on Re.
In Fig. 5.6b, the same quantity is presented at $z/h = -0.95$. In this region, the profiles remain unaffected by the Reynolds number for $y/h < -0.5$, where the riblet geometry predominantly dictates the location and intensity of the recirculation zone, as noted in [114]. However, the profiles show a significant Re dependence for $y/h > 0.5$. The vortex, already observed in Fig. 5.5, becomes stronger as the Reynolds number increases. The magnitude of this velocity component fluctuates between 2% and 4% of the bulk velocity near the core region while reaching peaks exceeding 15% close to the side wall. Moreover, this high value is sustained throughout the entire duct height rather than being a localised event. In contrast, smooth ducts exhibit a vertical velocity component that only reaches around 2% of the bulk velocity near the corners, with lower values elsewhere, as shown by the dotted lines in Fig. 5.6b, which appear negligible compared to the fluctuations present in the ribbed wall simulations.
These newly identified flow structures impact applications where momentum and energy transfer are critical, as they enhance mixing across the duct's cross-section.

To analyse the influence of these recirculations on bulk flow properties, the skin friction coefficient is used. Fig. 5.7 illustrates the effect of riblets on the flow's frictional behaviour. The local skin friction coefficient will be assessed based on the local wall shear stress, defined as $C_f = 2\tau_w/\rho u_b^2$, where $\tau_w = \mu \left(\left(\frac{\partial u}{\partial n}\Big|_{\text{wall}} \right)^2 + \left(\frac{\partial v}{\partial n}\Big|_{\text{wall}} \right)^2 + \left(\frac{\partial w}{\partial n}\Big|_{\text{wall}} \right)^2 \right)^{0.5}$, and n is the direction normal to the wall.

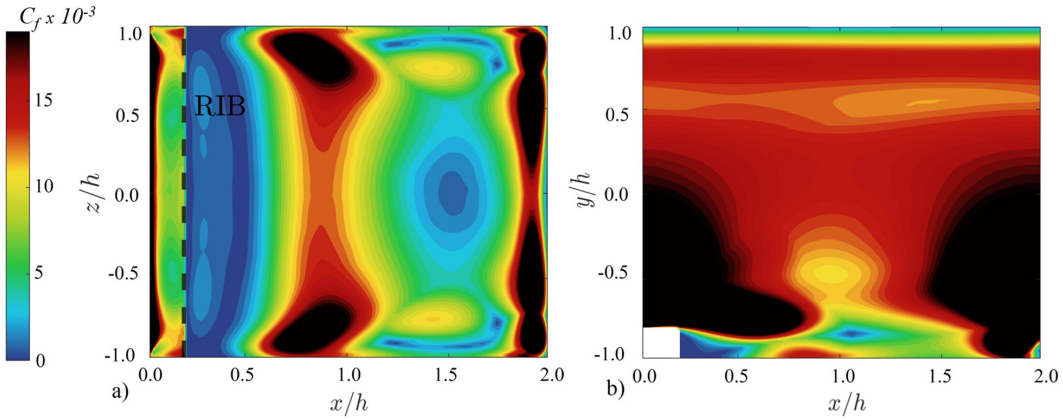

Figure 5.7: Skin friction coefficient C_f, presented on the a) bottom wall of the ribbed duct and b) smooth side wall of the duct.

With respect to a smooth duct, the distribution of the friction coefficient is remarkably non-uniform and depends on the periodical repetition of the riblets. Two different behaviours are found on the bottom wall and the side wall, which are affected differently by the recirculation of the flow.
Fig. 5.7a shows the bottom wall of the ribbed duct, where a direct connection between the recirculation region "A" and the area with the highest friction coefficient is evident. Comparing Fig. 5.3 and Fig. 5.7a, the skin friction coefficient appears to be larger where vortices "A" and "C" are present. Low friction is caused by the recirculation "B", where the flow detaches from the bottom wall. This is in agreement with the findings presented in [128], where the authors conducted an experimental study of turbulent flow in a square duct with square ribs and a high blockage ratio mounted on one wall in a wind tunnel. The behaviour of the C_f in Fig. 5.7 shows the main feature of k-type roughness, with a flow detachment after the riblet, which causes a low friction region and then the reattachment of the flow in a later stage.
On the side wall, the in-plane recirculation presented in Fig. 5.5a is accelerating the flow near the side walls, which then experience a significant increase in friction coefficient close to the riblet, as visible in Fig. 5.7b.
In a smooth duct, instead, the skin friction coefficient appears uniform in the streamwise direction and is only dependent on the distance from the side wall. In a square smooth duct, any side would present the same distribution, as documented in [129]. Therefore, it can be expected that the differences in friction factor between the bottom and side

walls, presented in Fig. 5.7, would not exist if the riblets were present on all four walls of the duct.

5.2.3 Turbulent Stresses

The fluctuations of the second order, or Reynolds stresses, are commonly used to quantify the turbulence of the flow. In Fig. 5.8, the profiles of normal and shear stresses along the same vertical lines used in the previous section, $z/h = -0.5$ (red lines) and $z/h = -0.95$ (black lines), are made dimensionless with respect to the turbulent kinetic energy, k. The same quantities from the smooth simulations at similar Reynolds numbers in [24] are compared.

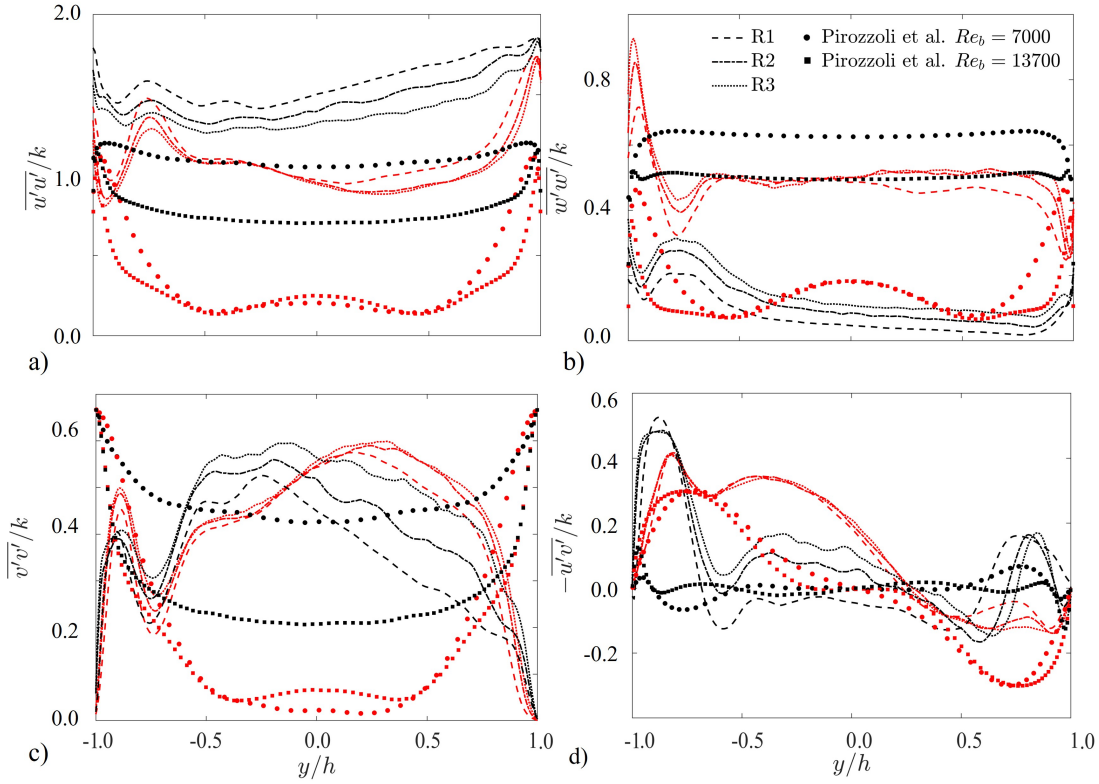

Figure 5.8: Normal and shear stresses along $z/h = -0.5$ (red lines) and $z/h = -0.95$ (black lines) on the plane passing through "A", for three different Reynolds numbers. Smooth simulations (squares and circles) from [24] are compared.

These stress profiles are illustrated for three distinct Reynolds numbers, enabling a comparative analysis of their impact on the flow and their relationship to the results obtained with a smooth wall. Both normal and shear stresses display a pronounced peak at the riblet's crest at $y/h < 0.5$. This occurrence is attributed to the disturbance generated by the riblet, as already shown in Fig. 5.3. The ri-

blet's protrusion into the flow induces unsteady eddies that reverse in the flow above the riblet and are subsequently transported downward, as supported by findings in [110].

For a smooth wall, normal stresses are symmetrical with respect to the duct's centerline, [130], while shear stress exhibits antisymmetry. However, introducing a riblet disrupts this symmetry, leading to an asymmetric flow. Moreover, the average stress magnitude on the vertical plane at $z/h = -0.5$ increases compared to the smooth wall case, driven by the strong motions previously shown in Fig. 5.5, which influence the entire duct cross-section. This effect is especially evident when considering the shear stress $\overline{v'v'}/k$, for which the red lines display values near zero for the smooth wall cases but significantly higher for the ribbed wall simulations.
All these observations are consistent with prior research based on PIV experiments involving ribbed duct flows with similar configurations, as described in [131, 132] [133], and with LES simulations discussed in [134].

Comparing the behaviour of the stresses when varying the Reynolds number, it can be noticed that closer to the core of the flow (red lines), the riblet influences the stresses, leading to notable differences compared to the smooth wall case, while the Reynolds number variation has minimal effect. In contrast, closer to the side wall (black lines), the Reynolds number has a significant influence, with all stresses, except for the streamwise normal stress, showing a noticeable increase as Re rises to 50%. This increase is particularly evident in $\overline{v'v'}$ and $\overline{w'w'}$, which are key contributors to the enhanced turbulence observed in previous sections. The term $\overline{v'v'}$ has a much higher value than in the smooth case throughout the vertical direction, with a peak at $y/h = 0$ that surpasses the initial riblet-induced peak. With increasing Re, this high value is kept over an increasingly broader region. This behaviour mirrors the production rate near the side wall, which has been investigated and presented in more detail in [76]. In the published work, an analysis of the vorticity behaviour is also conducted, which is strongly related to the side-wall fluctuations. For more details, refer to [76].

It is important to note that the observations illustrated in Fig. 5.8 refer to the behaviour of the Reynolds stresses at two distinct spanwise locations. For a complete evaluation of the impact of the side walls on the stress behaviour, Fig. 5.9 is presented.
The side walls have a significant influence on the flow statistics, resulting in two notable trends. For positions close to $y/h = \pm 1$, which correspond to the bottom and upper walls, the peaks in the Reynolds stress components tend to diminish as one moves from $z/h = 0$ (depicted by the solid black line) to $z/h = \pm 1$, as indicated by the arrow. This can be explained by the reduced velocity in these regions, stemming from their proximity to both walls.
Conversely, there is an observable increase in Reynolds stresses for vertical coordinates $0 < y/h < 0.75$. This results in a more flattened profile with a higher average stress value. Here, the flow transitions away from the core region and enters an area where the in-plane velocity components become more intense, as evidenced by the prominent near-wall secondary flow shown in Fig. 5.5. The near-side-wall region features more

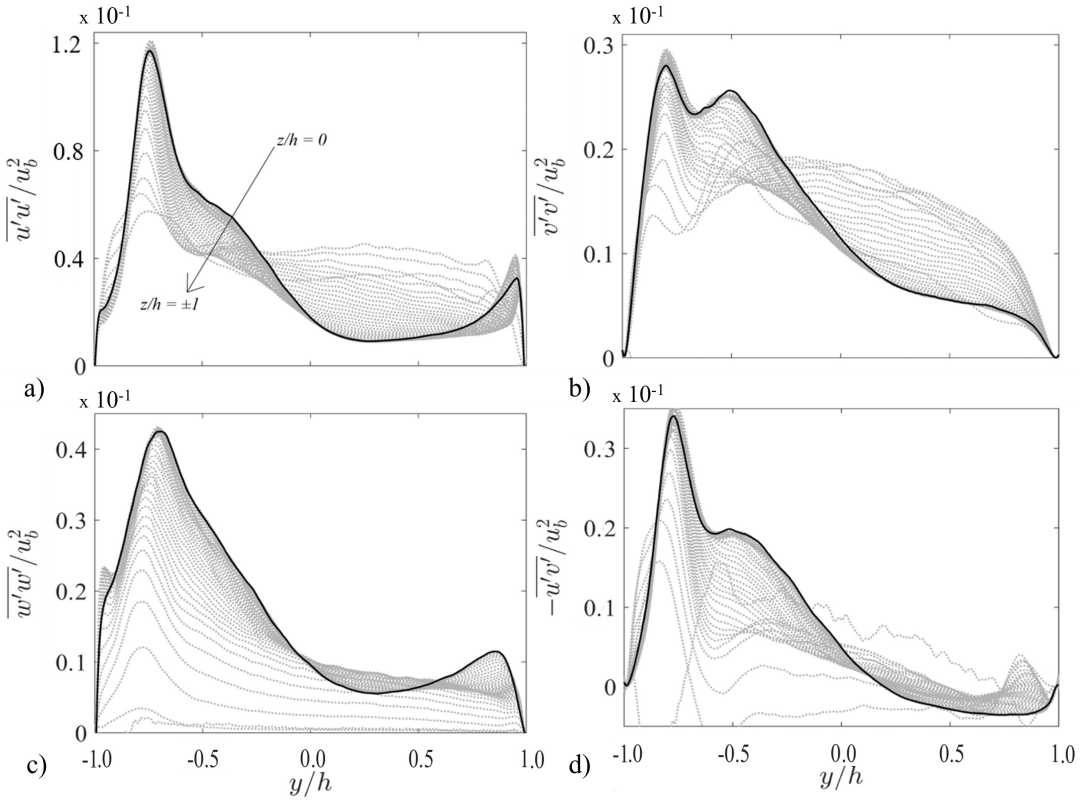

Figure 5.9: Normal and shear stresses on a plane passing through the vortex centre "A"; along the vertical line of the duct, at $z = 0$, with a solid black line, and in lighter grey for z coordinate going from the centre line towards both the side walls.

pronounced gradients in the in-plane velocity components, generating higher Reynolds stress values, particularly for $\overline{v'v'}/u_b^2$. These variations in flow behaviour would not be detectable if periodic boundary conditions had been used for the side walls.

5.2.4 Convergence Analysis

In smooth ducts, cross-flow velocities are characterised by a magnitude in the order of 2% of the bulk velocity. Consequently, their weak intensity requires a very long averaging time to be captured within DNS simulations. A first guess on the correct length of the time interval to get converged statistics was given in [135], in which a time interval of at least 6000 convective time units was suggested (defined as the ratio of boundary-layer thickness and free-stream velocity). Although other studies are available on the uncertainties in statistics in DNS [136], the work of Vinuesa et al. [34] is recalled as the first mathematical description of the converged state, which is based on the linearity of the total shear stress. They developed a convergence indicator and evaluated it using many previous simulations from different authors and research groups to compare the convergence of their work. For ducts, the indicator used to establish simulation

convergence is the residual of the momentum balance in the streamwise direction, as shown in Eq. (5.4).

$$\varepsilon_v = -\frac{\partial \overline{p}^+}{\partial x^+} + Re^{-1}\frac{\partial^2 \overline{u}^+}{\partial y^{+2}} - \frac{\partial \overline{uv}^+}{\partial y^+} - V^+\frac{\partial \overline{u}^+}{\partial y^+} + Re^{-1}\frac{\partial^2 \overline{u}^+}{\partial x^{+2}} \tag{5.4}$$

In [34], the authors analysed the decay of the convergence indicator as a function of two parameters, C_t and C_s. These parameters represent the averaging time interval and the time at which averaging begins, respectively, both normalised by the convective time, which is based on the bulk velocity and duct half-height. The lowest convergence rate for square ducts, approximately $1 \cdot 10^{-3}$, is found at $C_s = 200$ and $C_t = 5400$. However, they also demonstrated that after the first 1000 convective times, the decay rate in the convergence index becomes very slow. In other studies, more moderate convergence values have been reported. For instance, the same indicator is calculated in various works, yielding a value of around $3.3 \cdot 10^{-3}$ in [102] and $2 \cdot 10^{-3}$ in [120]. These values are more affordable, offering significant savings in computational time.
A second convergence indicator was introduced in [24], which focuses on the required flow symmetry. This second indicator is presented in Eq. (5.5), and its value is also listed in Tab. 5.1.

$$\varepsilon_p = \frac{1}{u_b}\left[\frac{1}{d_H^2}\int_0^{2h}\int_0^{2h}\left(\overline{u}\,(z,y) - \overline{u_{sym}}\,(z,y)\right)^2 dxdy\right]^{0.5} \tag{5.5}$$

In Eq. (5.5), $\overline{u_{sym}}\,(z,y)$ represents the mean velocity, averaged over the four quadrants of the smooth square section or over the two vertical halves of the ribbed duct. In their study, Pirozzoli et al. [24] developed a simulation database for smooth square ducts at various Reynolds numbers and evaluated both convergence indicators. It appears that the indicator in Eq. (5.5) converges more slowly than the other. This led us to adopt a conservative approach, using the indicator in Eq. (5.5) as the benchmark for convergence in our simulations. Nevertheless, the turbulent structures in a ribbed duct and generally over rough surfaces differ significantly from those in a smooth duct. As observed in previous sections, the classic corner eddies with an intensity of 2% of the bulk velocity are replaced by stronger vortices that depend on the rough geometry. Moreover, no study in the literature currently addresses convergence criteria for simulations of ribbed or rough wall ducts. Therefore, the first assessment of the convergence indicator proposed by [24] in the context of a ribbed duct is presented here. Given the stronger recirculation regions highlighted in this chapter, it is reasonable to expect that the convergence of these simulations would be faster than that of a smooth duct. The results are shown in Fig. 5.10.
First, the impact of the time interval C_t and the starting time C_s, as defined in [34], is analyzed. As shown in the contour plot on the left in Fig. 5.10, a darker region is reached when the averaging period begins after $C_s > 200$, which is consistent with the findings of [34].
Next, the convergence indicator is compared between the smooth and ribbed cases. As anticipated, the stronger eddies characteristic of a rough duct positively influence

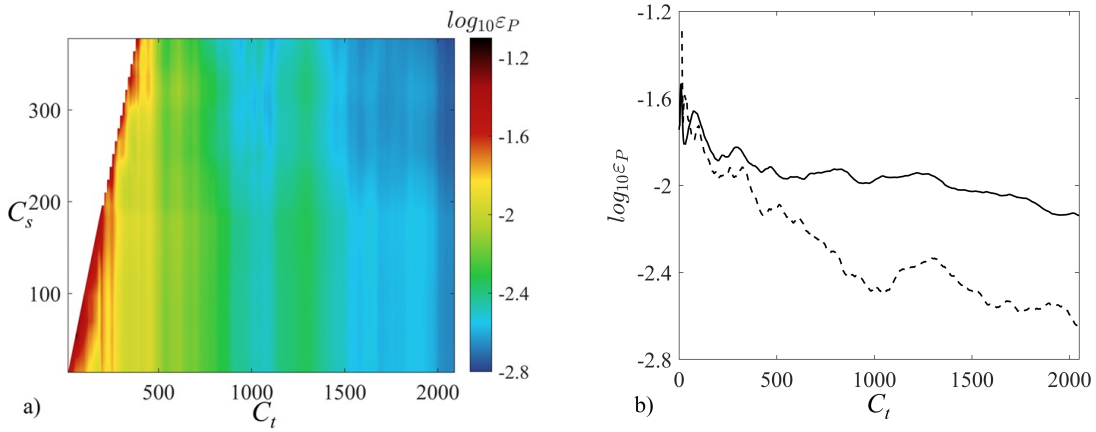

Figure 5.10: a) Contour plot of ε_P as a function of C_t and C_s for the simulation R1. b) convergence of the indicator ε_P compared between the simulation S (continuous line) and R1 (dashed line).

the behaviour of the indicator ε_P, enabling the simulation to reach a converged state faster. As presented in Tab. 5.1, the smooth and ribbed simulations achieve convergence indicators of 0.0069, 0.0017, 0.0015, and 0.0020, respectively, after a similar number of convective timesteps. This suggests that rough wall simulations require shorter simulation times compared to the longer, resource-intensive smooth simulations, where the weak secondary flows necessitate more than 4000 convective timesteps to fully resolve and reduce the convergence indicator to as low as 10^{-3}.

5.3 Flow over Rough Surfaces in a Duct

5.3.1 Methodology

The simulations presented in this section follow the same methodology described in section 5.2.1, which means that the solver, governing equations, and spatial and temporal discretisations remain unchanged. The fluid is considered again incompressible, while the mesh is selected respecting the same criteria, checking the Kolmogorov length as in Fig. 5.2.

Three differences are reported: the geometry of the domain, the activation of the energy equation, Eq. (3.14), which was not considered in the previous section, and the streamwise averaging procedure.

The geometry of the domain is built starting from real roughness measurements performed on the combustion chamber surface of the 3D printed segment presented in Chapter 4 and shown in Fig. 4.6. Several measurements are taken in different positions, and the resulting profiles are filtered to eliminate high frequencies that would make the

simulation too expensive. Indeed, as shown in [108], surface modifications and filtering are usually required to satisfy the boundary conditions and the expense of the computation.
The cutting frequency is chosen to retrieve a profile with an average roughness and a peak-to-valley distance that differ by less than 5% from the original measurement. The profiles are then interpolated into a rough surface, as shown in Fig. 5.11.

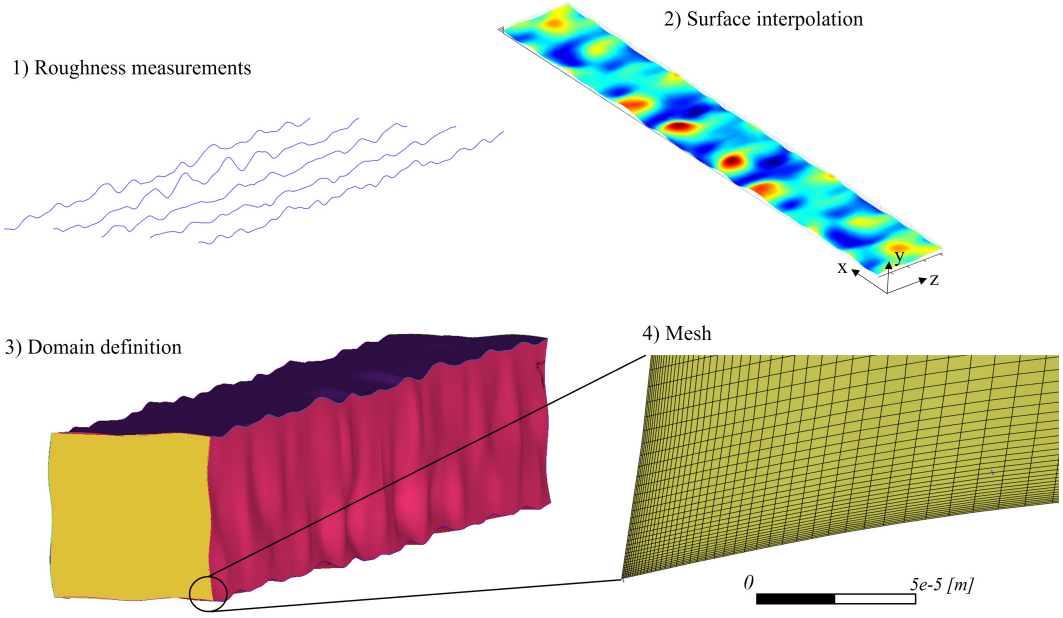

Figure 5.11: Steps for the creation of the simulation domain.

The interpolated surface is then used to create a rough square duct, which is meshed with a structured mesh. A zoom in the structured mesh is shown in Fig. 5.11. It is evident that the characteristic mesh spacing is significantly finer than the amplitude of the surface roughness, rendering the wall curvature barely perceptible. All the steps that are required for the construction of the domain are reported in Fig. 5.11. The energy equation is activated to calculate the flow temperature, and constant heat flux is applied only to one wall, which is referred to as the "bottom" wall.
The reference system is placed in the centre of the inlet section, as in the previous case. The domain is presented in Fig. 5.12, where the contour presents the streamwise velocity value, at a distance of $0.01h$ from the walls and in the inlet and outlet sections.

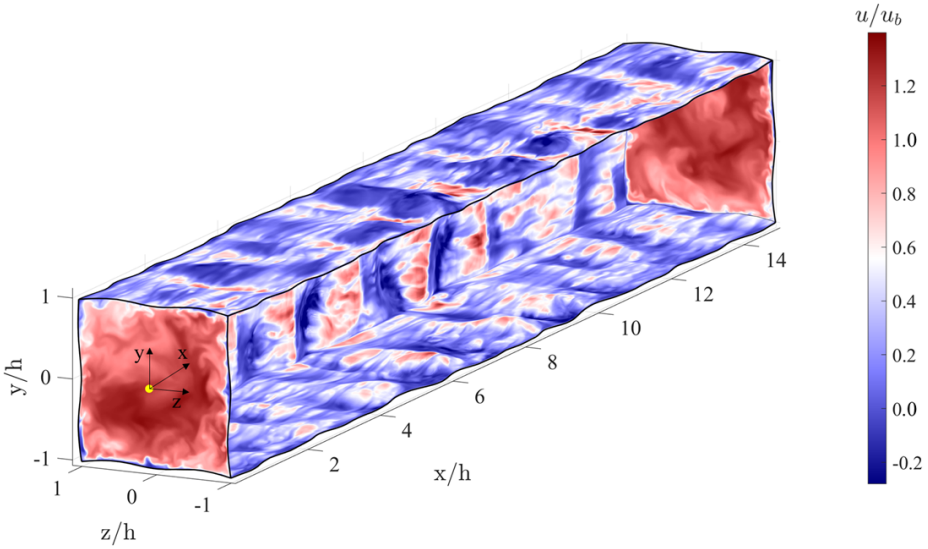

Figure 5.12: Three-dimensional view of the simulation domain and contours of the instantaneous streamwise velocity.

As visible, the rough surface has crests and valleys where the flow is distorted. Following what is done in [137], the flow is averaged in the streamwise direction, which was not possible in the simulations with a ribbed wall.

Simulation	Re_b	N_x	N_y	N_z	$(\Delta x/\eta)_{avg}$	$t_s/(h/u_b)$
S1	4410	120	120	600	2.91	1850
S2	7000	140	140	700	3.11	1620
S3	17800	240	240	1200	2.98	1710
RR1	4410	120	120	600	3.12	1650
RR2	7000	140	140	700	3.94	1590
RR3	17800	240	240	1200	3.19	1590

Table 5.2: Simulations data: N_x, N_y, N_z are the number of cells in each direction, ε_p is a convergence indicator defined in [34], η is the Kolmogorov length; t_s is the simulation time.

A total of six simulations are performed, with details reported in Tab. 5.2. Three different Reynolds numbers are simulated, which are the same as those in the work by Pirozzoli et al. [24], to facilitate comparison with data available in the literature. Smooth duct simulations are performed with the same boundary conditions for comparison.

5.3.2 Velocity Field

In section 5.2 it has been shown how the presence of a riblet influences the flow pattern. However, the riblet is a steep and sudden protrusion into the flow, used to simulate the effects of a rough surface. The riblet is seen as a single obstacle by the flow, which generates recirculation regions, as shown in Fig. 5.3.
On the contrary, the filtered rough surface presented in Fig. 5.11 does not exhibit sudden discontinuities, allowing the flow to follow the wall more uniformly. This does not mean that there are no recirculation regions. In Fig. 5.12, regions with a negative streamwise velocity can be seen in between the crests of the walls, but none of them is as big as the vortex "A" in Fig. 5.3.

A streamwise average is performed. The resulting velocity contours are reported from Fig. 5.13 to Fig. 5.15, while the vertical velocity component is shown from Fig. 5.16 to Fig. 5.18. In every figure, the velocity contour of the rough simulation is compared with the result from the respective smooth case.
In the rough duct, the presence of crests and valleys has to be taken into account when averaging in a streamwise direction. Thus, in each contour, two continuous black lines are used to mark the deepest valleys and the highest crests, which limit the flow domain. The dashed line marks the mean duct surface.

In the smooth simulations, complete symmetry is not reached because of the too short time interval used in the averaging process. However, the structures are fairly similar in all four quadrants of the section.
In the rough duct, due to the large relative roughness and the lack of full randomisation, the streamwise velocity does not exhibit symmetry. The velocity isolines exhibit clear sensitivity to the roughness geometry and variations in the Reynolds number. Indeed, the isolines are pulled toward the deepest valleys, and the boundary layer is much thicker than in the smooth case due to the non-uniform wall.
The Reynolds number's influence is more evident when examining the other components of the velocity vector. In Fig. 5.16 to Fig. 5.18, the vertical component is presented and compared with the smooth case. In smooth square ducts, secondary flows are induced by counter-rotating vortices created near the corners. The vortices are well visible in the figures, and their position moves towards the corners as the Reynolds number increases. When scaled with the bulk velocity, their intensity remains relatively constant, around 2% of the bulk velocity, u_b. This behaviour aligns well with previous studies, [24].
In the rough duct, the intensity of the vortices increases with the Reynolds number from 3% to 6% of the bulk velocity, within the range of Reynolds number analysed in this work, as visible when comparing Fig. 5.16a and Fig. 5.18a. Additionally, with increasing Re, their motions move toward the centre of the duct and affect almost the entire cross-section. Thus, two differences are highlighted when comparing the effects of the Reynolds number increase for a flow over a smooth and a rough surface.

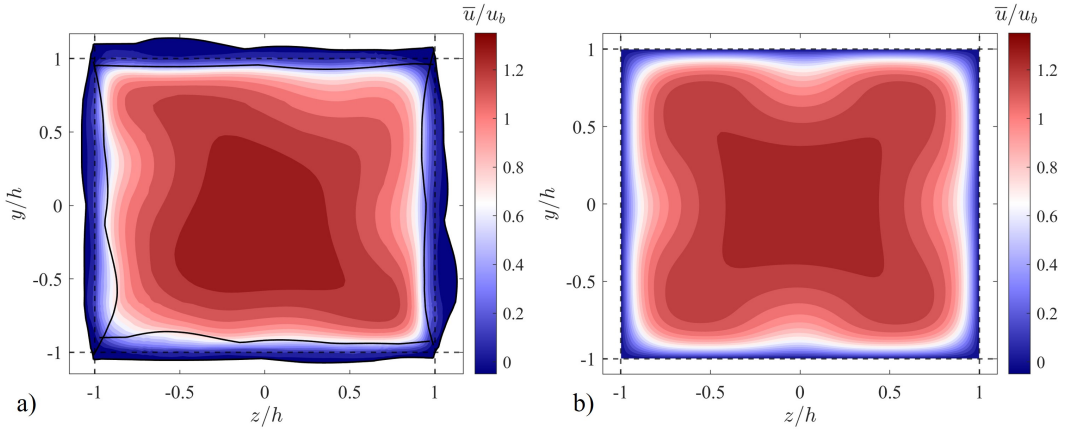

Figure 5.13: Streamwise velocity contours at $Re_b = 4410$ for a) rough domain and b) smooth domain.

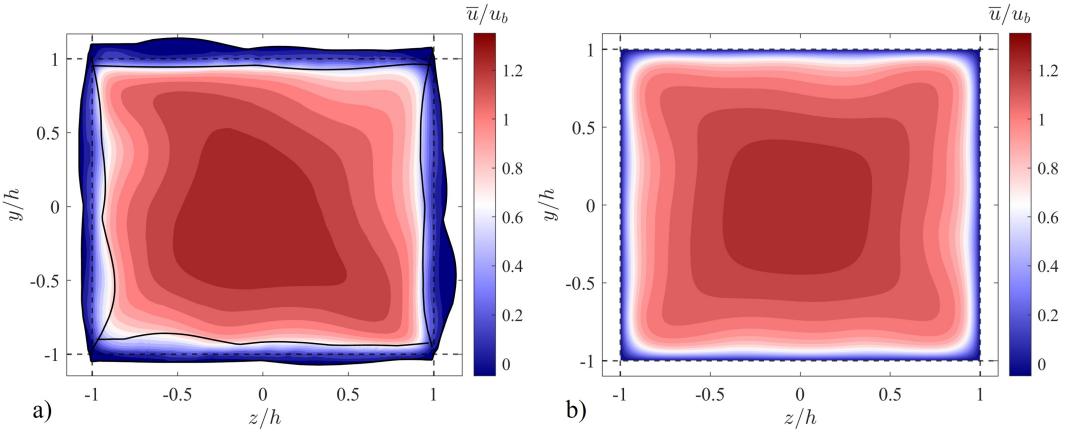

Figure 5.14: Streamwise velocity contours at $Re_b = 7000$ for a) rough domain and b) smooth domain.

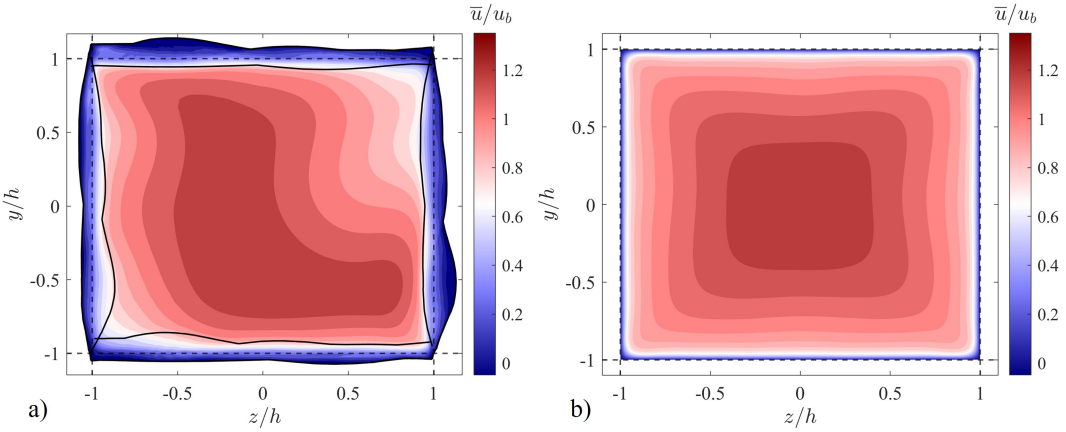

Figure 5.15: Streamwise velocity contours at $Re_b = 17800$ for a) rough domain and b) smooth domain.

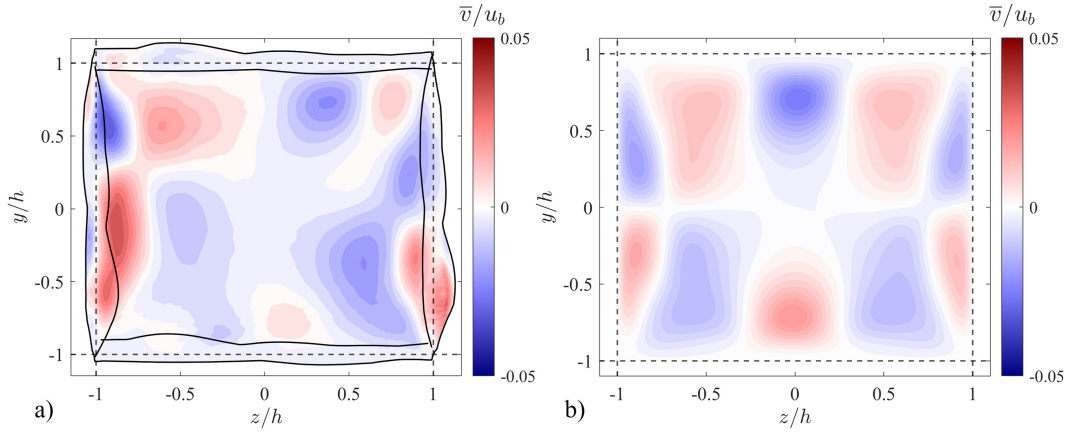

Figure 5.16: Vertical velocity contours at $Re_b = 4410$ for a) rough domain and b) smooth domain.

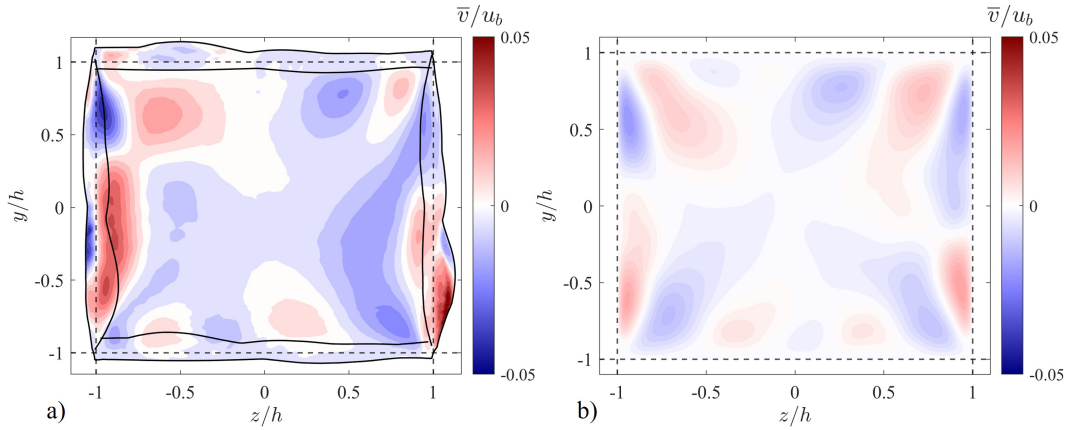

Figure 5.17: Vertical velocity contours at $Re_b = 7000$ for a) rough domain and b) smooth domain.

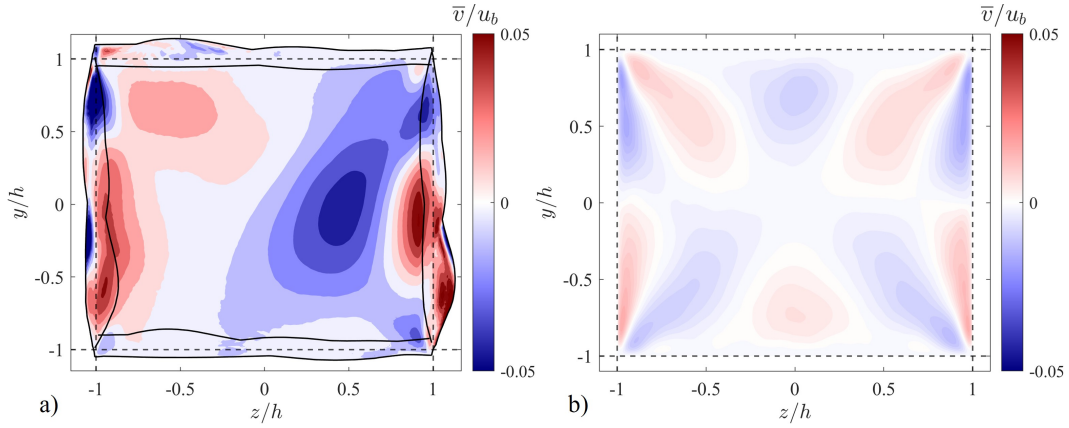

Figure 5.18: Vertical velocity contours at $Re_b = 17800$ for a) rough domain and b) smooth domain.

First, within a rough duct, the vortices are generated from the crests and valleys of the wall and extend to the centre of the duct when the Reynolds number is increased. Secondly, their intensity increases as well. In a smooth duct, instead, the vortices are generated in the corners and become increasingly confined to this region as the *Re* increases. Their magnitude stays the same.

These differences are more visible in Fig. 5.19, where the streamwise and vertical velocity profiles are shown on a vertical plane at $z/h = 0$. The profiles are always non-symmetric in the case of rough walls, and for the vertical velocity, they present more pronounced peaks, which are twice as intense as those of the smooth wall counterpart.

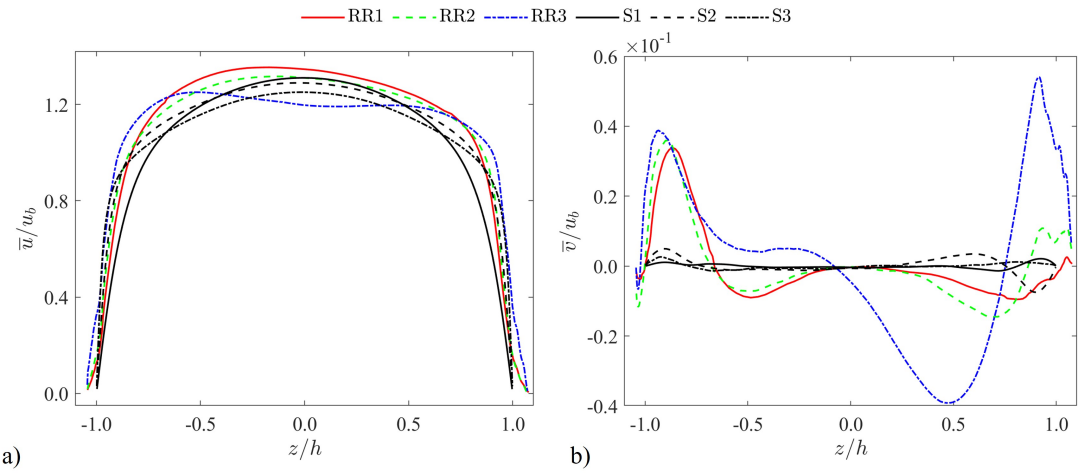

Figure 5.19: Velocity profile in $z/h = 0$ divided by the bulk velocity. a) streamwise velocity profile and b) vertical component of the velocity.

The increment in Reynolds number makes the difference between the two cases even larger.
As the vertical component of the velocity vector, the horizontal component is also stronger in the rough duct than in the smooth one, but it is not reported here because of its similarity with the vertical component. These two values are used to define the cross-velocity magnitude $\sqrt{\overline{v}^2 + \overline{w}^2}$, which was shown in Fig. 5.5 to present the secondary flow induced by the presence of a riblet. As in the ribbed wall case, it can be stated that stronger in-plane velocity components lead to stronger secondary flows, which enhance momentum and energy transfer. This enhancement results in both higher friction losses and increased heat transfer, as will be demonstrated in the following sections.

Fig. 5.20 shows the dimensionless velocity profile against the non-dimensional wall distance. As expected, a logarithmic layer is also found in the rough case, which proves the Reynolds analogy. However, the velocity profile is shifted further away from the wall of a distance ΔU^+, [57, 137].

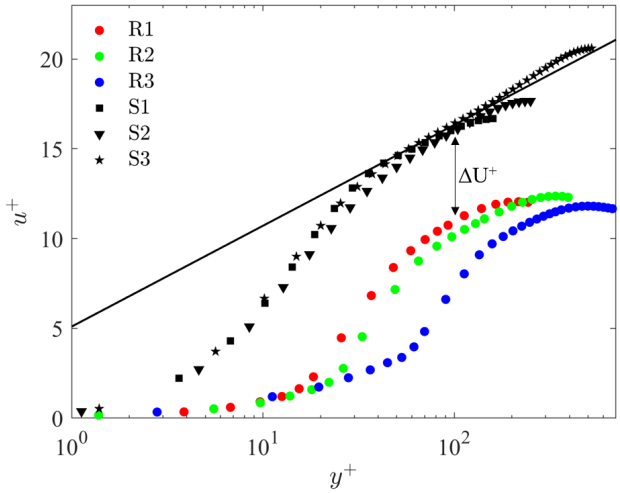

Figure 5.20: Dimensionless velocity profile for smooth and rough simulations. The black continuous line is the theoretical log-law, [18].

The shift is due to the reduced momentum near the wall, due to the roughness, [66], which consequently induces higher losses. As done for the riblet wall, the skin friction coefficient C_f is computed and compared with the smooth case. Fig. 5.21a presents the skin friction coefficient for the bottom wall of the rough duct, while Fig. 5.21b shows the same quantity on a smooth wall. On the rough wall, friction losses are not uniform and are distributed across the wall, following the velocity gradient dictated by the roughness features. On the contrary, on the smooth wall, the skin friction coefficient only varies in the spanwise direction, as the velocity profile.

The magnitude of the C_f appears to be higher on the rough surface, as many dark red regions are visible. A surface average is then computed, and the results are presented in Fig. 5.22. The skin friction coefficient is larger on the rough wall than on the smooth one. An increase of 20% is observed at the lowest Reynolds number and of 30% at the highest, highlighting the dependence on the flow conditions.
Although the Reynolds number used in these simulations is not representative of real operating conditions in engineering applications, these results provide insight into the physical interaction between a rough wall and the flow, the alteration of the flow pattern, and the resulting effect on pressure losses. These insights expand the existing database in the literature regarding rough ducts and are useful for a better understanding of the experimental results in the next chapter, where representative flow conditions are employed.

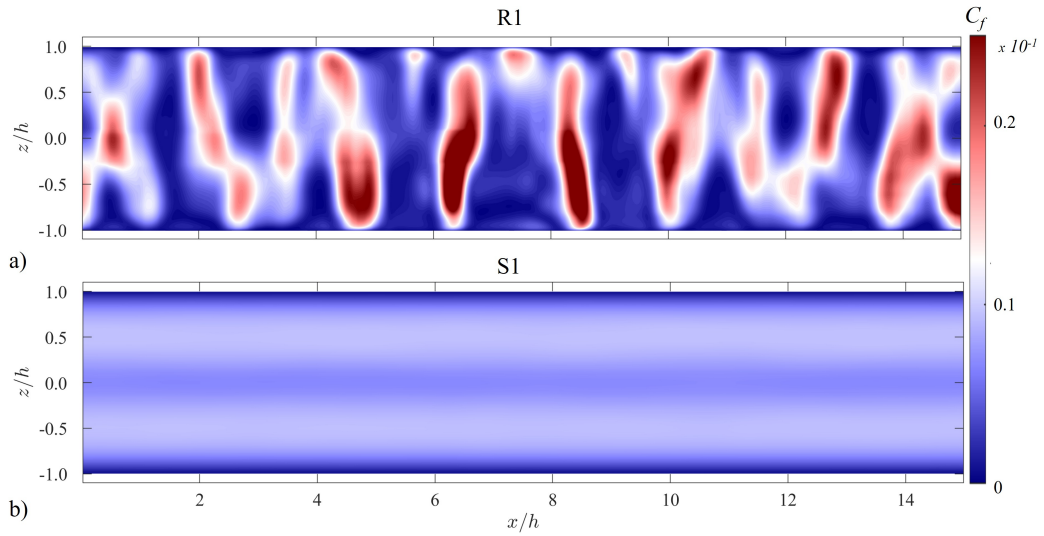

Figure 5.21: Contour of the skin friction coefficient on the wall for a) the rough domain and b) the smooth domain, simulated with $Re_b = 4410$.

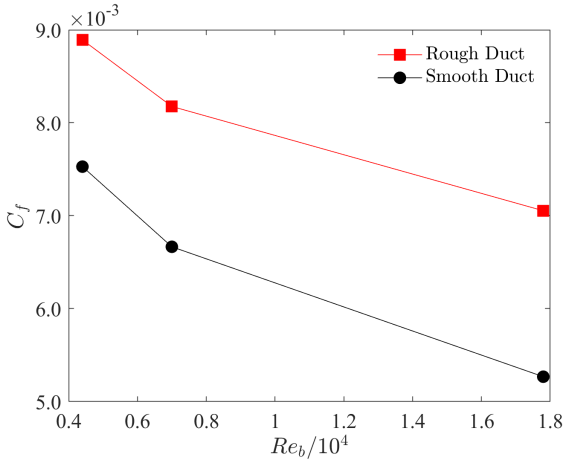

Figure 5.22: Averaged skin friction coefficient as a function of the bulk Reynolds number for rough and smooth simulations.

5.3.3 Thermal Field

The energy equation is activated to calculate the temperature of the flow and to resolve the heat exchange with the wall. In the previous section, the stronger secondary motions that are expected to enhance momentum and energy transfer across the duct section were presented. In this section, the temperature field is investigated to quantify this effect. Fig. 5.23 presents the temperature contour in the duct section for $Re = 4410$. As for the velocity field, the continuous black lines indicate the deepest valleys and the highest crests in the domain, while the dashed line represents the nominal square duct.

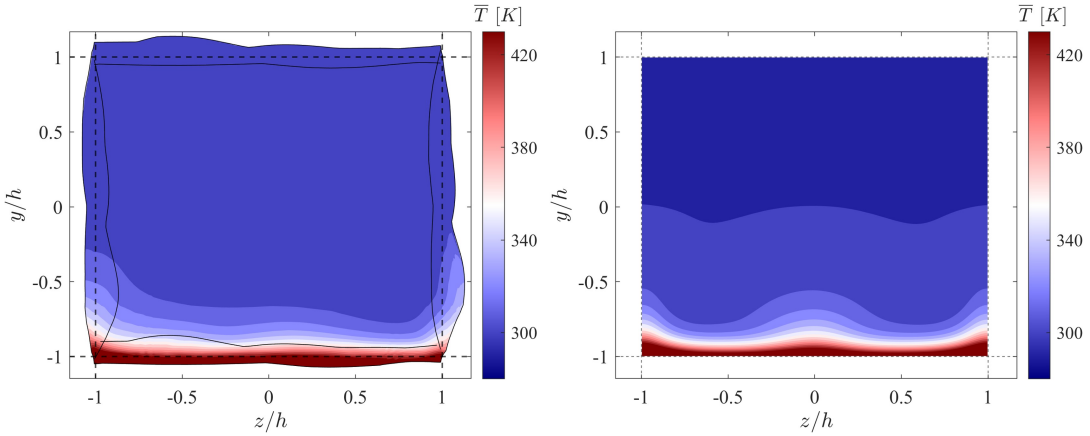

Figure 5.23: Temperature contours at $Re_b = 4410$ for a) rough domain and b) smooth domain. 10 contour lines are used, uniformly distributed between 280 and 430 K.

At first glance, the most noticeable difference is that the core flow in the rough case is warmer than in the smooth case. The effect is primarily due to the different energy input in the two simulations when using a constant heat flux boundary condition. In fact, in the rough case, a larger area of exchange is present, which then leads to a higher energy input to the flow. The difference in this case is only a few degrees, but it is important to note that this difference is measurable, even though the Reynolds number and heat flux are low compared to real applications. Scaling the simulations up to rocket engine conditions, the presence of roughness on the walls would lead to a significant difference in the bulk temperature of the flow with respect to the smooth case.

Furthermore, it can be observed that the temperature gradient near the wall differs in the two cases. For the rough case, the in-plane velocity components, as shown in Fig. 5.19, lead to better energy transfer in the corners, where these components are stronger. Generally, a less steep temperature gradient is observed for the rough simulations compared to the smooth case. This is shown in Fig. 5.24, where the temperature profiles are presented.
In Fig. 5.24a the temperature profile in the middle of the duct is shown. The rough and smooth profiles agree fairly well far from the wall, but the temperature from the simulations RR1, RR2, and RR3 always remains above the corresponding smooth result. Close to the wall instead, the profiles diverge: the temperature in the smooth wall simulations steeply increases towards the wall, while on the rough wall, the temperature shows a deflection point, which is due to the near-wall recirculation regions.
This effect is better visualised where the recirculations are stronger, namely near the corners. Fig. 5.24b shows the same comparison but at $z/h = -0.8$. Here, the near-wall difference between smooth and rough wall results is more evident and leads to a flow temperature difference up to 30 K in the first cell close to the wall. Additionally, the temperature profiles from rough and smooth simulations cross each other as the distance from the wall increases.

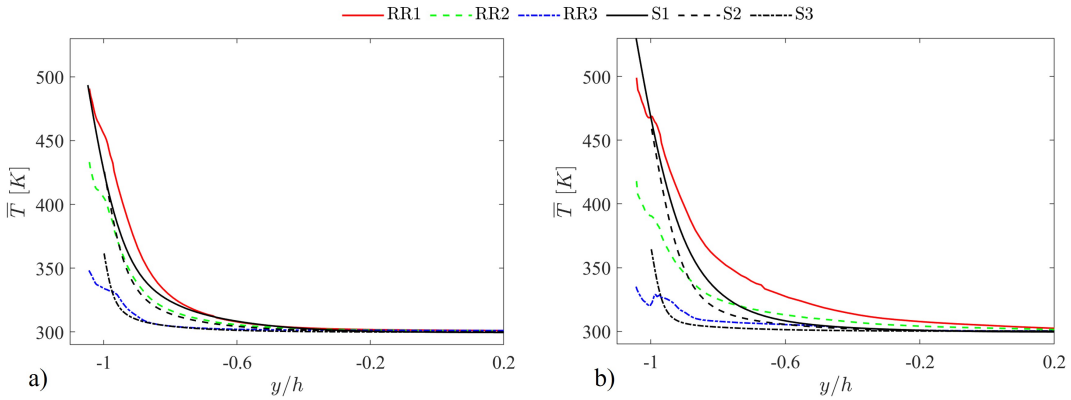

Figure 5.24: Temperature profiles at a) $z/h = 0$ and b) $z/h = -0.8$.

5.4 Conclusion

The results presented in this chapter contribute to the understanding of the impact of a single rough element, such as a riblet, on flow behaviour and provide insight into the overall effects of a rough wall on bulk flow properties.

First, a ribbed wall is used to amplify the roughness effects, making the induced in-plane motions more visible. Unlike the smooth case, where these motions are confined to the corners, they now extend throughout the flow domain. The intensity of these motions reaches approximately 15% u_b, which is much higher than in the smooth case, where cross-flow velocity intensities are limited to around 3% u_b.
The influence of the side wall is examined through Reynolds stress components, which show an increase near the walls, indicating stronger fluctuations and faster flow dynamics in that region. All stress components are higher than in the smooth case, particularly the vertical stress, $\overline{v'v'}$, which is up to ten times stronger near the side wall.

A rough duct is then modelled based on real roughness measurements. Simulations are performed at three Reynolds numbers, up to 17800, for both rough and smooth ducts, allowing for a direct comparison. While the rough case exhibits more moderate increases in intensity, the same general trend is observed. The in-plane motion intensity ranges from approximately $5\% - 6\%$ u_b and tends to increase with the Reynolds number, whereas in the smooth case, it remains independent of the Reynolds number.
Additionally, the behaviour of vortices differs between smooth and rough cases. In the smooth case, vortices migrate toward the corners as the Reynolds number increases, while in the rough case, they spread throughout the cross-section.
In all cases, the skin friction coefficient is evaluated to assess the correlation between recirculation zones and drag increase. For both the ribbed and rough cases, the coefficient is found to be higher than in the smooth case, ranging from 20% to 30%, depending on the Reynolds number.

Lastly, the temperature field is compared between the smooth and rough cases. This analysis confirms the enhancement of energy transfer from the wall to the core flow. The temperature boundary layer is observed to be thicker in the rough case than in the smooth one.

6 Experiments and Performance Evaluation

6.1 Motivation and Objectives

In Chapter 5, the most fundamental features of the flow over rough surfaces were presented, and a few hints on how the macro quantities are influenced were discussed. However, the numerical capabilities are still not sufficient to simulate real engineering conditions with such a level of detail. Therefore, in this work, a dual approach is employed, combining the numerical results obtained in the previous chapter with laboratory experiments.

Experiments, unfortunately, cannot resolve the quantities of interest at every point of the domain as is done in numerical simulations, but aim at representing real applications and flow conditions in order to further validate industrial models, as will be shown in Chapter 7.
Therefore, experiments are conducted in this work to quantify the effects of surface roughness on cooling performance under representative operating conditions.

The experiments have three main objectives, each addressing a critical aspect of the thermal and fluid dynamic behaviour of additively manufactured cooling components.
First, the testing activities aim to isolate the impact of surface roughness from other influencing parameters, such as geometry, flow rate, and thermal load. This is essential for understanding how the internal surface quality, resulting from the AM process, affects pressure drop and heat transfer efficiency within the cooling channels.
Second, the study investigates the influence of various surface finishes on the overall performance of the component. This includes evaluating the thermal and hydraulic behaviour of parts subjected to different post-processing treatments, such as machining and EDM (see Tab. 4.4). The goal is to determine whether additional manufacturing steps can lead to measurable improvements in component efficiency, and if so, to what extent these enhancements justify the added cost and effort.
Finally, the analysis extends to assessing the impact of non-uniformities inherent in the additive manufacturing process, particularly how these affect the temperature distribution across the component during operation. Local variations in surface roughness due to print imperfections can lead to localised temperature peaks, which may compromise the component's integrity or reduce its lifespan.

In this chapter, the logic and methodology behind the tests are explained, clarifying the reasoning behind the test article's position, the duration of the tests, and the reliability of the data. Furthermore, the test conditions are presented and explained, and a wide

discussion is given on the frictional behaviour and heat transfer capabilities of the additively manufactured test articles. Comparisons with the traditional setup and other results from the literature are used to discuss the outcomes. Part of the results have been presented in [138].

6.2 Methodology

Two sets of tests are performed in the frame of this work: the first set is run with the traditional setup presented in Fig. 4.2. This configuration is traditionally manufactured and features very smooth surfaces ($R_a \approx 1\ \mu m$) on both the combustion chamber and cooling channel surfaces. The results of the tests done with this configuration are presented in section 6.3 and are used as a reference database to assess the performance of a traditionally manufactured thrust chamber.
A second set of tests is performed with the test setup presented in Fig. 4.4, where the segment in position "R" perfectly substitutes the segments in positions "B" and "C" of the reference setup. This test article is manufactured with laser powder-bed fusion. Several segments are produced and tested in this second test campaign, as reported in Tab. 4.4. The results of the second campaign are presented in section 6.4 and compared against the traditional setup in terms of pressure losses in the coolant circuit, heat pickup, and temperature field distribution.

A significant difference between the two setups is the type of fluid used in the cooling circuit. In the reference setup, water is used for the entire length of the combustion chamber, while in the second configuration, the water passes through the segments "A" and "D" but not in the test article in position "R". The investigated article is cooled down with gaseous nitrogen such that the ratio between the measured quantity and the measurement error is as large as possible.

6.2.1 Thermal Field

This work mainly focuses on the heat transfer capabilities of the combustion chamber. In both setups, the coolant circuit is equipped with several thermocouples that record the temperature of the coolant at every transition manifold (between the segments) and at several azimuthal positions. By averaging along the azimuthal direction, a "calorimetric approach" is used to evaluate the heat flux absorbed by the coolant flowing in every segment, with the relation shown in Eq. (6.1).

$$\dot{q} = \dot{m}_{coolant}(h_{coolant,out} - h_{coolant,in}) \tag{6.1}$$

where "in" and "out" mean the axial position where the coolant enters and exits each segment. The calorimetric approach requires the enthalpy of the coolant at the inlet and outlet sections of the segment, which is retrieved using REFPROP [139] with the temperature and pressure measurements inside the transition manifolds.

However, this method relies on one important assumption: the evaluated heat flux, $\dot{q}$, which represents the heat absorbed by the coolant, is equal to the total heat flux only in the case of a thermally steady and insulated system. This means that the first derivative of the chamber wall's temperature must be zero (or very close to zero) when this method is applied. If that is not the case, the total heat flux is the sum of the heat flux absorbed by the fluid and the heat flux that has caused an increase in the temperature of the solid material of the combustion chamber.
In order to sustain this assumption, every test is run long enough to reach a thermal steady state of the material, which is reached when:

$$d(T_1 - T_2)/dt < \varepsilon \tag{6.2}$$

where T_1 and T_2 are the temperature readings of two generic thermocouples placed in the same chamber location but at two different radial depths, dt is the time derivative, and ε is a generic small number.
As further explained in the next sections, the required minimum duration of each operating point is set to 12 s.

The test campaigns described in this work aim to characterise the temperature distribution in the chamber in the azimuthal direction and compare the results of the traditionally manufactured hardware with those of the additively manufactured test articles.

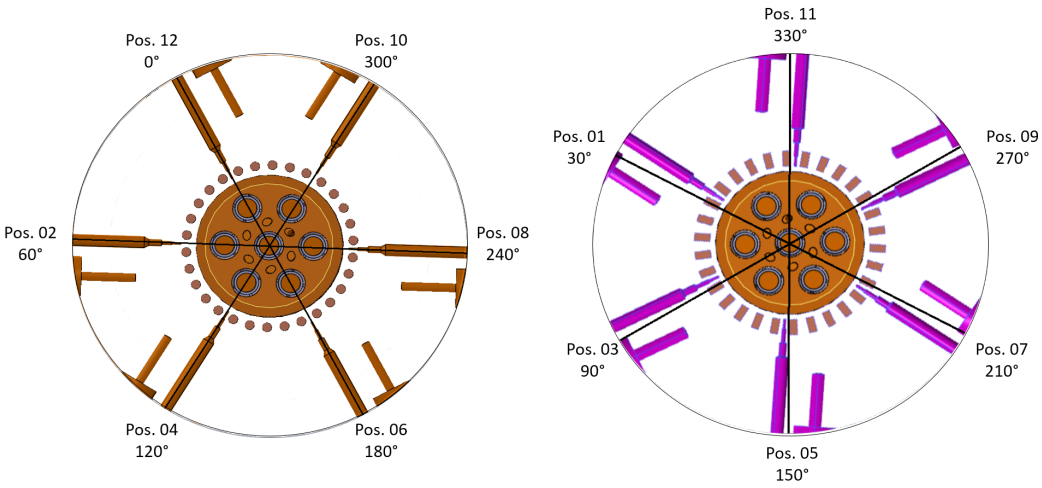

Figure 6.1: Cut section of the segment showing the thermocouples pattern in azimuthal direction for the traditionally manufactured segment (left) and the additively manufactured segment (right).

As said, the test articles are installed in the exact position of the previous segments "B" and "C". In this way, when the test article is tested, the difference in temperature field distribution can only be due to the different hardware and, thus, the different surface qualities. To map and quantify the temperature field distribution, recordings from thermocouples placed within the wall around the circumference of the segment are

used. Fig. 6.1 shows the thermocouples pattern for the traditional segment on the left and the LBM segment on the right. Additionally, at each position, two thermocouples are placed at 1 and 3 mm from the hot wall. Due to the limited space in the mounting process, the six thermocouples are misplaced in the two configurations of 30 degrees. The recorded values are used to define the temperature deviation θ as in Eq. (6.3).

$$\theta = 100 \cdot \frac{T_{az,i} - T_{az,avg}}{T_{az,avg}} \tag{6.3}$$

This temperature deviation measures how much the local temperature in a certain azimuthal position, $T_{az,i}$, deviates from the averaged temperature $T_{az,avg}$, which is defined as the arithmetic mean among the readings of the six thermocouples.

The same thermocouples are used to estimate the temperature of the hot wall. A 1-D conduction problem is solved as in Eq. (6.4).

$$\rho c_p \frac{\partial T}{\partial t} = k \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \tag{6.4}$$

Assuming the chamber is in a steady state and integrating between the position of the thermocouple and the hot wall, a simple conduction problem is derived, as shown in Eq. (6.5).

$$T_{wg} = T_m + \frac{Q}{2\pi L k} \ln(r_2/r_1) \tag{6.5}$$

where T_{wg} is the estimated hot wall temperature, T_m the measured temperature of the thermocouple, Q the total heat reaching the segment, k the thermal conductivity of the material, and r_1 and r_2 the radial position of the hot wall and the sensitive element of the thermocouple tip, respectively. In the following, the T_m is the averaged value among the six measurement points in the azimuthal direction.

6.2.2 Lifted Flame

For a proper comparison of the temperature distribution along the circumference of each test article, the combustion during each test must be uniform and stable. It is common practice to define combustion as “rough” if the intensity of the high-frequency excitation is equal to or higher than 5% of the combustion chamber pressure [140]. The FFT analysis of the dynamic pressure measurement is used in this work to ensure that none of the performed tests are affected by instabilities, as shown in the following section.

For what concerns uniformity instead, a phenomenon referred to as “lifted flame” is encountered in this setup. This phenomenon is attributed to the poor mixing of the propellant at injection, resulting in a maldistribution of the heat flux around the chamber. Thus, the tests presenting a lifted flame are discarded from the analysis.
To do so, a precise threshold to discard the test has to be defined. As said, the lifted flame influences the test results, causing a large non-uniformity in the temperature field in the azimuthal direction, particularly visible in the first segment of the chamber.

Fig. 6.2 shows the reading of five thermocouples of segment "A", which are placed at the same axial and radial position but at five different azimuthal positions around the circumference.

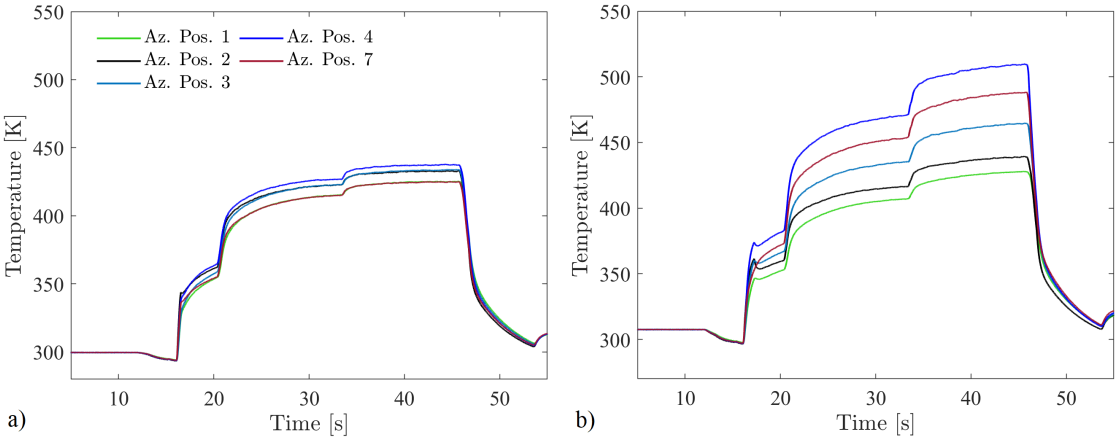

Figure 6.2: Temperatures in the first segment of the chamber at the same axial and radial position (72.5 *mm* from the faceplate and 1 *mm* from the hot wall) but at different azimuthal positions for a) anchored flame and b) lifted flame.

In Fig. 6.2a, the recording of some temperature sensors is shown for a test with an anchored flame. In this case, the temperature readings are very close to each other with only a limited variation. This shows uniform combustion, where all seven injection elements are working uniformly, and the heat flux reaching the hot wall is independent of the azimuthal coordinates. In Fig. 6.2b instead, the lifted flame is visible in the test results through a much wider spreading of the temperature values. Besides the spreading, the maximum temperature reached on one side is much higher than the maximum reached in the anchored case.
In Fig. 6.3, the calorimetric approach is used to evaluate the heat flux in each segment a compare different scenarios. Note that the calorimetric approach in Eq. (6.1) is applied in a prescribed time window of one second when a situation of thermal steadiness is reached. Besides, the computed heat flux level of each segment is assumed to be constant along the entire length of the same segment. Four tests are presented: in black, two operating points with anchored flame, and in red, the same two operating points in a test showing a lifted flame. It is easy to recognise that the lifted flame largely affects the results on the first segment "A". The heat flux calculated on segment "A" in a test with a lifted flame is almost double that of a test with an anchored flame. On the other hand, the differences between the red and black lines in segments "B" and "C" are of the same order as the experimental accuracy. Thus, it cannot be stated whether the lifted flame influences this part of the chamber or at least not significantly. Lastly, the nozzle appears less solicited in the case of a lifted flame.

It is concluded that the significantly higher heat flux delivered to the coolant in the first part of the chamber can be used as an indicator to recognise a test affected by a

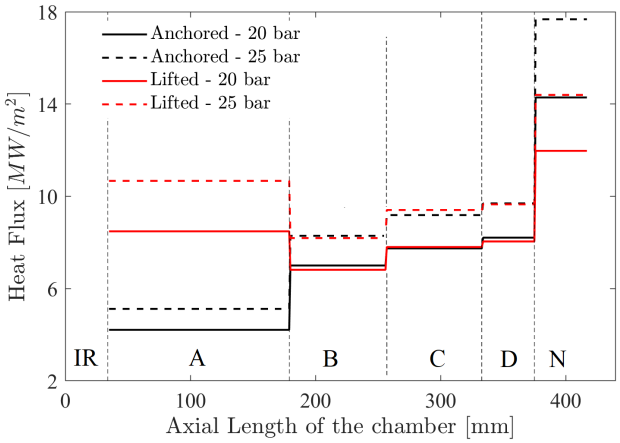

Figure 6.3: Heat flux per segment, evaluated with Eq. (6.1), in four tests with different conditions.

lifted flame. Once identified, these tests are discarded because it would be impossible to distinguish between the effects of the lifted flame and the surface roughness.

6.2.3 Comparison of ALM Specimens

Each segment is manufactured using the same printer and following the same building direction through the LBM process. The printing parameters are also kept the same in every printing job. Each segment is installed in the same axial position, at the centre of the chamber, as shown in Fig. 4.4, and with the same azimuthal orientation, which is ensured by a reference marker printed on each segment, as seen in Fig. 4.5. Preserving the same azimuthal orientation of each segment is crucial to assess whether and when features resulting from the printing process impact the test results.

Each segment is equipped with the same measurement sensors and tested according to the same test matrix. At the end of the test campaign, specifically when the fifth test article has been tested and dismounted, the first test article is reinstalled and retested to assess the repeatability of the setup and ensure that no changes have occurred in the setup during the testing period.

To fulfil the objective of the test campaign, the combustion chamber pressure is varied to test the performance with different heat flux levels. Additionally, the mass flow of the coolant is varied on three levels to assess the change in friction losses and heat transfer capabilities as the Reynolds number changes. The operating points are described in Tab. 6.2. The minimum mass flow rate of nitrogen has been selected to prevent damage to the combustion chamber structure. Despite the poor heat transfer capabilities, a gaseous coolant is selected for the tests to generate a larger pressure and temperature difference between the inlet and outlet of the channel, thereby minimising

the error-to-measurement ratio.

As previously described in Tab. 4.4, the combustion chamber and the cooling channel surfaces have been treated differently in every test article.
To properly correlate the change in performance between different test articles and between different azimuthal positions of the same test article, surface roughness is measured. For this task, a tactile profilometer DriveUnit GD 120 is used. Four different azimuthal positions of the inside surface of the combustion chamber are measured, as shown in Fig. 6.4, and in each location, five adjacent stripes are measured, for a total of twenty measurements. In addition, these measurements are performed both on the "groove" and the "extrude" side of the article (respectively, close and far from the printing plate). The measuring length is 40 mm per measurement.
The results of these measurements are shown in Tab. 6.1, only for the segment 2RR in a purely indicative way to point out the roughness non-uniformities on the as-built combustion chamber surfaces, which are due to the printing process. It can be noticed that both the arithmetic mean roughness, R_a, and the maximum peak-to-valley height, R_z, show a significant variation around the circumference.

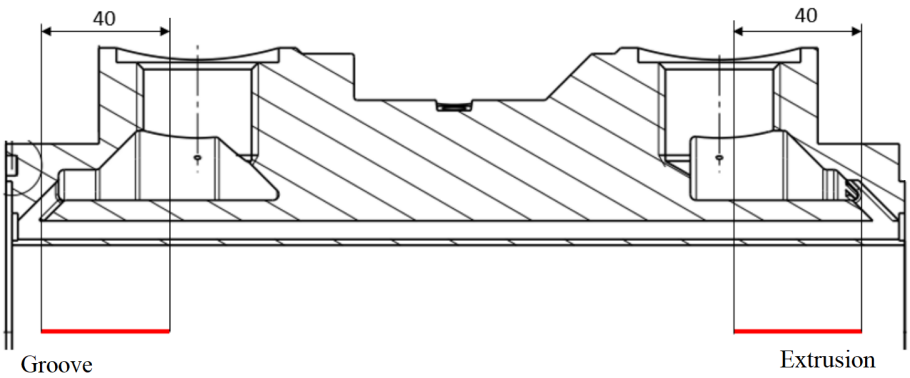

Figure 6.4: Measurement positions for the surface roughness measurements. Not to scale.

	Groove [μm]				**Extrusion [μm]**			
	R_a %	*Std*	R_z %	*Std*	R_a %	*Std*	R_z %	*Std*
30°	-7.34	0.28	-13.75	2.93	-13.54	0.76	-17.32	12.78
120°	+8.40	1.64	+12.59	17.85	-2.96	0.94	-6.39	12.65
210°	-15.45	0.34	-16.22	11.40	-5.59	1.38	-4.01	22.01
300°	+14.39	1.52	+17.39	18.10	22.10	1.28	+27.74	17.15

Table 6.1: Measured roughness statistics on the combustion chamber surface of the segment 2RR. For each position, the results are presented as a percentage deviation from the average value.

The surface located at an angle of 30° features a lower roughness than the side located at 300°, and the difference reaches the 22% for the arithmetic mean deviation and the 28% for the maximum peak-to-valley height. These differences play a significant role in the temperature distribution during the tests and will be investigated further in the following sections. A similar deviation is expected within the cooling channels. However, it could not be measured due to the destructive nature of the required tests.

6.3 Reference Database

6.3.1 Operating Point Definition

Before the start of the test campaign, the target operating points are defined in terms of combustion chamber pressure, P_c, oxidiser-to-fuel ratio, OF, and coolant mass flow, $\dot{m}_c$. Once the geometrical dimensions of the chamber are given, the CEA code, [141], is used to retrieve the required propellant mass flows, giving the desired combustion chamber and OF values as input.
When the required mass flows are known, a set of flow checks is performed at the test bench to properly tune the upstream pressure and the opening of the control valves.

As explained in section 4.2.3, the chamber's ignition procedure utilizes two torch igniters. The ignition stage is set at 10 bar and occurs 16.5 seconds after the sequence starts, lasting approximately 5 seconds.
After this stage, usually, two operating points are executed within the same test. Fig. 6.5a shows the behaviour of the pressure inside the chamber at various axial locations. The sensors are called "P_c100" to "P_c113" as the distance from the faceplate increases. As expected, the pressure decreases throughout the chamber because of the expansion and acceleration of the hot gases. In Fig. 6.5a, three stages are clearly visible: from seconds 16.5 to 22, the ignition stage at 10 bar is visible. Afterwards, the pressure in the chamber is ramped up to approximately 20 bar and maintained at this level for 12 seconds. Lastly, the pressure increases again up to 25 bar, and after another 12 seconds, the chamber is shut down, approximately at $t = 46\ s$.
This sequence illustrates an exemplary test.
Eq. (6.1) is used to compute the heat flux as a function of time for each segment, as shown in Fig. 6.5b.
In Fig. 6.5b, the three different stages of the tests are clearly recognised, as an increase in combustion chamber pressure directly leads to an increase in heat flux.
To assess the thermal steadiness of the chamber wall, the measurement of the thermocouples placed in the wall can be analysed. Fig. 6.6 presents the behaviour of the temperature readings of the sensors at 1 mm from the hot wall. It can be observed that the copper becomes hotter and hotter as the combustion chamber pressure increases, and it takes several seconds to adjust its temperature and reach a steady state. The required time is particularly long at the beginning when the test starts and when the pressure is increased to 20 bar, while it is faster when transitioning from the first to the second operating point. Several tests have been performed to assess that the ignition

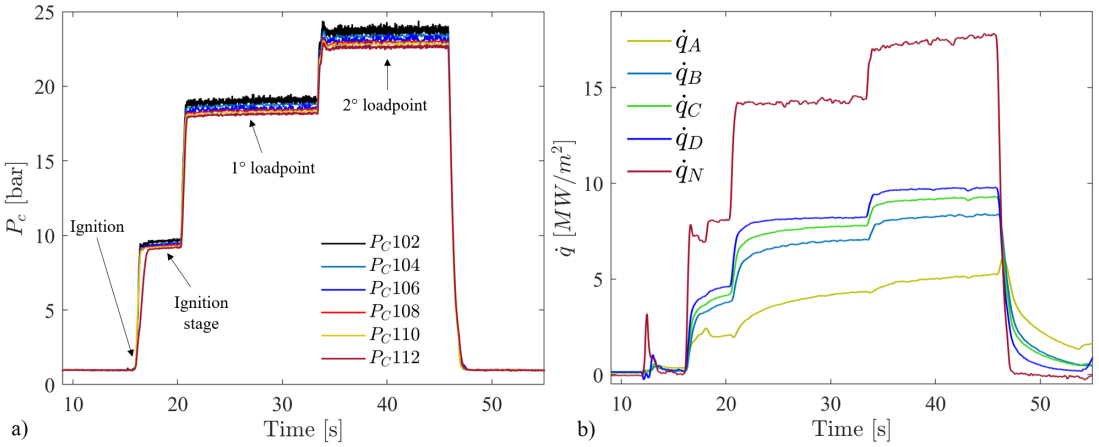

Figure 6.5: Sample test results: a) combustion chamber pressure behavior during the test; b) calculated heat transfer on each segment.

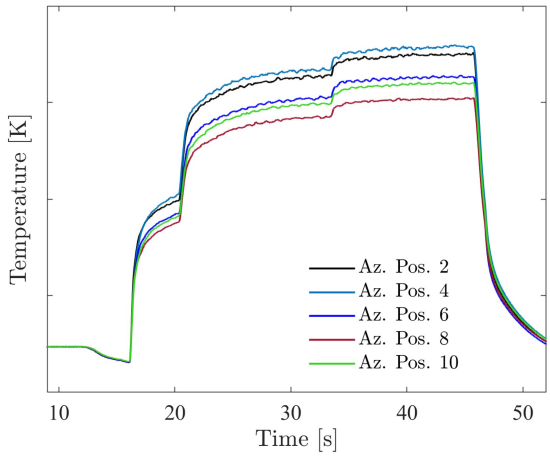

Figure 6.6: Temperature reading of the thermocouples at the different azimuthal positions.

procedure, shown in section 4.2.3, and the sequencing presented here do not affect the final thermal state of the chamber.

The thrust chamber is equipped with three dynamic pressure sensors, as shown in Tab. 4.3, placed in the GOx manifold, CH4 manifold, and at the beginning of the combustion chamber. These sensors are used to monitor the stability of the combustion. The high-frequency measurement enables us to perform a Fast Fourier Transform and determine which frequencies are excited during the test, as well as measure their intensity. In Fig. 6.7, the spectrum of one test is shown through a waterfall plot. An excitation of a well-defined frequency range is visible around 1500 Hz, which is the first longitudinal mode of the chamber. However, the magnitude of this oscillation is around 1% of the combustion chamber pressure, which satisfies the requirement on combustion stability defined in section 6.2.

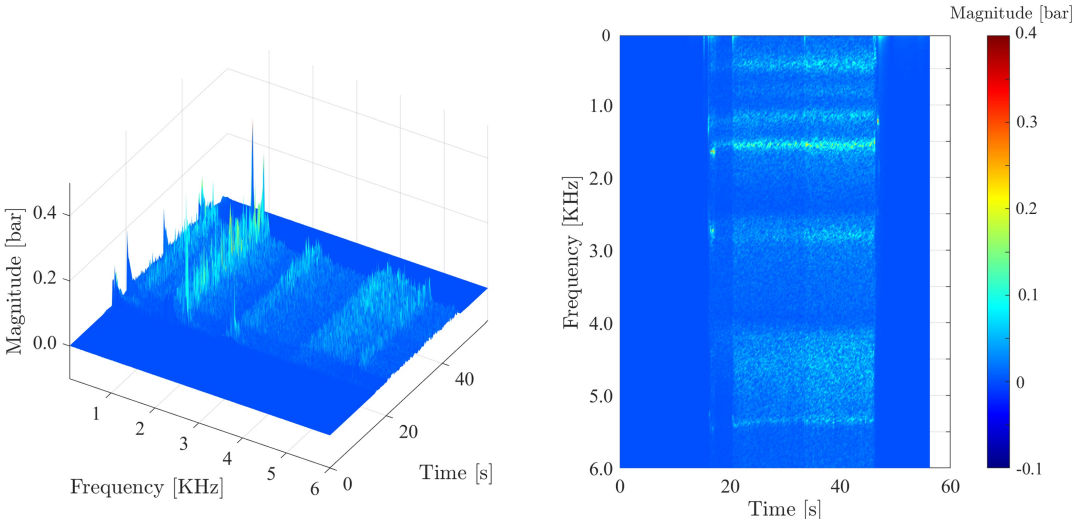

Figure 6.7: Waterfall diagram of the spectrum of one exemplary test.

6.3.2 Reference Setup Results

As explained in section 6.2, the calorimetric approach only gives one scalar value for each segment and assumes that the heat flux is constant over the length of the segment itself. Of course, this is not true: on the first segment, "A", the heat flux increases from almost zero close to the faceplate to a final value at the end of the segment, which is higher than the mean value predicted by the calorimetric approach.
This can be proved by the pressure behaviour along the axis of the chamber, presented in Fig. 6.8. It can be seen that the pressure evolution is very similar when made nondimensional at both pressure levels, 20 and 25 bar. Additionally, a significant peak at the beginning of the chamber is visible, followed by a rapid decrease, which indicates the expansion and acceleration of the hot gases. At around $x = 250\ mm$, a plateau is reached, which indicates that the combustion is in equilibrium.
In Fig. 6.8, dashed lines indicate the position of each segment. Since the results are already time-averaged, the error caused by measurement noise is not considered, and only the fixed accuracy error, as specified in the sensor datasheet, is reported in the plot with error bars.
It is shown that the combustion is almost completed in segment "B" and totally completed in segment "C". This consideration has ruled the position of the test article, which has been placed substituting the segments "B" and "C", as shown in Fig. 4.4. In this way, it is assumed that the test article is subjected to an approximately uniform heat flux in the axial direction.
With this setup, presented in Fig. 4.2 as the traditional setup and indicated here as "TRAD", a first test campaign is performed. A combustion chamber pressure varying from 18 to 25 *bar* is used, and an oxidiser to fuel ratio of 3.4.
Fig. 6.9a shows the average heat flux on segments "B" and "C" at varying combustion chamber pressure that has been evaluated with Eq. (6.1). As expected from a smooth

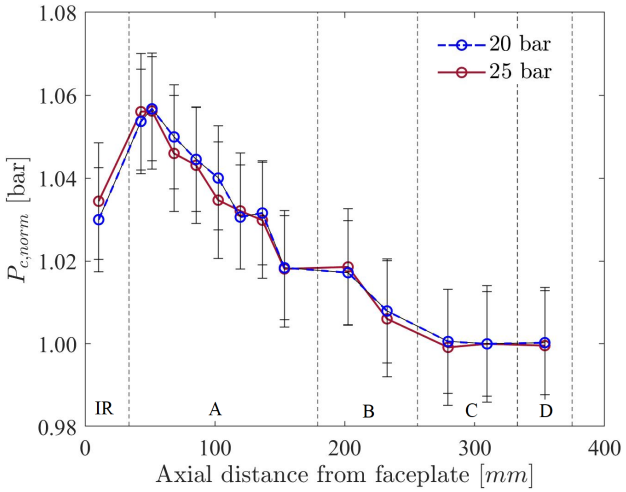

Figure 6.8: Pressure evolution along the axis of the chamber, normalised with respect to the last value.

chamber with smooth cooling channels, the resulting heat flux points can be interpolated as a function of the combustion chamber with a power law. In this case, the exponent is equal to 0.84, close to the theoretical value found in the Bartz correlation, [7].

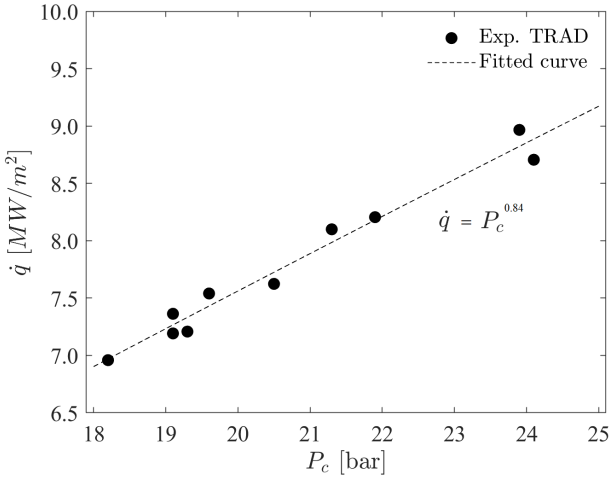

Figure 6.9: Heat flux as a function of the combustion chamber pressure in all the tests run with the traditional setup from Fig. 4.2 (black dots) and their interpolation (dashed line).

The quality of the fitted curve has been measured with the R^2 method (or the coefficient of determination), which delivered a value of 0.97, where 1 is the perfect fit.

The calorimetric approach provides a global result, measuring the average heat transfer absorbed by the coolant in a specific segment. In the following, great attention is given to the temperature field distribution around the circumference, highlighting any non-uniformity of the temperature field that is already present in the traditional setup.

The resulting deviation is presented in Fig. 6.10, where a polar representation is used to better display the position in the chamber of each value. Each curve represents the results of a single test.

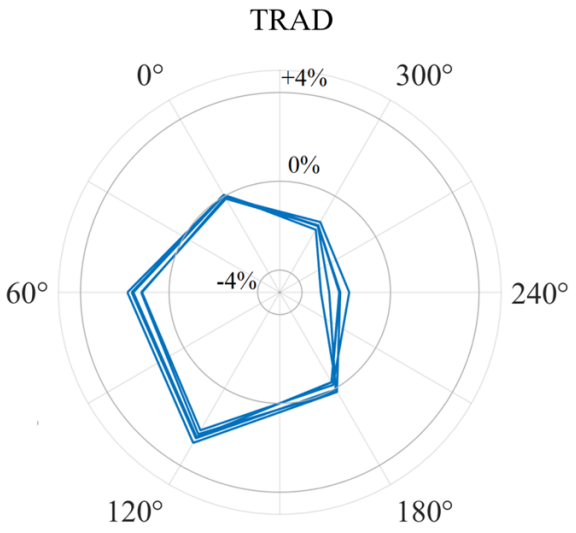

Figure 6.10: Azimuthal distribution of the temperature deviation θ. Darker grey lines mark steps of 4%.

A trend is clearly visible, with higher temperatures in the range of azimuthal positions between 60° and 120°. On the opposite side of the chamber, a colder side is visible. However, the magnitude of these deviations is only in the order of $2 - 3\%$ of the mean temperature.

6.4 Test Campaign

6.4.1 Introduction

In this section, the test results of the five test articles presented in Tab. 4.4 are discussed and compared against the reference database presented in the previous section.

The test matrix used for the test campaign is presented in Tab. 6.2: every operating point is repeated at least twice, and the same sequences are used to run the control software.

The objective of the tests is to assess the influence of the surface roughness on the hydraulic performance of the test articles, namely, pressure losses and heat pick up.

LP	$p_c[bar]$	$\dot{m}_{GOx}[g/s]$	$\dot{m}_{CH4}[g/s]$	OF	$\dot{m}_{N2}[g/s]$	$p_{N2}[bar]$
REF	22.5	269.4	79.2	3.4	1000	100
LP1	25	299.3	88.0	3.4	1000	100
LP2	22.5	269.4	79.2	3.4	1200	120
LP3	20	240.1	70.6	3.4	1000	100
LP4	22.5	269.4	79.2	3.4	800	80

Table 6.2: Test matrix used for every test article.

6.4.2 Friction Enhancement

The first output of the tests is the increase in friction losses resulting from the higher surface roughness present in the cooling channels. To present the results in a general frame that is not dependent on specific test conditions, the friction factor and the Reynolds number are used. The friction factor is evaluated as in Eq. (6.6).

$$f = \Delta p \frac{2 d_H}{\rho u^2 L} \tag{6.6}$$

where L is the cooling channel length, ρ and U are the fluid density and velocity. d_H is the hydraulic diameter of the channel evaluated as $d_H = 4A/P$, where A is the cross section area and P the perimeter. The nominal design values have been used to define the hydraulic diameter, without accounting for the presence of roughness.
To use this equation, an average density ρ between the inlet and outlet manifolds has been calculated with REFPROP, [139], by using the pressure and temperature readings in the inlet and outlet manifolds. The geometrical dimensions of the channel and the measured mass flow are used to retrieve the velocity of the flow. For the pressure difference, $\Delta p = p_{in} - p_{out}$, the pressure sensor's reading has to be corrected. As can be seen in Fig. 6.4, the connection between the inlet manifold and the beginning of the cooling channel is made through a sharp turn, and the same happens at the outlet side. This causes a pressure loss that is not due to the surface roughness and thus affects our measurements. In order to take this into account, the correction described in appendix A.4 is used.

The resulting friction factor is shown for all five test articles in Fig. 6.11. This figure displays the results of multiple tests conducted under various conditions. Some tests are performed in "cold" conditions, meaning that the gaseous nitrogen flows in the cooling channels, and no heat source is present. Other tests (at $Re > 4 \cdot 10^5$) are conducted in "hot" conditions, meaning that the combustion takes place in the combustion chamber while the nitrogen flows in the channels.
In Fig. 6.11, it can be seen that for each test article, the friction factor stays approximately constant over a wide range of Reynolds numbers, from $1 \cdot 10^5$ to $9 \cdot 10^5$.
This was expected, as already demonstrated from the experiments of Nikuradse, [55]. The average friction factor value calculated for each test article is summarised in Tab. 6.3,

along with the standard deviation of the measurement.
Error bars are reported for each measurement. A trend is easily recognised, with the error bars being much larger on the left side of the plot and disappearing on the right side. This is explained by looking at how the friction factor is evaluated in Eq. (6.6). With a high mass flow rate, the Δp becomes larger and larger, minimising the measurement error. At a low mass flow rate, the small pressure difference between the inlet and outlet is more influenced by the measurement accuracy.

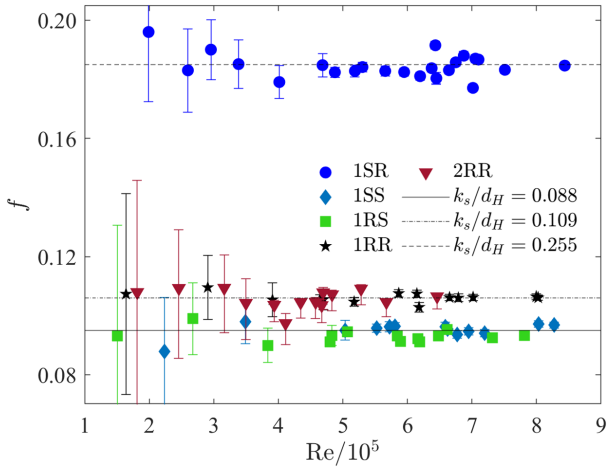

Figure 6.11: Friction factor of the test articles calculated with Eq. (6.6).

Article	f_{avg}	Std
1SR	0.185	0.004
1SS	0.095	0.002
1RS	0.093	0.002
1RR	0.106	0.001
2RR	0.106	0.003
Smooth Eq. (6.7)	0.012	-

Table 6.3: Average friction factor and its standard deviation.

Larger error bars are particularly visible for the results at $Re < 4 \cdot 10^5$, which are all cold tests. The hot tests described in Tab. 6.2 and that will be analysed in the next sections are all characterised by higher Reynolds numbers and, thus, negligible measurement error.

To compare the values in Fig. 6.11 with open literature, black lines are used to show the k_s/d_H that has to be used in the Colebrook relation, Eq. (6.7), to get the same results, where k_s is the sand-grain parameter discussed in section 3.5.5.

Overall, the two segments with post-processed cooling channels, the 1SS and 1RS, have the lowest friction factor among the investigated specimens, proving that the EDM process smoothens the channel surfaces. However, this value is still much higher than the ideally smooth surface, as predicted from the Colebrook relation, in Eq. (6.7), from [58], and reported in Tab. 6.3 for comparison.

$$f = \frac{1}{\sqrt{f}} + 2\log\left(\frac{k_s/d_H}{3.72} + \frac{2.51}{Re\sqrt{f}}\right); \tag{6.7}$$

The segments 1RR and 2RR, which have cooling channels as-built, show the same friction factor, which is higher than that of the post-processed segments. This highlights the increased friction losses resulting from rougher surfaces within the cooling channels.

The last segment, the 1SR, exhibits a significantly higher friction factor, even though the cooling channels are left as built and should have similar performance to segments 1RR and 2RR. This may indicate some off-nominal conditions of the test article. For example, some channels might be smaller than the nominal design, or some powder might not have been properly removed during the depowering phase.
It is of great importance to underline that the difference in the friction factor between the segments 1RR and 1SS might not seem very large. However, a difference in friction factor of 12% is quite significant because it directly correlates with the pressure drop occurring in the cooling channels.

The friction factor shown in Fig. 6.11 is a dimensionless parameter that well describes the friction losses in the investigated flow path. An additional comparison of the performance is presented in Fig. 6.12, where the corrected pressure difference between the inlet and outlet is shown for each segment and operating point. Error bars are reported, although the measurement error is negligible with respect to the absolute value of the pressure difference.

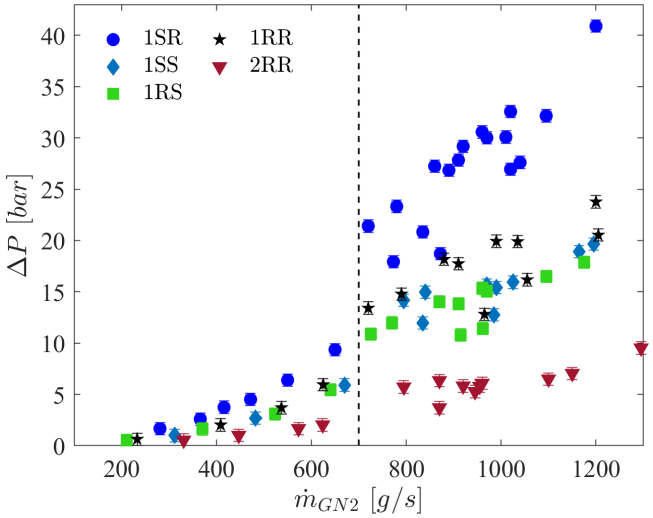

Figure 6.12: Pressure difference between cooling channels' inlet and outlet for each test article. The black dashed line divides cold tests (on the left) and hot tests (on the right).

A clear separation is immediately visible between the cold tests and the hot tests. The distinction is clear because, for the cold tests, the data for each specimen are very well aligned and would be easy to fit. On the other hand, when the combustion gases heat up the coolant, the pressure difference between the inlet and outlet is no longer uniquely dependent on the mass flow of the coolant, but also on the amount of heat absorbed, which influences the coolant temperature. This is why, on the right of the black dashed line in Fig. 6.12, a more pronounced spreading of the data point is visible. Despite the spreading, the trend of each segment can still be distinguished: the 1SR is the one with the highest pressure drop; the 1RR follows, with a less steep behaviour. The two segments with EDM cooling channels exhibit a very similar pattern. Lastly, the lowest

pressure drops are recorded in the 2RR due to its larger cooling channel section.
Apart from segment 2RR, the other four articles feature the same geometrical dimensions of the cooling channels. However, the pressure drop, when compared at the same operating points, is very different. This difference is purely due to the quality of the surfaces, highlighting the importance of accounting for surface roughness in the design of additively manufactured components.

6.4.3 Heat pickup

The second objective of this test campaign is to characterise the heat transfer capabilities of the rough surfaces in rocket-like conditions.
The heat transfer performance is strictly related to the friction behaviour of the flow, as well documented in [142] and in recent studies [143, 144], which focus on how to optimise the two phenomena.
In this case, no optimisation of the geometry is carried out. However, the study examines how an additional finishing step, specifically the EDM in the cooling channel, can influence the heat transfer performance while also reducing friction losses, as discussed in the previous section.

Eq. (6.1) is used to quantify the enthalpy increase of the coolant and obtain the heat flux. Fig. 6.13 displays the results for the five test articles at the same operating point, specifically the operating point REF from Tab. 6.2. Since the calorimetric approach only gives one scalar value of heat flux for each segment, no axial variation within the same segment can be calculated.

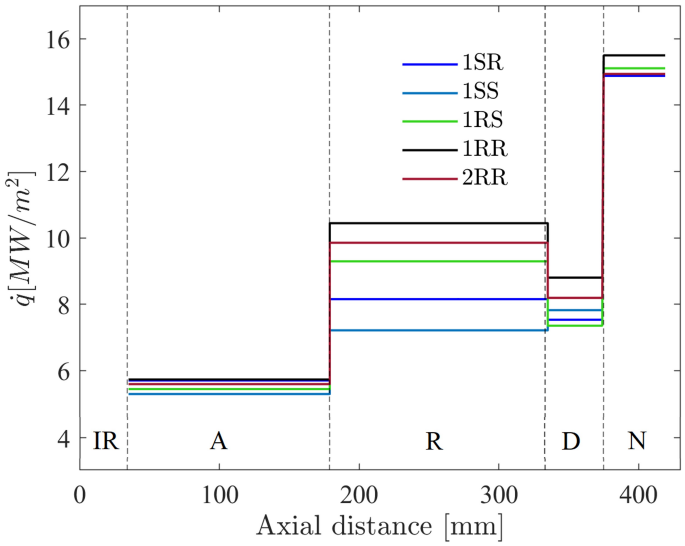

Figure 6.13: Heat flux calculated with Eq. (6.1) for each segment and for each test article. All curves have been evaluated at the same operating point, REF in Tab. 6.2.

This assumption is proven to be reasonable in the case of segment R by checking the completeness of the combustion at the axial coordinate, Fig. 6.8. Additionally, the method is applied after 10 s of testing the operating point, when thermal stability is reached, according to the criteria in Eq. (6.2).

Fig. 6.13 shows that the investigated segments have very different heat pickup performance. Segment 1SS, with a machined chamber and post-processed cooling channels, has the lowest heat flux value. This segment features the closest surface quality to the currently used standards for space components, which are always machined, obtaining a very low roughness level.
For segment 1SR, it can be seen that the increased roughness in the cooling channels is beneficial for the heat transfer performance since the heat pickup is approximately 12% higher. This effect is attributed solely to the roughness in the cooling channels, as segments 1SR and 1SS have equally smooth combustion chambers.
Another significant enhancement of the heat transfer process is recognised by looking at the performance of segment 1RS, whose heat flux is approximately 30% higher than that of segment 1SS, which has the same cooling channel surface quality but a smooth chamber. The comparison between the two highlights the remarkable effect of a rough combustion chamber.
Lastly, the two segments featuring as-built surfaces for both the combustion chamber and the cooling channels exhibit the highest heat pickup, showing a total heat flux increment of 40%, which is the sum of the enhanced heat transfer capabilities on both the hot and cold sides of the wall.

As expected, comparing the same operating point for all test articles results in approximately the same heat flux level on segment “A”, indicating no difference in the setup. On the other hand, segment “D” shows larger differences in the results. The heat pickup in this segment exhibits greater variability compared to segment “A”, while following the trend observed in the investigated test article. This can be attributed to the interaction of the hot gases with the rough chamber surface. The increased turbulence generated in this region affects the heat transfer to the smooth surface of segment “D”, leading to varying levels of heat flux, as illustrated in Fig. 6.13.
In addition, both segments “A” and “D” are in contact with the specimen under investigation, meaning conduction must also be considered. Heat is likely transferred from segment “R” towards segments “A” and “D”, as these segments are better cooled by their respective water circuits and therefore maintain lower temperatures during testing. This mechanism could also explain the recurring similarity observed between segments “R” and “D”, where the black and red lines stand out above the others. It is likely that due to the very high wall temperatures reached in segment “R” under these conditions, a strong conduction influences the heat pickup in the neighbouring segment as well. Nevertheless, this effect has not been quantified and will not be discussed further.

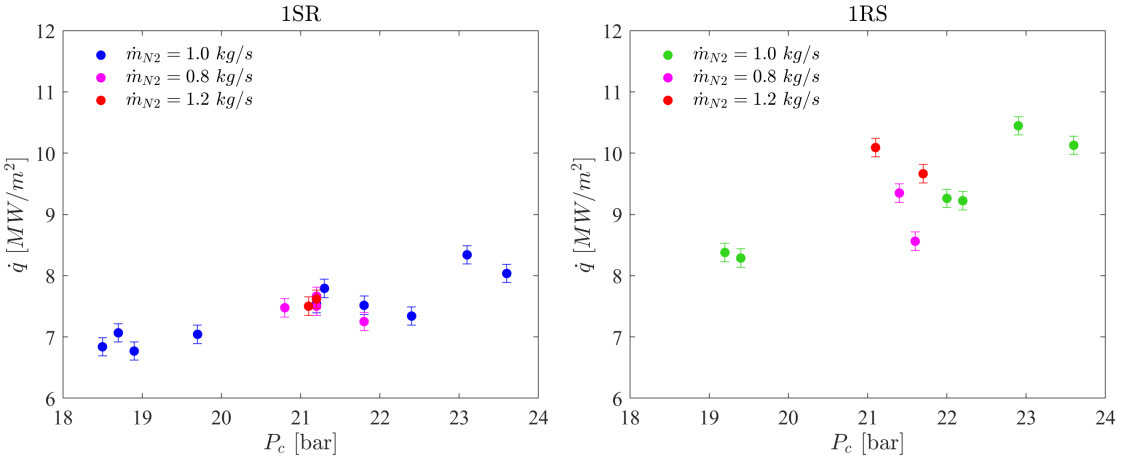

Figure 6.14: Heat flux calculated with Eq. (6.1) for segments 1SR and 1RS and for every operating point.

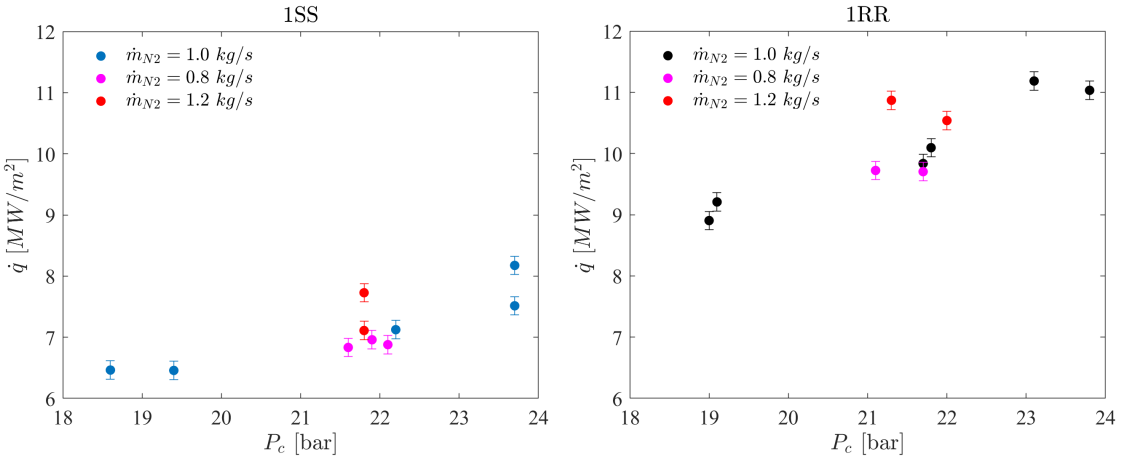

Figure 6.15: Heat flux calculated with Eq. (6.1) for segments 1SS and 1RR and for every operating point.

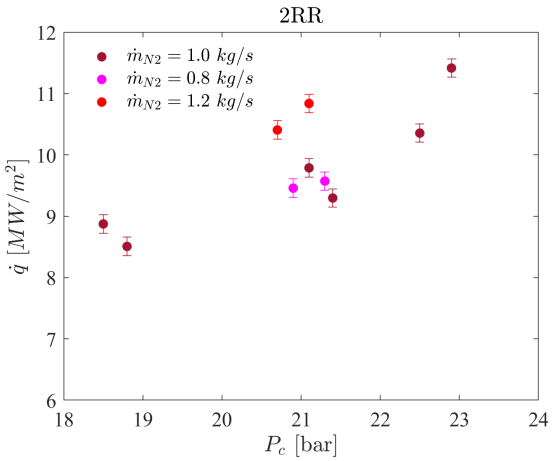

Figure 6.16: Heat flux calculated with Eq. (6.1) for segment 2RR and for every operating point.

By repeating the same procedure, it is possible to evaluate the heat flux on every test article and at every operating point tested during the test campaign. This is presented in Fig. 6.14, Fig. 6.15, and Fig. 6.16, where the heat flux is shown against the tested combustion chamber pressure. In each plot, the results of one specimen are shown, and different colours are used to highlight the different mass flows used for the coolant during the test campaign. The trend is evident and consistent with theoretical expectations, showing an increase in heat flux with rising combustion chamber pressure. Additionally, the heat pickup is known to scale almost linearly with the Reynolds number of the coolant, [145]. Accordingly, the points corresponding to a lower mass flow (magenta) are consistently located below those with a higher mass flow (red). However, the magnitude of this difference is highly variable, indicating that coolant mass flow is not the dominant parameter in this case and that surface roughness can enhance heat pickup to a similar extent as an increase in mass flow.
It should also be noted that the scatter is not negligible, as additional parameters, such as the oxidiser-to-fuel ratio (OF), influence the heat flux. Although the OF is nominally set to 3.4 in all tests, oscillations in valve behaviour introduce small deviations, which in turn affect the measured heat flux.
The scatter is further amplified in combustion chambers with rough walls (1RS, 1RR, and 2RR). This suggests that wall roughness not only increases the absolute heat pickup but also intensifies sensitivity to small variations in OF or other influencing factors.

The results in Fig. 6.14, Fig. 6.15, and Fig. 6.16 are then merged together in Fig. 6.17, ignoring the difference in mass flow and only focusing on the differences among the specimens.

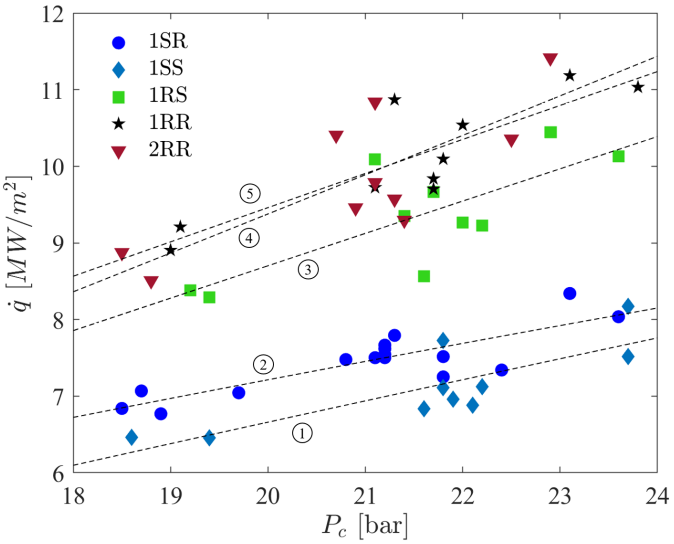

Figure 6.17: Heat flux calculated with Eq. (6.1) for each segment and each operating point of the test campaign. Fitting lines (black dashed lines) are added as interpolating functions.

At first look, two well-separated groups of data points are visible: the lower one, which involves the test articles with a smooth combustion chamber surface (curves 1 and 2), and the upper one, with the segments featuring a rough chamber surface (curves $3 - 4 - 5$).
As shown in Fig. 6.13, the comparison between the segments 1SS and 1SR points out the enhancement of the heat performance due to the roughness in the cooling channels, which is confirmed to be in the order of $10 - 12\%$ across the tested combustion pressure range. Examining 1RR, 1RS, and 2RR, a significant increase in heat pickup is evident, which is attributed to the roughness in the combustion chamber. The enhancement in heat pickup capabilities is significantly higher than the one due to the rough cooling channels and is confirmed in the range of $30 - 35\%$ when comparing the segment 1SS and 1RS.
Despite the different shapes of the cooling channels, the distribution of data points belonging to segments 1RR and 2RR appears to be similar, highlighting that the surface quality of the chamber and/or the channels has a much stronger effect on heat pickup than the change in aspect ratio.

The data points of each test article are interpolated with a power law, as was done for the traditional setup, in Fig. 6.9a. The parameters of each curve are given in Tab. 6.4.

		$\dot{q} = b \cdot P_c^m$		
Article	**Curve N.**	b	m	R^2
1SS	1	0.54	0.83	0.66
1SR	2	0.97	0.68	0.72
1RS	3	0.47	0.98	0.60
1RR	5	0.56	0.92	0.62
2RR	4	0.35	1.08	0.59

Table 6.4: Parameters of the interpolation curves presented in Fig. 6.17, and their goodness, measured with the R^2 method.

Although the present correlation between the heat flux $\dot{q}$ and the chamber pressure P_c follows an exponential trend similar to that of the convective heat transfer coefficient proposed by Bartz, it should be emphasised that this analogy is only approximate. In a real multilayer heat transfer process, the overall heat flux depends not only on the gas-side convection but also on the conductive resistance of the wall and the convective resistance on the coolant side. Therefore, the pressure dependency of $\dot{q}$ may differ from that of the gas-side heat transfer coefficient $h(P_c)$, as reflected by the different slope observed in Fig. 6.17.
Nevertheless, the empirical fit $\dot{q} \propto P_c^{0.84}$ found for the traditional setup in Fig. 6.9 provides a useful engineering correlation within the investigated operating range, capturing the general pressure dependence of the measured heat flux. This relation is thus retained as a practical interpolation of the experimental data, while acknowledging

that it does not represent a fundamental scaling law of the heat transfer coefficient itself.

In Tab. 6.4, it can be seen that the best-fitting power law for the results of segment 1SS has an exponent of 0.83, which is very close to the theoretical value of 0.8 for the Bartz correlation [7]. This correlation is usually applicable for smooth combustion chambers, and indeed, segment 1SS features the smoothest surfaces among the investigated specimens. Segment 1SR, however, has a lower exponent, indicating that the roughness in the cooling channels slows the increase in heat flux with respect to the combustion chamber pressure. On the contrary, the segments with a rough combustion chamber fit with a larger exponent, always above 0.9.
This effect has never been reported in open literature and has important implications. On a wider combustion pressure range, the variation with respect to the theoretical value can become significantly larger than the 30% difference found in this work, which is limited to a specific combustion pressure range. Of course, this has enormous implications for the design of components and their performance predictions, which deviate significantly from the smooth wall empirical laws commonly used in the design phase.
It is clear from Tab. 6.4 that the accuracy of the power laws used to fit the data points is limited due to the scatter of the points, as can be seen by the R^2 value, which ranges between 0.6 and 0.7. Despite the spreading of the results, it can be stated that the surface roughness changes the slope with which the heat flux varies with respect to the chamber pressure. The surface roughness in the cooling channels seems to decrease the exponent of the power law while the roughness inside the chamber increases it.

It is then found that the rougher the surface, the more enthalpy can be absorbed by the coolant. However, this trend is not enough to predict the behaviour of the combustion chamber wall temperature. A rough wall exchanges more heat with the coolant, thus becoming colder. At the same time, the rough inner wall of the combustion chamber exchanges heat more effectively with the hot gases, becoming hotter.
To assess these behaviours, the readings of the thermocouples placed in the chamber wall are used to estimate the temperature of the hot wall. A 1-D steady-state system is assumed, as shown in Eq. (6.5). The temperature difference of the coolant between the inlet and outlet is presented in Fig. 6.18 against the resulting estimated wall temperatures.
The results of each test article can be interpolated with a linear function, reported in Fig. 6.18. As expected, segment 1RS reached the highest wall temperatures due to the rough chamber walls, which efficiently exchange heat with the hot gases, and the smoother cooling channels, which are less effective in dissipating the heat. On the contrary, segment 1SR features the lowest wall temperature due to the smooth chamber walls and the rough cooling channels.
Once again, two groups can be distinguished, but this time, the ruling factor is the surface quality of the cooling channels. The first group includes the segments 1SS and 1RS, which are the two articles with EDM cooling channels. The results of these two segments align perfectly on a common straight line, and the only difference between

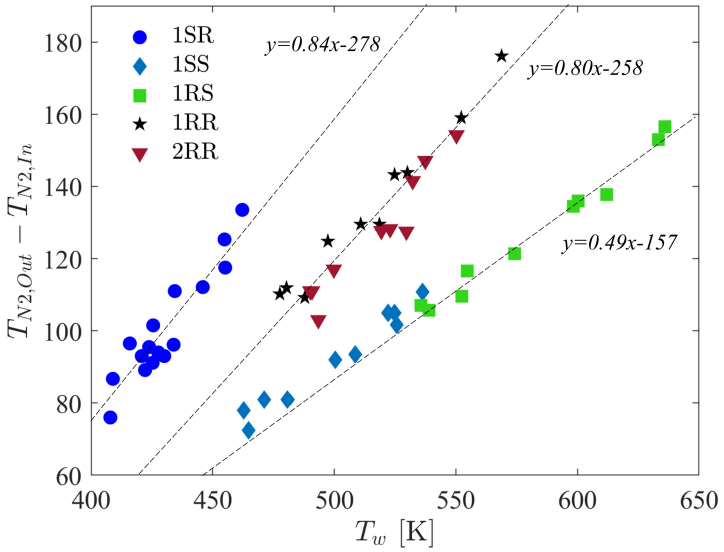

Figure 6.18: Temperature difference between outlet and inlet of the cooling channels, against the estimated wall temperature, with Eq. (6.5).

them is a shift of the results of the 1RS towards the top-right of the graph due to the roughness on the combustion chamber surface.
The second group, on the other hand, consists of segments with as-built cooling channels, specifically the 1SR, 1RR, and 2RR. The results of these segments, even if not aligned on the same straight line, align on lines with almost the same slope, independently of the surface quality of the chamber walls.
These results indicate a linear correlation between the chamber wall temperature and the increase in coolant temperature. The slope of this linear function is set by the surface quality of the cooling channels, while the roughness on the combustion chamber surfaces increases the wall temperature. Consequently, it linearly increases the coolant heat pickup.

Since the slope of the second group is steeper than the first one, it can be concluded that the roughness in the cooling channels helps the coolant absorb more heat while keeping the combustion chamber walls colder. This is important when the wall temperature has to be kept low. However, if possible or necessary, using a rough combustion chamber would result in even higher heat pickup.
These considerations become significant when considering a wider system level, rather than just the investigated component. Indeed, these results could find a perfect application in an expander cycle, where the heated coolant must drive the turbine, and the greater the heat pickup in the cooling channels, the better the turbine performance. However, the wall temperature is often used as an indicator to select the best trade-off between the chamber's performance and structural integrity. Therefore, knowing about the change in slope of the curves in Fig. 6.18 becomes crucial.

6.4.4 Nusselt number

The Nusselt number is typically used to provide a dimensionless quantification of heat transfer performance. However, in an experimental setup, the calculation is not always straightforward and involves several assumptions.
In this work, the Nusselt number is evaluated as in Eq. (6.8).

$$Nu = \frac{h d_H}{k} \qquad \text{with} \quad h = \frac{\dot{q}}{T_{ch,wall} - T_b} \tag{6.8}$$

where h is the convection coefficient inside the cooling channels, $T_{ch,wall}$ the temperature of the channel's wall and T_b the bulk temperature of the coolant.
Obviously, all these parameters have significant variations depending on the exact position of evaluation. In our calculations, we evaluate these quantities at the exact position of the thermocouple, specifically at the midpoint of the cooling channel. In the assumption of uniform heat flux, a linear increase of the coolant temperature is expected along the channel. Thus, we assume the bulk temperature to be the arithmetic average between the outlet and inlet temperatures that are recorded in the manifolds, $T_b = (T_{N2,in} + T_{N2,out})/2$.
Additionally, assuming one-dimensional heat transfer along the radial direction, the channel wall temperature is assumed to be identical to the reading of the thermocouple. A precise description of the positioning of the thermocouple and the assumed channel wall temperature is given in Fig. 6.19. $T_{ch,wall} = T_m$ is assumed. Bearing in mind these assumptions, the Nusselt number is evaluated and presented in Fig. 6.20. Once again, the test articles with post-processed cooling channels exhibit overlapping performance and are well aligned across the investigated Reynolds number range. It appears that the EDM process is well repeatable and that, when made dimensionless, the cooling channels of the test articles 1SS and 1RS have the same performance. The roughness of the combustion chamber does not affect the evaluated performance indicator of the cooling circuit. The three segments with as-built cooling channels keep the same slope as the other two segments, but their curves are shifted to higher Nu values. The increase is quite significant.
The segments 1SR and 2RR exhibit an increase in the Nusselt number by approximately 80% to 100% across the investigated range of Reynolds numbers compared to 1SS or 1RS.
The presence of surface roughness in the cooling channels is highly beneficial to the component design when the objective is to maximise heat transfer capabilities. However, the increase in heat transfer performance must be carefully weighed against the increased pressure losses, as shown in Fig. 6.11, and the higher wall temperatures, as depicted in Fig. 6.18, which result in increased pressure budget requirements and a shorter operating life, respectively.
For this reason, in literature, many indices can be found to evaluate how much the heat transfer performance is increased with respect to the friction losses. For example, from the energy equation Eq. (3.3), Osborne Reynolds derived the "*Reynolds Analogy*", that

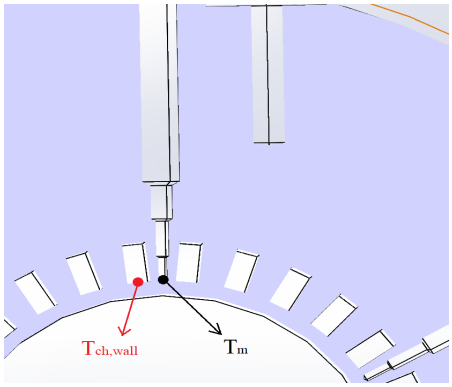

Figure 6.19: Thermocouple position: difference between the measurement point (black) and the channel wall position (red). Not to scale.

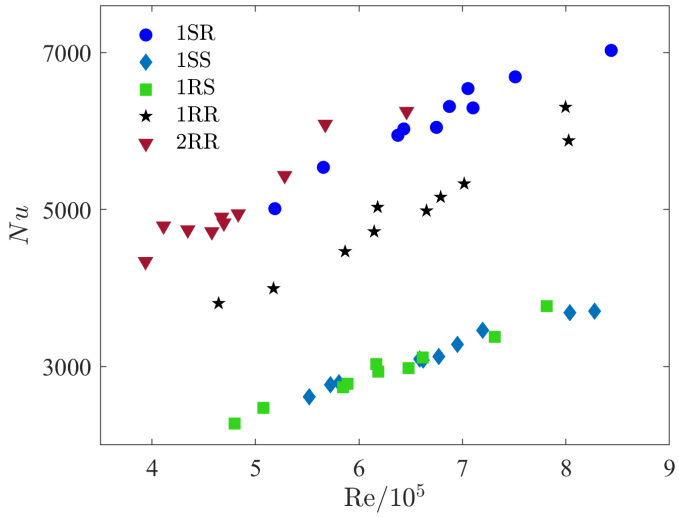

Figure 6.20: Nusselt number evaluated with Eq. (6.8) for the five different test articles.

in its most basic form can be written as:

$$\frac{2St}{C_f} \approx \frac{1}{Pr} \tag{6.9}$$

where St is the Stanton number and C_f the skin friction coefficient. Even if this relation should not be used to provide precise quantitative measures, the logic behind it is applied in the design of laboratory experiments and industrial components, despite the numerous assumptions and warnings about its limitations.
Bons investigated the reliability of this correlation in the case of rough turbine blades, [62, 77], and defined a "Reynolds analogy factor" as $(2St/C_f)_{rough}/(2St/C_f)_{smooth}$, showing that the friction losses increase more than the heat transfer.
Norris provides another example of the correlation between friction and heat performance [78], which utilises the Nusselt number and the friction factor to evaluate the ratio

between the two increments, as shown in Eq. (6.10).

$$\frac{Nu}{Nu_0} = \left(\frac{f}{f_0}\right)^n \qquad \text{with} \qquad n = 0.68Pr^{0.215} \tag{6.10}$$

where the subscript $(-)_0$ stands for the smooth case values.
The author states that this correlation should be valid for $f/f_0 \leq 4$. For ratios greater than 4, the Nusselt number increments, Nu/Nu_0, should not increase anymore. However, this statement has been found not to be true in another work, [146], where f/f_0 higher than 4 have been tested.

In this work, the Nusselt number and the friction factor are used to evaluate the heat transfer and friction loss enhancement, respectively. For the smooth case values, empirical relations from Colebrook [58] and Gnielinski [145] have been used. The results are presented in Fig. 6.21. In the same figure, the published data from [147] and [146] are reported. From [146], results for rectangular channels manufactured in several dimensions and materials are taken. From [147], instead, results for channels with angled grooves are extracted. More information on the nomenclature used in the legend of Fig. 6.21 can be found in the original works, [146, 147].
It is worth noting that the data published in other works were recorded under transitional conditions, meaning with a Reynolds number between 100 and 20.000. That is the reason for which the data from [146] and [147] shows a significant variation in the friction factor.
In this work, however, the experiments are carried out at a much higher Reynolds number, which results in a constant friction factor. Nevertheless, the test conditions were varied enough to present remarkable variations in the Nusselt number. This explains why in Fig. 6.21, the data from the present experiments show a vertical spreading, while the data from previous works feature a stronger horizontal spreading.

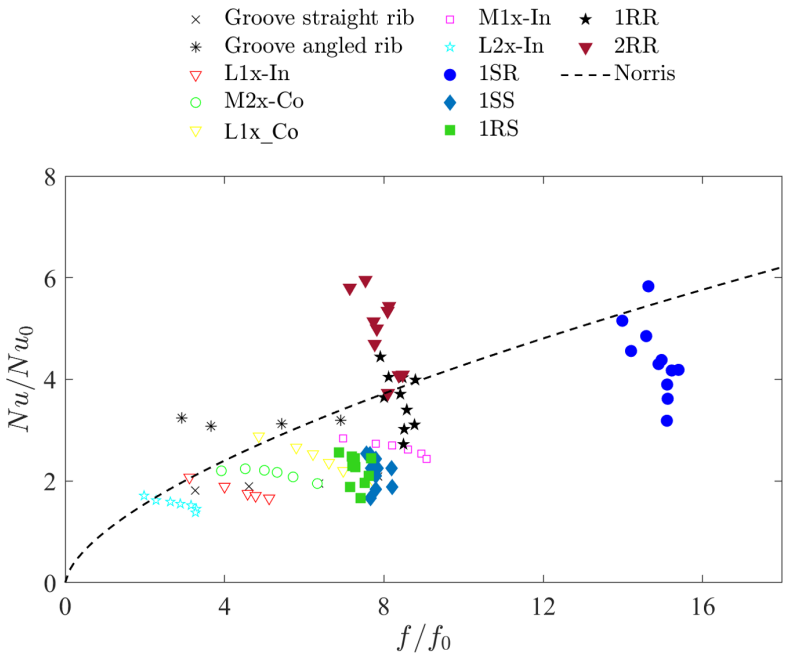

Figure 6.21: Heat transfer enhancement against friction loss increase. The correlation from Norris, Eq. (6.10), is reported with a dashed line. The data from [146] and [147] are reported with empty colored symbols. The data from this work are presented with filled-colored symbols.

As shown in Fig. 6.21, the Norris correlation (dashed line) does not reproduce the experimental results with satisfactory accuracy. While it reflects the general increasing trend of Nu/Nu_0 with f/f_0, large discrepancies are evident. Some data sets, such as the 2RR segment, lie significantly above the correlation, whereas others fall below it. This shows that the experimental data are too widely scattered to be represented by a single curve. The spread arises from the variety of test conditions and test specimens, which leads to markedly different behaviours that the Norris relation cannot capture.

Therefore, it can be stated that the deviation from the empirical relation remains important and that the correlation is of limited applicability to the present experiments. It is noted that testing the performance of components with rough surfaces remains necessary. However, in this work, the database of available test data is extended, and a comparison is given between the Norris correlation and the test data. This provides a broader experimental foundation for future refinement of correlations capable of capturing the effects of surface roughness and varying operating conditions.

6.4.5 Temperature Distribution

As explained in the previous section, section 2.2, the printing direction, the heat of the laser, the powder dimension, the overhang, and many other parameters strongly influence the final surface quality of the product. It follows that the surface roughness

distribution is not uniform at every point of the printed component. As can be seen from Tab. 6.1, a significant variation has been measured on the segment 1RR, not only along the axial direction (comparing "groove" and "extrusion" side) but also and especially in the azimuthal direction.
In these experiments, great attention is given to the temperature distribution and how this is affected by the different surface qualities. Unfortunately, only one axial position is measured and no data about the axial influence of the roughness can be extracted. However, six thermocouple positions are available, uniformly spaced around the chamber's circumference at 60-degree intervals.

The temperature deviation θ, defined in Eq. (6.3), is used to quantify the deviation of each measurement with respect to the azimuthal average. The results are shown in polar coordinates in Figs. 6.22 to 6.24. For the definition of θ, a negative deviation does not mean that the chamber is colder than the smooth case. This only means that, in that position, the temperature is lower than the average temperature recorded during the same operating point on the test article.

As can be seen, a more or less clear pattern is found for every specimen. Each line represents a different test, indicating that the pattern is consistently observed under various working conditions.
Contrary to what has been shown in Fig. 6.10, the deviation is now much more pronounced. In the traditional setup, temperature deviations are in the order of a few per cent, approximately 3% of the average temperature. In the case of additively manufactured segments, deviations higher than 10% are encountered. Additionally, the specimens with cooling channels post-processed with EDM exhibit a pronounced trend in the temperature profile, which is as evident as that of the as-built surfaces. This is explained by the fact that the EDM process does not utilise a sharp cutting tool, unlike common CNC (Computer Numerical Control) machines. The smoothing of the surface is obtained through electrical arcs generated in a dielectric fluid. Thus, the crests of the roughness profile are expected to be cut, significantly lowering the peak-to-valley distances R_z. However, the non-uniformity around the circumference of the segment remains consistent.

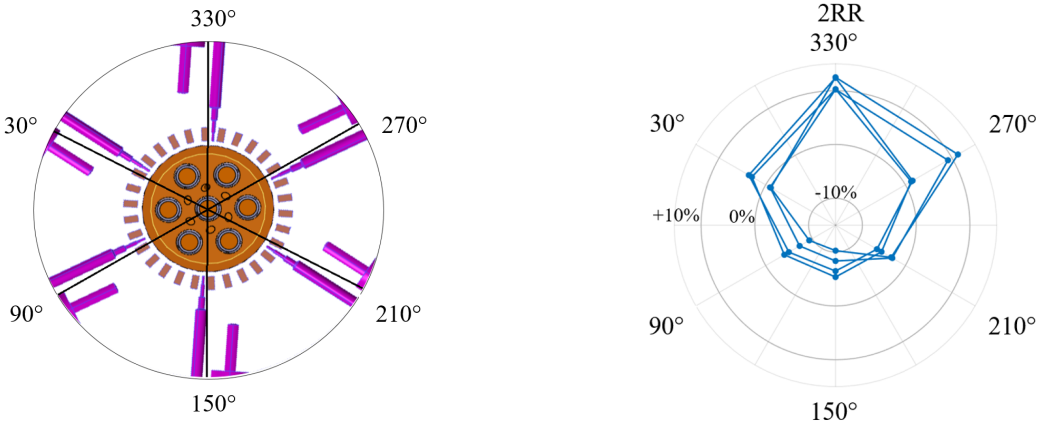

Figure 6.22: Thermocouples positions on the left, temperature deviation θ, from Eq. (6.3) for segment 2RR on the right. Darker grey lines mark steps of 10%.

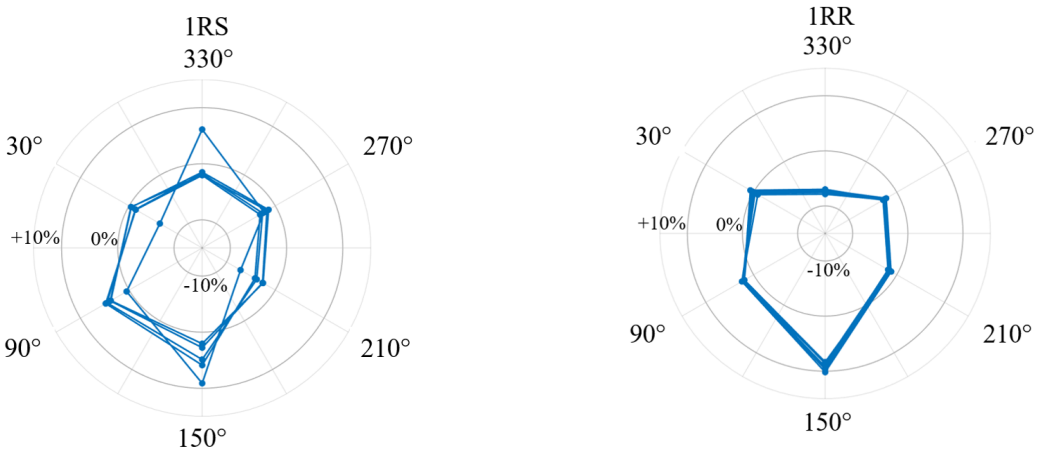

Figure 6.23: Temperature deviation θ, from Eq. (6.3) for segment 1RS, left, and 1RR, right. Darker grey lines mark steps of 10%.

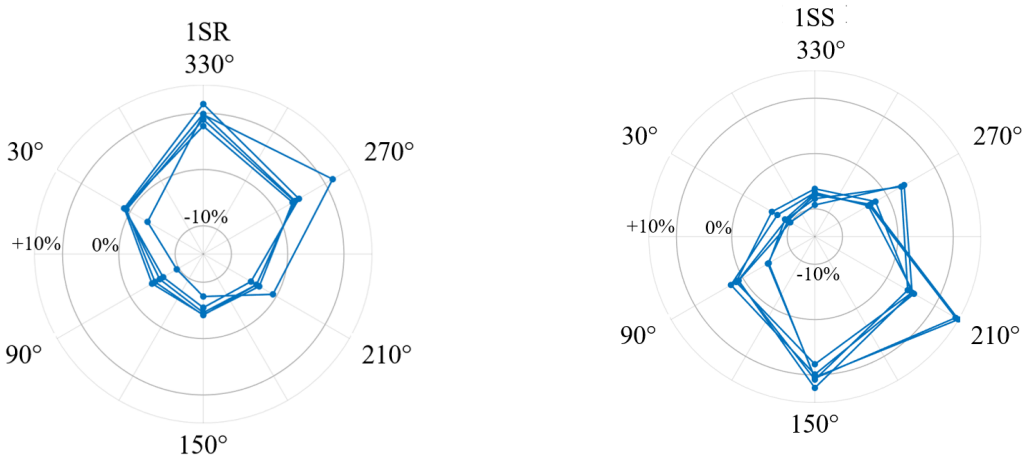

Figure 6.24: Temperature deviation θ, from Eq. (6.3) for segment 1SR, left, and 1SS, right. Darker grey lines mark steps of 10%.

Focusing, for example, on the temperature distribution found for the segment 2RR, in Fig. 6.22, a relation with the roughness distribution presented in Tab. 3.1 can be extrapolated. The roughest azimuthal sector is at 300°, as shown in Tab. 6.1, where the highest positive temperature deviation is measured, approximately +10%. It is concluded that the hot wall exchanges heat more effectively with the hot gases due to its higher roughness, becoming hotter than the opposite side of the chamber. In the sector at 120° indeed, the deviation θ is negative and reaches the lowest value, around -10%.

6.5 Conclusion

Within this test campaign, the effects of surface roughness due to the 3D printing process have been quantified. By comparing the five LBM test articles tested at the TUM Space Propulsion Laboratory, new trends are discovered and discussed in this chapter.
The friction factor increases from 8 to 15 times the theoretical smooth value, as shown in Fig. 6.21. The post-processing of the cooling channels through EDM is proven effective in reducing pressure losses.
It has been found that the heat pickup of the coolant increases by $10 - 12\%$ thanks to the roughness in the cooling channels, and approximately by $30 - 35\%$ if the combustion chamber presents a rough surface. The two effects are also cumulative.
When extrapolating the heat-flux results with respect to the combustion chamber pressure, it is observed that the slope of the fitting function increases for rough combustion chambers. In a large P_c range, this effect becomes significant, as the actual heat flux would substantially exceed the theoretical value predicted by empirical models developed for smooth walls. Additionally, the impact on the hot wall temperature is evaluated, showing that rough cooling channels help keep the wall temperature low and absorb larger quantities of heat with respect to the EDM channels. It is found that, for the same coolant heat pickup, the different configurations exhibit a temperature difference of up to 200 K on the hot wall, as shown in Fig. 6.18.
Lastly, an absolute difference of 20% of the mean temperature is recorded on two opposite sides of the chamber in the azimuthal direction during the same test. In a real combustion chamber, where the wall temperature is in the order of 900 K, this result would imply a difference in wall temperature of 180 K during the operation. Since the rocket combustors often operate near their structural limit, such a difference could be critical and lead to catastrophic failures.
These differences and non-uniformities are due to the printing process. Several printing parameters can be adjusted to enhance the surface quality, including the powder layer thickness, hatch distance, and laser power. Since these parameters significantly impact the surface quality of the component, they also directly affect its performance. Therefore, considerable resources must be invested in investigating these parameters to ensure the product is safe and reliable.

7 Comparison of Results with RANS Simulations

7.1 Motivation and Objectives

Unlike the DNS presented in Chapter 5, RANS simulations require significantly less computational effort and a shorter computation time. This makes these simulations the favourite approach in the industry. The lower computational cost of RANS simulations is given by the increased modelling, which is used to simplify unknown or complex terms in the equations. Of course, modelling requires a lot of experience to be performed in a meaningful way, leading to large errors otherwise.

Several roughness corrections have been developed to be integrated into the most common turbulence models used in the aerospace sector, as already presented in section 3.5.5. The objective of this chapter is to investigate the accuracy of roughness models available in the open literature. As in the previous chapter, attention is paid to the fluid dynamic and thermal properties of the flow, as well as the heat exchange with the wall. When available models are found to be inaccurate in predicting performance, suitable quantities are tuned to find a more realistic model configuration that better predicts the experimental results obtained in Chapter 6.

In this chapter, the extensive experimental database created and presented in the Chapter 6 is used as a reference. All simulations are performed using the software Ansys Fluent, version 2021 R2 [148], and consider a CHT setup that includes simulation of heat transfer through the walls.
Part of the results presented in this chapter have already been published in [149].

7.2 Methodology

This section presents the logic that has been followed when preparing and analysing the simulations. In the next paragraphs, the numerical setup is explained, and a validation case is presented, together with a mesh sensitivity analysis and a description of the implemented thermodynamic model.
Since the equations are intrinsically time-averaged in RANS, the notation with the overline for the time-averaged quantities used in Chapter 5 is hereby dropped, $\overline{u} = u$.

Numerical simulations are performed for the test articles presented in Chapter 4. The resulting 3D domain is analysed and compared with the experimental data gathered in

Chapter 6.

The friction factor is evaluated as in Eq. (6.6), using the derivative of the pressure along the axial direction instead of the integrated dimensions, namely dp/dx instead of $\Delta p/L$. This allows us to calculate the friction factor at every axial position. It is then extracted at an axial length $x = 0.75L$, where a completely developed flow is found since $x/d_H >> 10$, and no influence from the outlet boundary condition is present (also $(L - x)/d_H > 10$. As it is commonly done for square and rectangular ducts, the hydraulic diameter, d_H, is used as the characteristic length, see Refs. [150, 151].

The friction factor is used as an indicator to tune the roughness model. The roughness correction needs to input the value of the sand-grain parameter, k_s, introduced in section 3.5.5. The tuning process involves varying the k_s of the simulation until the resulting friction factor matches the experimental value. When its value is found, three simulations are run for every test article. For each specimen, the loadpoint REF, LP2, and LP4 from Tab. 6.2 are simulated, namely those tests with a varying coolant mass flow. The objective of these simulations is to assess the effectiveness of the available roughness correction in terms of pressure losses and heat transfer, with varying Reynolds numbers.
Additionally, the same three simulations are repeated with the same boundary conditions, but with the wall roughness correction turned off. In this way, an analysis of the correction's effects is possible.
Lastly, two more simulations are performed at a lower Reynolds number to extrapolate the friction factor on a broader Re range. However, these two simulations are not compared with experiments in the following part of this work because of the high measurement uncertainty that affects the low-Reynolds-number range, as visible in Fig. 6.11. In total, at least eight simulations are performed for each test article. It will be shown in the following sections that additional simulations are needed for one of the specimens.

The boundary conditions of each simulation are prescribed using the experimental results from the previous chapter. The coolant mass flow, outlet static pressure, and heat flux, as calculated from the experiments, are used as input to the simulations.

Only four of the five test articles are simulated. Indeed, attention is primarily given to modelling the cold side of the cooling circuit. As visible in Fig. 7.2, the internal walls of the cooling channels are simulated, and their roughness is modelled. Segment 1RS is not included in this analysis, as it would differ from segment 1SS only by a different setting of the heat flux boundary conditions, resulting from the different combustion chamber surface quality (see Fig. 6.17). As already shown in Fig. 6.20, the cooling performance between the 1SS and 1RS are exactly the same when calculated in terms of the Nusselt number on the cold side. Thus, the analysis includes all the segments with rough cooling channels and only one of the two articles with post-processed channels.

7.2.1 Validation

The validation of a CFD model is a critical step in ensuring its accuracy and reliability. The validation process is used to assess how well the model represents real-world fluid flow phenomena by comparing simulation results with experimental data or analytical solutions. To do so, the reference test case from Yohann Torres' PhD dissertation is used [152]. In his work, a block of copper with a varying section and a heater was used to reproduce the high heat flux commonly encountered in rocket engines. The selected reference case is considered suitable for this analysis due to its very similar boundary and operational conditions. A smooth duct was used in the experiments. Thus, no roughness model is applied in the validation of the setup.
In Fig. 7.1a, a sketch of the experimental setup used in [152] is shown. The heater is attached to the bottom of a massive block of copper. Thanks to the change in surface area of the copper block, the heat flux reaching the channel has a magnitude that is comparable to the heat flux values in rocket engines. Seven thermocouples are positioned along the side of the channel, from bottom to top, to measure the temperature distribution along the vertical direction. The same pattern is repeated for fourteen axial positions, as shown in Fig. 7.1a.

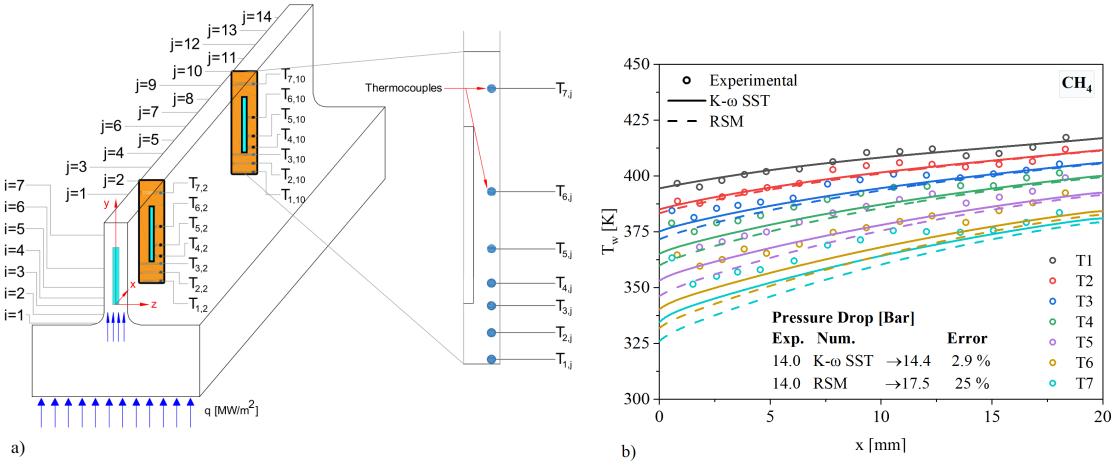

Figure 7.1: a) Sketch of the experimental setup used in [152]; b) Comparison of the simulation results and the reference experimental case.

To simulate this case, the domain shown in Fig. 7.2 is used. An inlet mass flow and a static outlet pressure boundary condition are used, taking their values from the experimental readings. The top wall is assumed to be adiabatic, while a uniform heat flux is applied to the bottom wall, as also observed in the experimental results in [152]. Exploiting the symmetry of the domain, only half the channel is simulated, and symmetry boundary conditions are used on the sides, as visible in Fig. 7.2.
The NIST real gas model, implemented in the software, is used to simulate the thermodynamic properties of the fluid, which in this case is cryogenic liquid methane. The $k-\omega$ SST turbulence model is used, as described in Eq. (3.20) and Eq. (3.21).

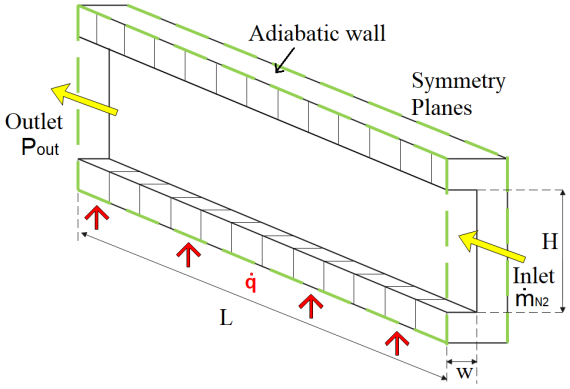

Figure 7.2: Simulation domain with boundary conditions.

A mesh sensitivity study is conducted to ensure the resolution of the near-wall region. However, the mesh study, the values of the boundary conditions, and many other parameters are not reported here but in the published work in [153].
In Fig. 7.1b, the final results are reported, showing the temperature of the wall at all seven heights and fourteen axial positions where the thermocouples are placed. The numerical and experimental results are compared. The pressure drop is also reported and compared with the experimentally obtained value.
A good agreement is obtained between the simulations and the experiments for measurements closer to the hot wall. A slight divergence is visible in the temperature values of the upper measurement points, whose origins need to be investigated in the experimental setup. In his work, [152], Torres states that the nitrogen environment in which the tests were carried out caused convective losses up to 6%.
Thus, the results obtained from our simulations are considered to be representative of the experimental setup, validating the numerical framework.

The effect of natural convection is minimised in the test setup presented in Chapter 4 because of the circumferential repetition of cooling channels. In this way, natural convection only affects a small portion of the domain. On the contrary, the channel in the setup from [152] had three of the four sides of the channel directly exposed to the external fluid.

7.2.2 Mesh Study

The simulation domain is adapted to the test article presented in the experimental setup in Chapter 4. The dimensions of the domain are changed accordingly to simulate the cooling channels of the test articles, therefore, a new mesh is necessary.
A structured mesh is chosen, with refinement close to the cooling channel walls, as shown in Fig. 7.3, and as was also done in the validation case of the previous section. A yellow dot is used to indicate the centre of the reference system, which uses x

for the flow direction, y for the vertical direction, and z is defined by the right-hand rule.

Three different meshes have been compared, whose details are reported in Tab. 7.1. The first mesh, called "Ref", has been defined in order to have y^+ and z^+ lower than 5, as in the validation case. Then, a coarser mesh and a finer mesh are created by reducing or increasing the number of cells in each direction and the dimension of the first cell near the wall.

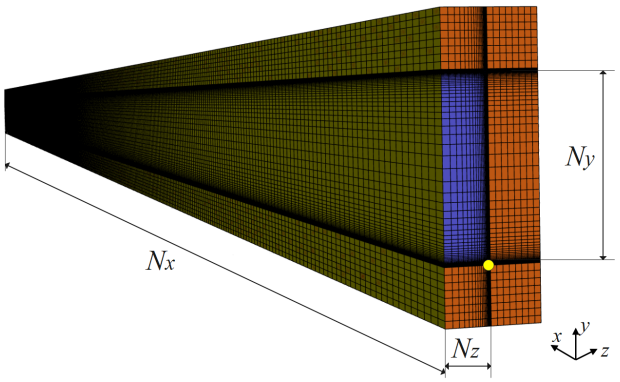

Figure 7.3: Visual representation of the mesh used for the simulations.

	Coarse	Ref.	Fine
N_x	750	1000	1000
N_y	40	70	100
N_z	20	35	50
y^+	>15	2-5	0.5-4
z^+	>15	2-5	0.5-3

Table 7.1: Mesh description.

Fig. 7.4 presents the results of the mesh study. On the left axis, the bulk pressure is divided by the inlet pressure. It is shown that the refinement of the mesh does not significantly impact the pressure drop calculation, which remains constant across all three meshes. The same happens when looking at the bulk temperature on the right axis.

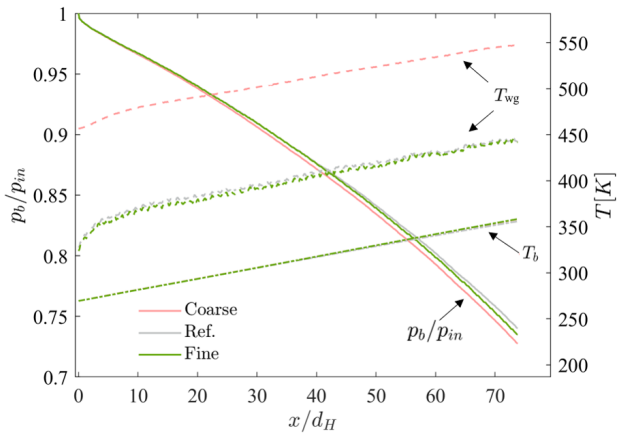

Figure 7.4: Results from the mesh convergence study. The dimensionless coolant bulk pressure, bulk temperature, and maximum wall temperature are reported.

However, the heat transfer near the wall is highly influenced, as seen from the behaviour of the wall temperature on the hot gas side T_{wg}. The coarse mesh cannot accurately

predict the near-wall heat exchange, whereas it has been proven that the fine mesh yields the same results as the reference mesh. Therefore, all subsequent simulations are performed using the reference mesh. Note that, for the simulations of segment 2RR, which has a higher aspect ratio, the number of cells in the vertical direction, N_y, is increased in order to obtain the same value of y^+ and the same dimension of cells in the core flow region.

7.2.3 Thermodynamic Properties

For the coolant's density and thermodynamic properties, lookup tables have been created using data from the NIST database [154]. Thus, a real-gas equation is implemented, allowing the thermodynamic properties and density to vary as a function of pressure and temperature. The equation of state used to create the tables is the published reference for nitrogen by Span et al. (see [21]).
This increased complexity is necessary due to the significant pressure losses and temperature increases encountered in this test setup, as shown in the previous chapter.
The Mach number is evaluated at the outlet section of every simulation, where the coolant moves the fastest because it heats up as it passes through the duct. The resulting Mach number is always lower than 0.4, apart from one simulation, where it locally reaches a peak of 0.43. Therefore, it is assumed that the results presented in the following sections are not affected by compressibility effects, as the flow conditions are sufficiently far from the transonic or sonic regimes.

7.3 Results

7.3.1 Friction factor

In his experiments, [55], Nikuradse found that for fully turbulent flows, the Reynolds number does not have any influence on the friction factor. Indeed, when the flow is fully rough, the roughness parameter k_s should be the only variable influencing the friction factor, which otherwise stays constant when the Reynolds number is increased.
In their work, Mazzei et al., [155], use the $k-\omega$ SST model and the correction from Aupoix, presented in Eq. (3.29). In this way, they adjust the sand grain parameter k_s and match the experimental data. Their experimental database spans a range of Reynolds numbers up to $5 \cdot 10^4$, and the authors demonstrate that a single k_s value exists, which enables the simulations to match the experimental data. However, discrepancies are visible when moving to higher Reynolds numbers. It can be noticed that the predicted friction factor for a given roughness value k_s and at Reynolds numbers higher than $2 \cdot 10^4$ is no longer constant. Instead, it decreases as Re increases, underestimating the value found in the experiments.

In this work, the sand-grain parameter is iterated in order to have the simulation results match the experimental ones.
In Fig. 7.5a, the friction factor obtained from the simulations with the $k-\omega$ SST and

its roughness correction, Eq. (3.29), are presented along with the experimental values coming from the segment 1SR. In Fig. 7.5a, simulations at Reynolds numbers spanning from $0.4 \cdot 10^5$ to $9 \cdot 10^5$ and with varying roughness parameters are compared. It is found that, for a given k_s, the friction factor significantly drops when moving to a higher Reynolds number, which is incorrect, as can also be seen from the experimental values, which stay roughly constant across the entire range of analysis.

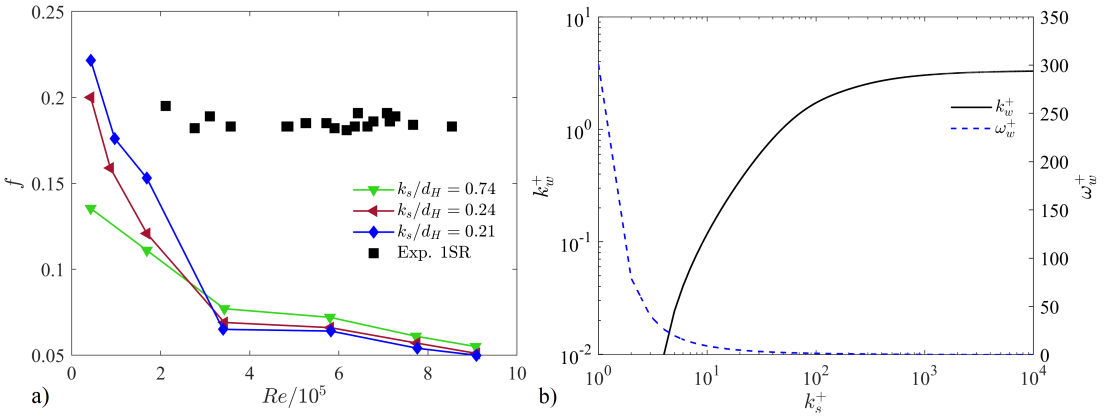

Figure 7.5: a) Friction factor results coming from experiments and simulations with the $k - \omega$ *SST*; b) modified boundary conditions from Eq. (3.29).

Furthermore, three different values of k_s are used, and the simulations are repeated over the same Reynolds number range. Fig. 7.5a shows that the variation of the sand-grain parameter does not help with the tuning process. Apparently, the increase in k_s leads to an increase in the friction factor only at Reynolds numbers lower than $2 \cdot 10^5$. Afterwards, the resulting friction factors collapse to a more or less constant value, independently of the parameter used.
To better understand this behaviour, the modified boundary conditions for the turbulence model are presented in Fig. 7.5b, against the roughness Reynolds number, defined in Eq. (3.23). It is found that for $k_s^+ > 10^3$, the value of these boundary conditions no longer changes, leading to similar results even when the value of k_s is altered. This means that if the real hardware presents very high roughness (or better, it is simulated at very high roughness Reynolds number), this model is not capable of reproducing the same results because the effects of the only input parameter are completely saturated.

On the other hand, when using the Spalart-Allmaras turbulence model and its correction, as presented in Eq. (3.28), the results significantly improve, as shown in Fig. 7.6a. With a given value of k_s, the resulting friction factor does not drop down with increasing Re, as it happened in the previous case. A slight variation is still visible when comparing the lowest and highest Reynolds numbers in this analysis, indicating that a dependence on the Reynolds number remains. However, what is most relevant is that when varying the k_s, the friction factor curve shifts up and down, giving the user greater flexibility when adjusting this parameter. For a perfect fit of the simulation on a wider range of Re, the parameter k_s should be varied with the Reynolds number, as $k_s = k_s(Re)$. However,

this work only focuses on the Reynolds number values between $6 \cdot 10^5$ and $9 \cdot 10^5$, see Chapter 6. Thus, a constant k_s is used for each specimen.

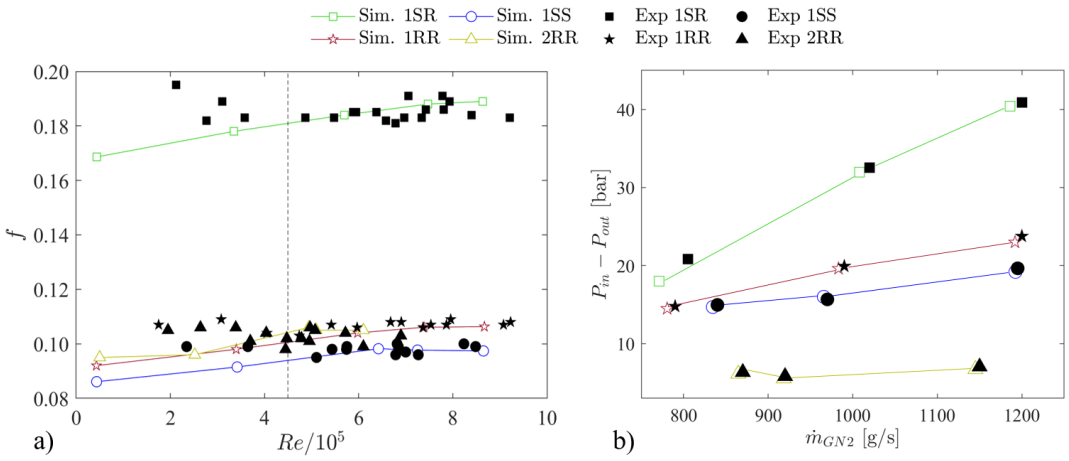

Figure 7.6: a) Friction factor results coming from experiments ("Exp.") and simulations ("Sim.") with the Spalart-Allmaras; b) Pressure difference between inlet and outlet in the experiments and in the simulations.

Fig. 7.6a shows the friction factor retrieved from the experimental results, black symbols, and the results from the CFD simulations with colored symbols. The tuning process of the sand grain parameter leads to the selection of $k_s/d_H = 0.21$ for the segment 1SS, $k_s/d_H = 0.24$ for the segment 1RR, $k_s/d_H = 0.74$ for the segment 1SR and $k_s/d_H = 0.22$ for the segment 2RR.

Fig. 7.6b shows the pressure drop across the duct predicted by the CFD simulations against the one obtained in the experiments for the three loadpoints and the four test articles. A good agreement is obtained for all specimens and almost every condition, indicating that the choice of k_s was successful and that the roughness correction of the SA model can accurately predict the pressure drop in rough cooling channels. It is useful to remember that specimen 2RR has cooling channels with a larger section area than the other segments, which feature a much lower pressure drop.
Given these justifications, the results that are presented from now on are all obtained with the Spalart-Allmaras turbulence model and its relative correction due to the impossibility of properly adjusting the available parameter with the $k-\omega$ SST.

Looking at the fluid-wall interaction, the wall shear stress can be investigated. This quantity, already defined in section 5.2.2, is mainly responsible for the pressure losses. However, its effects are commonly evaluated through the skin friction coefficient, $C_f = 2\tau_w/\rho u^2$. Fig. 7.7 shows the contour of this coefficient on the bottom wall of the simulated duct for a rough wall (left) and a smooth wall (right) for the loadpoint REF.

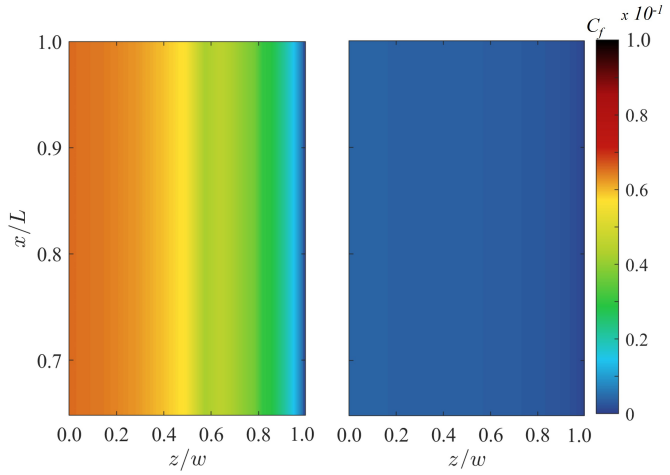

Figure 7.7: Skin friction coefficient for the segment 1RR with and without the roughness correction, for the loadpoint REF.

Close to the side wall, at $z/w \approx 1$, the coefficient is close to 0 because the velocity goes to zero. On the other hand, the different magnitudes of this coefficient, in the rough and smooth cases, become evident in the middle of the channel. The roughness correction increases the shear stresses on the walls, which are the cause of the increased pressure drops observed in Fig. 7.6.

For a better comparison, the resulting skin friction coefficient is averaged for each specimen and test condition, with a surface average on the bottom wall. The results are then compared in Fig. 7.8 as a function of the Reynolds number.

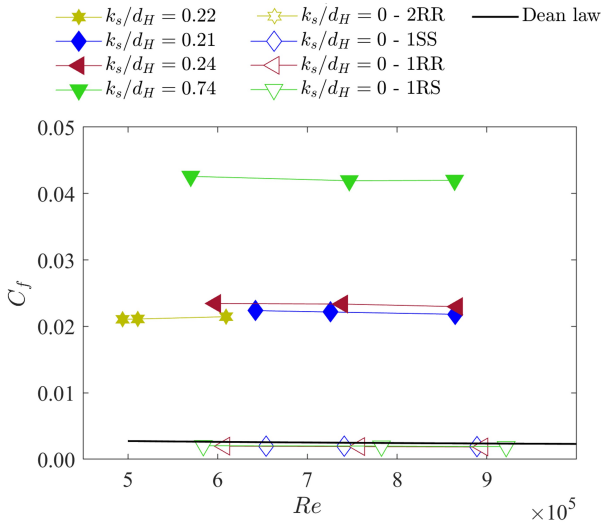

Figure 7.8: Skin friction coefficient for all the segments and at every loadpoint, with and without the roughness correction. The correlation for smooth pipes from Dean is added for comparison, [156].

As is well known, the skin friction coefficient is strictly correlated to the friction factor presented in Fig. 7.6a, as $C_f = 1/4f$. Thus, the simulations show once again that the three articles exhibit similar frictional behaviour, while the fourth one, with $k_s/d_H = 0.74$, has a significantly higher skin friction coefficient. In Fig. 7.8, it can be seen that when the roughness correction is deactivated, $k_s/d_H = 0$, the predicted C_f is almost one order of magnitude lower than the rough case. The smooth results are also compared with the empirical relation for smooth circular pipes from Dean, [156], and the good agreement proves once more the worthiness of the simulation setup.
It is understood that the roughness correction significantly increases the shear stresses on the wall to simulate the increased turbulence generated by the rough surface and to match the real pressure losses in the duct.

7.3.2 Velocity Profile

As mentioned, eight simulations are performed for each test article, varying the operating point as shown in Tab. 6.2 and switching the roughness correction on and off.

Fig. 7.9a presents the streamwise velocity on the symmetry plane, along the vertical coordinate of the duct. The velocity profile for the smooth cases is mostly flat and fairly symmetric, as expected for high Reynolds number flow, [18]. Besides, the different mass flow of the three loadpoints does not affect the profile shape when it is scaled with respect to the bulk velocity, calculated as $u_b = \int \rho U dA / \int \rho dA$. When the ratio k_s/d_H is set differently to 0, the roughness correction is activated, and the effects on the velocity profile become visible: the boundary layer becomes thicker, with respect to the smooth case, and the velocity has a higher value in the core region. This effect becomes

increasingly pronounced with the increase of the sand-grain parameter, as visible for the case of $k_s/d_H = 0.74$.
In all the cases where the roughness correction is activated, the velocity profile is evidently asymmetric. This occurs because the correction modifies the heat transfer near the wall, resulting in a faster flow near the hottest wall.

This effect is not visible in the smooth wall simulations because the heat exchange near the wall is low with respect to the conductivity of the solid walls. The thermal conductivity of the wall is high enough to spread the heat uniformly through the walls and thus heat up the coolant from every side.
When the roughness correction is activated, the near-wall heat transfer increases, and stronger differences become visible between the top and bottom of the duct.

The change in velocity profile can be visualised even better when the quantities are scaled in wall units, as introduced in section 3.2.2. To do so, the friction velocity is defined as $u_\tau = \overline{\nu}/\kappa(d_w + 0.03k_s)$, with d_w being the distance of the first cell center from the wall, as derived from the Boeing extension of the Spalart Allmaras model, [79]. The wall unit in Eq. (3.7) and Eq. (3.8) are evaluated and used in the representation in Fig. 7.9b.
The smooth simulation results clearly distinguish between the viscous sub-layer, a buffer layer, and a logarithmic layer, as the smooth wall theory expects, [18][73]. This distinction is lost when examining the simulations with roughness correction, as the viscous sublayer is destroyed by the presence of roughness and replaced by a so-called roughness sublayer. This is correctly simulated and presented in Fig. 7.9b, where the dimensionless wall velocity profile grows very slowly but eventually reaches a region where the logarithmic law still stands.

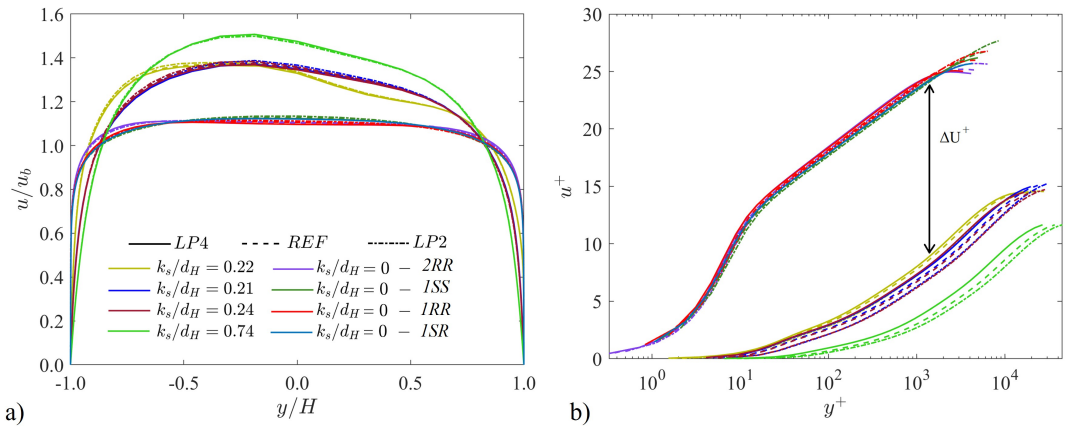

Figure 7.9: a) Velocity profile scaled with respect to the bulk velocity for smooth and rough simulations; b) velocity profile scaled in inner-wall units for smooth and rough wall simulations. Both plots present the results for the test conditions REF, LP4 and LP2, Tab. 6.2

The roughness function ΔU^+, Eq. (3.25), can then be calculated by difference with respect to the theoretical value. As discussed in section 3.5.3, numerous attempts have been made by researchers to establish analytical relations that estimate the value of the roughness function based on the sand-grain parameter. In this work, we evaluate the roughness Reynolds number of every simulation and plot the value of the resulting roughness function in Fig. 7.10, along with other works from the literature.

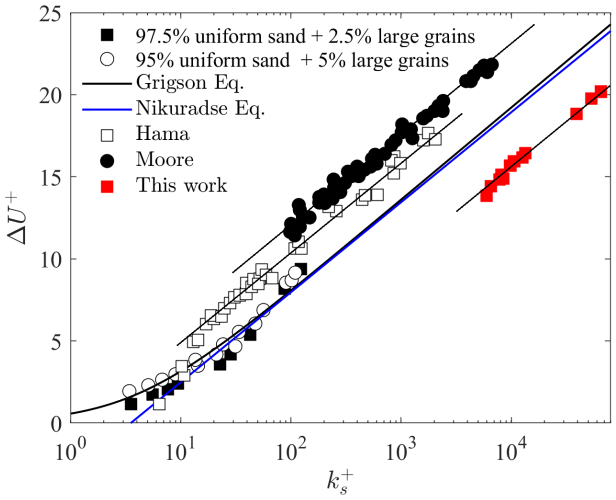

Figure 7.10: Roughness function for different types of roughness, adapted from Clauser, [66]. Grigson and Nikuradse equations are also presented, [56, 68].

It is clear from Fig. 7.10 that one single relation cannot fit data from different real roughness shapes. A good agreement with empirical relations is found at relatively low roughness Reynolds numbers ($10 < k_s^+ < 100$). When moving to higher values of k_s^+, the amount of data is insufficient to extrapolate an accurate relation. According to Jimenez [126], for $k_s^+ > 100$, the roughness functions should collapse altogether. However, in that work, the available data did not cover a wide range of k_s^+, with a focus on the transitional regime and reaching values close to 200. In Fig. 7.10, the results from this work are compared with other experimental data that reached $k_s^+ > 1000$. Every batch of data reaching the fully rough regime, including the results coming from this work, follows a relation in the form $\Delta U^+ = 1/\kappa \ln k_s^+ + 5.1 - B$. However, the shift B appears to vary according to the real shape and geometry of the rough surface. For example, the shift of the roughness function for the uniform sand depends on the percentage of the large grains included in the sand, Fig. 7.10. In this work, the slope of the data is the same as that of every other batch of data, but a value of $B = 12$ is found.

7.3.3 Heat Transfer Prediction

Primary importance is given in this work to the evaluation of the heat transfer from the simulations. To measure the reliability of the results, every simulated case is compared against the reference experimental results. As explained in Chapter 4, the experimental setup allows us to measure the fluid's inlet and outlet temperature and the copper's

temperature at the same radial distance from the hot gas wall as the bottom wall of the channels, as shown in Fig. 6.19. The comparison between CFD simulation results, with colored symbols, and experimental results, with black symbols, is presented in Fig. 7.11. In Fig. 7.11a, the difference between the outlet and the inlet temperature of the coolant is shown, where the bulk temperature is evaluated as $T_b = \int \rho U T dA / \int \rho U dA$. The same symbols as in Fig. 7.6 are utilised. Once more, a good agreement is found for all four specimens and the three simulated loadpoints (REF, LP2, and LP4 in Tab. 6.2). A maximum deviation of +6.5% is found in case LP4 for the 2RR specimen, while all the other predictions have an average error close to ±3%. A sum up of the deviations with respect to the experimental values is given in Tab. 7.2. These results demonstrate that the correct amount of energy is transferred to the fluid during the simulation and that the events observed during the experiments are accurately reproduced.
On the other hand, looking at Fig. 7.11b, a larger deviation is found.

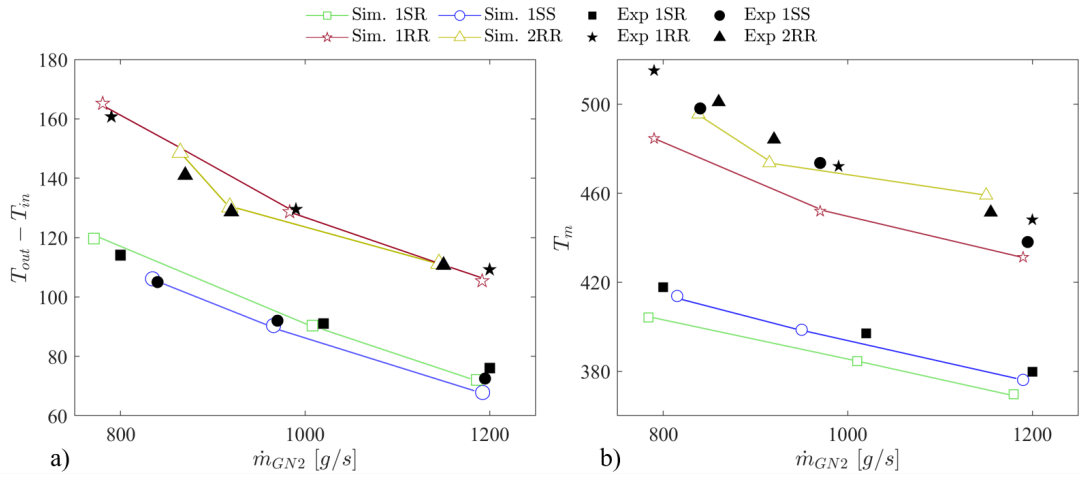

Figure 7.11: a) Coolant temperature difference between outlet and inlet of the cooling channel for simulations and experiments; b) wall temperature in the measuring point, Fig. 6.19, for simulations and experiments.

In Fig. 7.11b, the wall temperature recorded by the thermocouples is compared with the simulated temperatures at the same position. The prediction of the wall temperature is fairly accurate for the simulations of the test articles 1SR and 2RR, with an error within the ±3%. Worse predictions are found for the segment 1RR, with an average error of 6%. Lastly, severely incorrect results are obtained for the 1SS (blue line and black circular symbols), resulting in a deviation of approximately 15% from the experimental data.
For segment 1SS, a deviation between 50 and 70 degrees is visible, although the bulk temperature of the fluid was correctly predicted. To better investigate this deviation, the Nusselt number is evaluated as in Eq. (6.8). The comparison with the experimental setup is possible by evaluating the Nusselt number at the same axial position of the thermocouple.

The Nusselt number comparison between the simulations and experiments is shown in Fig. 7.12, along with empirical relations for smooth, [145], and rough pipes, [72].

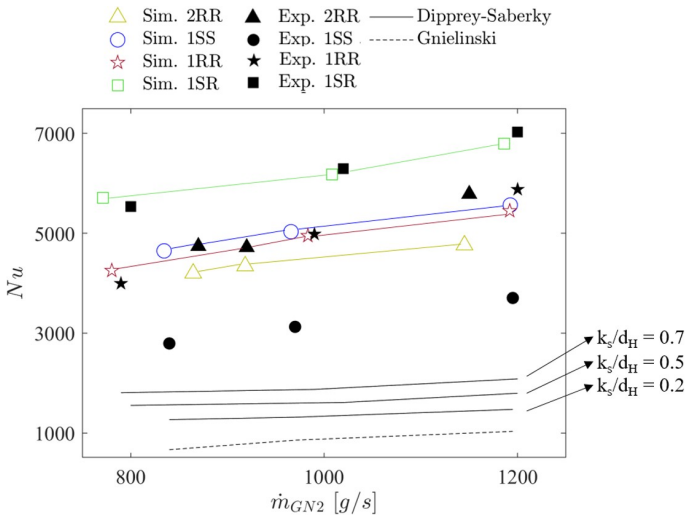

Figure 7.12: Nusselt number calculation from Eq. (6.8) for CFD simulations and the experimental results, [138]. The empirical correlations of Gnielinski for smooth pipes, [145], and of Dipprey and Saberski for rough pipes, [72], are also reported for comparison.

Firstly, it can be observed that, with respect to the empirical laws from Gnielinski and Dipprey-Saberski, the CFD simulations provide a significantly better estimation, demonstrating the superior prediction of the roughness model under analysis compared to empirical relations. Indeed, the use of the empirical relation from Dipprey-Saberski (plotted in Fig. 7.12 three times, for $k_s/d_H = 0.2$, 0.5 and 0.7 values) would lead to a wrong prediction, underestimating the experimental values of approximately $200-300\%$. More specifically, Fig. 7.12 shows that the results for specimens 1SR, 1RR, and 2RR match the experimental values with a discrete accuracy, albeit slightly underestimating them. These three segments are the ones whose cooling channel surfaces have not been post-processed, and the roughness is purely due to the printing process.
On the other hand, the results of the simulations for segment 1SS significantly overestimate the Nusselt number when compared with the experimentally obtained values (approximately +70/80%). The deviations from the experimental values of each test case are reported in Tab. 7.2.

An overestimation of the Nusselt number was already found in literature when using the Boeing roughness correction, [83, 157, 158], and several authors tried to adjust the predictions by modifying the turbulent Prandtl number as explained in Eq. (3.30). Here, the turbulent Prandtl number correction is applied only to the simulations of specimen 1SS to improve the wall temperature prediction. The coefficients A and B in Eq. (3.30)

are taken from the model developed by Latini et al., [83], as reported in Eq. (7.1).

$$\begin{aligned} A =& (-2.346 \cdot 10^{-4}) Pr^2 + (2.102 \cdot 10^{-3}) Pr + 3.542 \cdot 10^{-3} \\ B =& (-2.303 \cdot 10^{-3}) Pr^2 + (5.588 \cdot 10^{-2}) Pr - 3.043 \cdot 10^{-3} \end{aligned} \tag{7.1}$$

This procedure is then implemented in Ansys Fluent through a UDF (User-Defined Function), and the simulations for segment 1SS are repeated, incorporating this correction while maintaining the same boundary conditions. In order to estimate the roughness function and feed it into Eq. (3.30) in the UDF, the Grigson relation has been used, [68]:

$$\Delta U^+ = \frac{1}{\kappa} \ln \left(1 + \frac{k_s^+}{e^{1.3325}} \right) \tag{7.2}$$

The results are presented in Fig. 7.13, and summed up in Tab. 7.2.
In Fig. 7.13a, it can be seen that the difference between outlet and inlet temperature is not heavily affected by the change in turbulent Prandtl number, as heat flux and mass flow stay the same.
On the contrary, the effect of the correction function is evident in Fig. 7.13b, where the wall temperature at the thermocouple position is presented. The predicted temperature in this case is between 30 and 40 degrees higher than in the uncorrected case, even though it still does not match the experimental values. It is essential to consider that the implemented correction, as shown in Eq. (7.1), is developed for circular pipes and adjusted according to the Dipprey-Sabersky correlation [72]. For this reason, a perfect agreement was not expected, and a new tuning process is needed for rectangular ducts and rocket operating conditions.
Furthermore, the correction changes the Nusselt number on the wall towards the right values, as seen in Fig. 7.13c, and improves the simulation results. A quantitative comparison between the uncorrected and corrected cases is given in Tab. 7.2.

This comparison shows that the tuning of the turbulent Prandtl number should not be carried out only with respect to the Nusselt number, as done in numerous works, [83, 146]. Indeed, it is shown that with the applied correction, the resulting wall temperature remains far from the experimental values, although the Nusselt number becomes similar to that evaluated from the experimental data.

To understand how the roughness modification affects the temperature profile, the dimensionless temperature T^+ is used. This temperature is defined as $T^+ = (T_w - T)/T_\tau$, as proposed in Ref. [71], and $T_\tau = -\dot{q}_{f,s}/(\rho c_p u_\tau)$. It is important to notice that the wall heat flux implemented in this relation is not the imposed boundary condition on the bottom of the solid wall but rather the resulting heat flux at the fluid-solid interface on the bottom of the duct.
The resulting profiles are presented in Fig. 7.14.

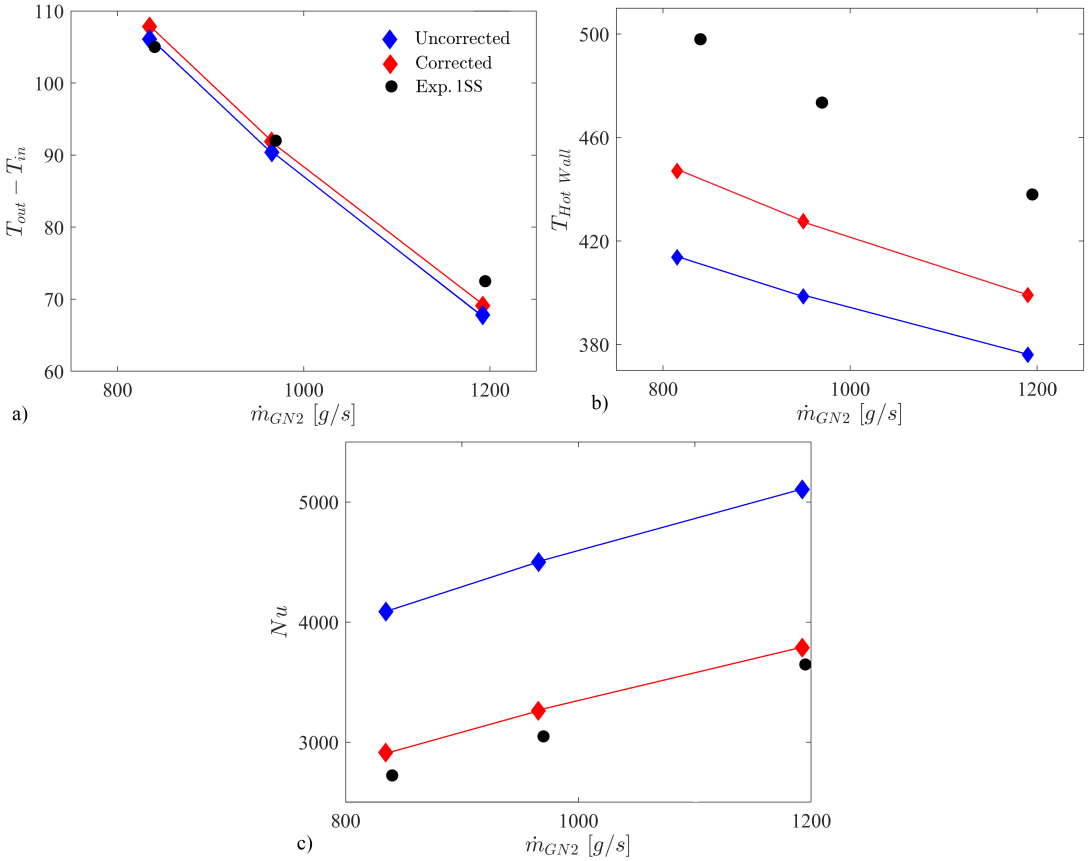

Figure 7.13: Comparison between the experimental results of the segment 1SS, the uncorrected and corrected (with Eq. (7.1)) simulations. a) Temperature difference between inlet and outlet, b) wall temperature at the thermocouple position, c) Nusselt number.

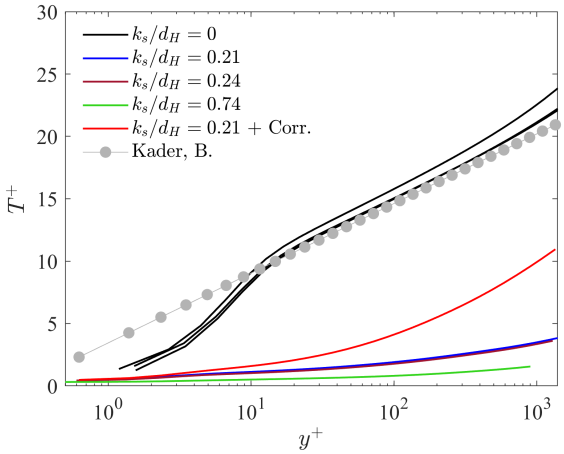

Figure 7.14: Dimensionless temperature profile with respect to wall units for smooth and rough simulations. The theoretical profile for a smooth wall, as presented in Ref. [159], is shown for comparison.

It can be seen that the temperature profile of the smooth simulations (in black lines) well aligns with the data from [159]. When the roughness correction is activated (colored lines), the profile shifts downwards as the velocity profile. The same behaviour was found in [83], where similar k_s/d_H ratios were simulated, and in the recent work in [160]. Lastly, the red line shows the effect of the turbulent Prandtl number correction, which moves the profile closer to the smooth ones. Indeed, the shift in T^+ indicates that, close to the wall, there is a lower difference between the wall and the fluid temperature, simulating the higher mixing due to the roughness. This effect is well simulated for segments with as-built cooling channels, which feature rough surfaces. However, the channels of segment 1SS, being post-processed with EDM, are smoother. From this discussion, it is clear the need for an additional correction to damp the strong temperature shift that is applied with the roughness correction in Eq. (3.28).

The Pr_t correction has been shown to be required only to match the data coming from segment 1SS, whose cooling channels were post-processed with EDM. On the contrary, the heat transfer results are not too far from the experimental values in the simulations concerning the other segments, whose cooling channels were left as printed.

Article	**LP**	$\mathbf{k_s/d_H}$	$\mathbf{Pr_t}$ **corr.**	$\mathbf{k_s^+/10^3}$	$\mathbf{\Delta U^+}$	$\eta_{\mathbf{f}}$	$\eta_{\mathbf{T_b}}$	$\eta_{\mathbf{T_m}}$
1SR	LP4	0.74	No	39.0	18.8	-0.5	5.8	-3.5
1SR	REF	0.74	No	53.6	19.7	3.7	-0.11	-3.4
1SR	LP2	0.74	No	65.2	20.2	4.8	-3.7	-2.9
1SS	LP4	0.21	No	8.1	15.1	0.2	3.7	-17.3
1SS	REF	0.21	No	9.8	15.7	1.8	1.7	-16.0
1SS	LP2	0.21	No	12.6	16.2	-1.6	-2.4	-14.4
1RR	LP4	0.24	No	7.8	14.8	-1.9	4.5	-6.3
1RR	REF	0.24	No	10.6	15.9	-0.8	-0.4	-4.7
1RR	LP2	0.24	No	13.4	16.5	-0.6	-3.2	-4.1
2RR	LP4	0.22	No	5.9	13.8	1.0	6.6	-1.0
2RR	REF	0.22	No	6.5	14.4	0.1	2.0	-3.5
2RR	LP2	0.22	No	8.5	14.9	-1.0	-1.6	1.6
1SS	LP4	0.21	Yes	7.3	14.5	-4.1	3.0	-10.3
1SS	REF	0.21	Yes	8.8	15.1	-2.1	0.3	-9.7
1SS	LP2	0.21	Yes	11.6	16.1	-5.1	-3.5	-8.9

Table 7.2: Simulation details sum up. Every raw indicates a simulation: the test article it refers to and the reference loadpoint in the experimental database. The k_s/d_H value used, the roughness Reynolds number and velocity shift, k_s^+, ΔU^+, are also reported. The error η is defined as $\eta = 100 \cdot (u_{sim} - u_{exp})/u_{exp}$, (with u generic quantity) and it is reported for, f, T_b, T_m

.

7.3.4 Hot-wall Temperature

Although the corrected simulations of segment 1SS still show inaccuracy in predicting the wall temperature, this deviation is approximately 10%. Thus, it is possible to compare the global effects of rough surfaces on the maximum temperature reached on the hot wall side. As will be shown, the 10% of error is much lower than the difference between the rough and the smooth cases.
In Fig. 7.15, the maximum wall temperature on the hot gas side is presented for every simulation run in this work, both applying the roughness correction and leaving the smooth wall assumption. The maximum temperature occurs on the bottom wall, at the symmetry plane that passes between two channels.

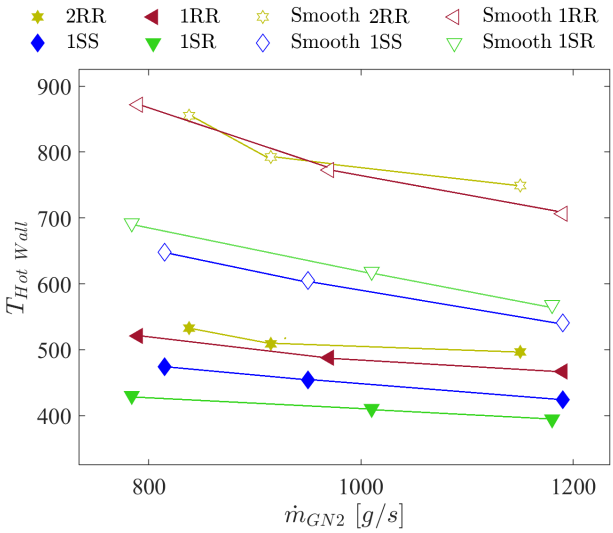

Figure 7.15: Comparison of the maximum hot-wall temperature at the three simulated operating points for the smooth and rough cases. All the results come from simulations whose boundary conditions are set to replicate the given test article conditions.

Even accounting for the error in the wall temperature calculation, a clear division between the results of the simulations with smooth and rough walls stays evident. With rough walls inside the cooling channel, the increased capability of exchanging heat with the fluid leads to significantly lower wall temperatures across the entire simulated range. Additionally, the absolute difference between the maximum temperature on smooth or rough walls becomes even more significant for higher values of heat flux, as visible in the case of the segments 1RR and 2RR, which have a rough combustion chamber that leads to higher heat flux, Chapter 6.

The two segments with a smooth chamber present another important feature: when simulated without roughness correction, the hot wall temperature is slightly higher for segment 1SR because the heat flux imposed at the boundary is higher. When the roughness correction is activated, the two curves switch their relative position, and segment 1SR now shows lower hot wall temperatures even though it

receives a higher heat flux. This happens because segment 1SR features a higher level of roughness in the cooling channels. This roughness helps the fluid absorb more heat than in segment 1SS, which results in a hotter wall, even receiving a lower heat flux.

The same influences and effects can be visualised when examining the contours of the temperature fields of the fluid and the wall at the outlet section of the duct. This is presented in Fig. 7.16. This figure presents the temperature field for the loadpoint REF, Tab. 6.2, of the four investigated specimens, both in the case of rough, Fig. 7.16a-d, and smooth walls, Fig. 7.16e-h.
The first immediately visible difference is the average resulting temperature, which is remarkably higher in the smooth case than in the rough case when the roughness correction is activated. Secondly, another important difference can be highlighted: in the smooth wall simulations, a very strong temperature gradient is observed in the fluid close to the walls, indicating that the fluid heats up significantly but cannot dissipate that energy from the near-wall region to the core of the flow.
On the other hand, when the roughness correction is activated, Fig. 7.16a-d, a significantly smoother transition is visible from the near-wall temperature to the core temperature of the flow. The change in the wall boundary conditions of the turbulence quantity appears to accurately reproduce the increased near-wall mixing, which is expected from rough surfaces, thereby leading to enhanced diffusion and more uniform temperatures within the fluid.

As expected, the roughness on the walls leads to enhanced heat transfer performance since the coolant can absorb more heat under the same flow conditions, but also results in higher pressure losses.

7.4 Conclusion

In this work, the Spalart-Allmaras model and the roughness correction from Aupoix [79] are employed in CFD simulations to evaluate the model's reliability in simulating liquid rocket engine conditions. It can be concluded that the model gives good flexibility, even featuring only one parameter, the sand-grain roughness, contrary to what happens with the $k-\omega$ SST and the roughness correction presented in [81].
The k_s can be adjusted to reproduce the test results regarding pressure losses. In this way, the heat transfer predictions are found to be satisfactory in terms of the bulk temperature of the fluid, with an average error of 3%. However, the correction yields a lower performance in predicting the wall temperature, which is found to be within 6% for as-built surfaces. For post-processed surfaces, an additional correction is found to be necessary. In this case, the turbulent Prandtl number correction from Latini et al. [83] improves the simulation results but requires further work, specifically in properly tuning the coefficients and applying it to the specific case.

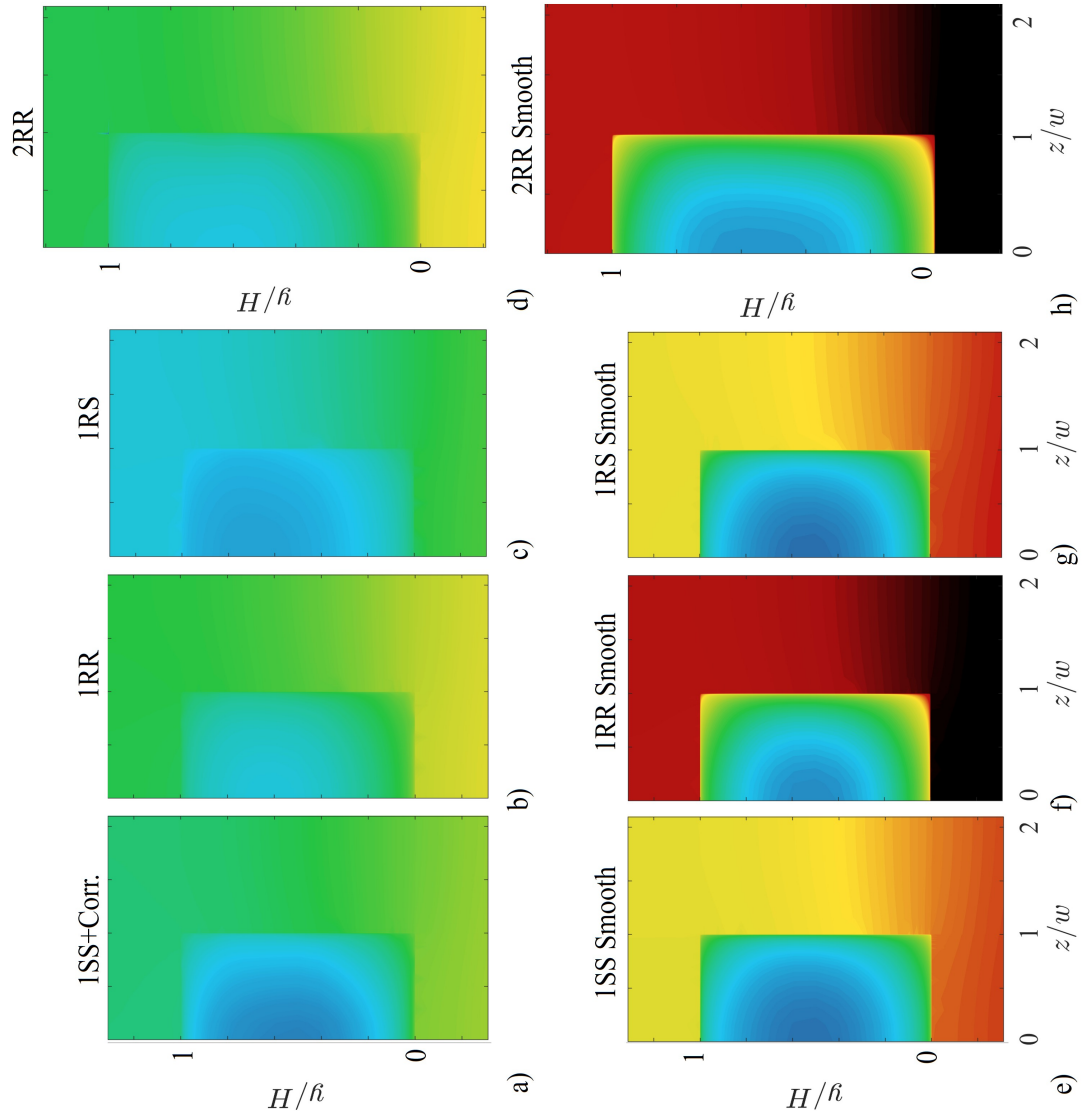

Figure 7.16: Temperature contours at the duct outlet section for a-d) rough simulations and e-h) smooth simulations. For all representations, the loadpoint REF is used, as shown in Tab. 6.2. Dimensions not to scale.

8 Conclusion & Outlook

This research has examined a central challenge in the application of Additive Manufacturing to space propulsion components, namely the influence of AM-induced surface roughness on aerodynamic and thermal performance. Through a multi-scale methodology integrating Direct Numerical Simulations, experimental testing, and Reynolds-Averaged Navier-Stokes simulations, the study has provided detailed insights into roughness-induced flow phenomena and evaluated the capabilities and limitations of current modelling approaches.
The study was guided by five central research questions, which are addressed here in light of the findings:

1. To what extent does surface roughness induced by AM influence flow dynamics and heat transfer within propulsion components?
   The results consistently demonstrate that surface roughness has a significant impact on both flow and thermal fields. DNS analyses revealed that AM-induced roughness increases skin friction due to enhanced in-plane vortices and turbulence, especially near sidewalls. The thermal field is also significantly affected, with roughness thickening the temperature boundary layer and increasing the core flow temperature. Experimental tests confirmed these trends, showing friction factors up to 15 times higher than theoretical smooth values, and heat transfer enhancements ranging from $10 - 12\%$ in cooling channels to $30 - 35\%$ in combustion chambers. These findings confirm that surface roughness is not a secondary feature, but a critical parameter that influences performance.

2. Does the effect of surface roughness remain consistent across varying operating conditions, particularly with changes in Reynolds number?
   The study found that the impact of roughness is not constant across all conditions. DNS demonstrated that at low Reynolds numbers, the interaction between roughness and the flow evolves as the Re is changed, affecting the flow dynamics. Experimentally, it is found that in a fully rough regime, the friction behaviour does not change when the Reynolds number is increased. Heat transfer rates, however, are significantly affected. This reinforces the necessity for roughness models that are sensitive to the operating regime, rather than relying on fixed empirical correlations.

3. What is the impact of small-scale, process-induced surface non-uniformities on the overall performance of propulsion components?
   Investigations into printed components revealed significant azimuthal temperature

variations, up to 20% of the average wall temperature, linked to surface non-uniformities. These irregularities, although localised, present a potential risk to structural integrity due to thermal fatigue and material degradation. This underscores the importance of detailed surface characterisation and strict process control in AM fabrication, as even minor geometric deviations can propagate into major performance variations.

4. How accurately do current industrial roughness models capture the physical behaviour observed in experiments and high-fidelity simulations?
   RANS simulations showed that while industrial models based on sand-grain roughness can be calibrated to match experimental pressure losses, their accuracy in predicting heat transfer, especially wall temperatures, remains limited. Predictions of bulk fluid temperatures achieved average errors of 3%, but wall temperature deviations were significantly larger, particularly for post-processed surfaces. Incorporating turbulent Prandtl number corrections improved the accuracy, but a broader applicability across surface types and flow regimes remains a challenge.

5. In what ways can simulation models be refined or calibrated to more accurately predict the influence of surface roughness?
   This work demonstrates that model refinement must be grounded in experimental validation. Printing test samples, measuring surface profiles, and calibrating models against real performance data are essential steps. Furthermore, tailored corrections, such as flow-dependent Prandtl number modifications, show promise in bridging gaps between simulations and measurements. A pathway forward includes developing data-driven models using high-resolution surface topographies and exploring machine learning techniques to generalise predictions across varying AM conditions.

Beyond the immediate context of rocket propulsion, the insights gained here are widely applicable to any high-performance system, such as gas turbines and heat exchangers, that is affected by surface roughness. The demonstrated sensitivity of aerodynamic and thermal performance to surface features suggests that accurate roughness modelling is essential in any application where thermal efficiency and reliability are critical.
Looking ahead, several directions for future work emerge:

- Development of advanced roughness models that incorporate surface statistics and flow regime sensitivity.
- Exploration of mitigation strategies, including post-processing techniques, and assessment of their trade-offs between surface quality and mechanical performance.
- Application of machine learning approaches to build predictive roughness models informed by experimental data and surface measurements.

From an industrial perspective, the findings highlight the importance of considering surface roughness effects early in the design process. By incorporating roughness-aware

design principles, engineers can improve both the performance and durability of AM components. Close collaboration between research institutions and industry will be key to translating these insights into practical tools and design standards.

In conclusion, this work has provided a rigorous assessment of AM-induced surface roughness effects in propulsion components, addressing all initial research questions through a combination of numerical, experimental, and modelling approaches. The results not only deepen our understanding of roughness-related phenomena but also lay the groundwork for improved predictive tools and design strategies, ultimately supporting the advancement of reliable, high-performance AM technologies for space and beyond.

A Appendix

A.1 AM Processes Comparison

As mentioned in section 2.2, several AM techniques are nowadays available on the market. Although L-PBF is probably the most mature and utilised in the aerospace sector, other techniques are introduced quickly here and compared, in Tab. A.1.
EB-PBF is similar to L-PBF but uses an electron beam within a vacuum atmosphere. Within the electron beam gun, a tungsten filament is heated to achieve thermionic emission of electrons that are then accelerated through lenses and, upon impact with the powder, achieve melting
The LP-DED does not use a powder bed. Instead, it uses concentrated thermal energy to fuse metals by melting them locally while they are deposited. This process allows for both the freeform fabrication of components and the addition of material to components for the purposes of coating, repair, or modification.
The AW-DED process uses metal inert gas (MIG) welding or pulsed MIG welding. The arc-based deposition process does not fabricate precise features but only net-shape components with surfaces with high waviness that require final machining.
In the CGS process, metallic powders are accelerated to supersonic velocity by injection into a high-temperature and high-pressure gas jet and then impinged onto a substrate. The high-velocity gas jet is generated by use of a converging-diverging de Laval–type nozzle. The physical description of this process is based on aerodynamics and the dynamic plasticity behaviour of metals.
Several other techniques exist and are commonly used worldwide for different applications. A detailed comparison of the advantages and disadvantages of each process is given in Tab. A.1.

Process Attribute	Metal AM Process				
	L-PBF	**EB-PBF**	**LP-DED**	**AW-DED**	**CGS**
Melted material	Y	Y	Y	Y	N
Feedstock	Powder	Powder	Powder	Wire	Powder
Techno maturity	High	High	High	Medium	Medium
Layer thickness	$20 - 100\mu m$	$20 - 100\mu m$	$250\mu m$	$2 - 3mm$	Variable
Feature resolution	High	Med-High	Medium	Low	Med
Deposition rate	$2 - 30\ cc/hr$	$50 - 100\ cc/hr$	$20 - 200\ cc/hr$	$200 - 1500\ cc/hr$	$300 - 1500\ cc/hr$
Availability	Very common	Uncommon	Common	Rare	Rare
Equipment cost	Very high	Very high	High	Medium	Very high
Density ($> 99\%$)	Y	Y	Y	Y	$95 - 99\%$
Support material	Y	Y	Y	Y	Y
Surface Finish	Fine	Fine	Coarse	Very Coarse	Coarse
Simultaneou Materials	No	No	Y	Y	Y
Post-proc. difficulty	High	High	Medium	High	Medium

Table A.1: Comparison of main characteristics of the presented AM techniques. Table readapted from [3].

A.2 Testbench P & ID

The gaseous oxygen is fed from high-pressure bundles in the gas storage with a maximum pressure of 300 *bar*. The feedlines are equipped with pressure and temperature sensors to record the fluid's thermodynamic behaviour and a mass flow meter to measure the mass flow. A dome-loaded pressure regulator is used to set the desired pressure from the control room, and a control valve is used to set the desired mass flow to deliver to the combustion chamber.
A similar setup is found on the methane feeding line.
The same gases are used to feed the two torch spark igniters, which ignite the propellant entering the chamber from the main injector.
Two 2000 L high-pressure air tanks are also in the gas storage. Using a dome-loaded pressure regulator, the air is used to pressurise the 500 L water tank, which feeds the cooling channels of the engine. Orifices are used to regulate the mass flow of water, and differential pressure measurements are used to measure the mass flow during the tests.
High-pressure nitrogen is used to flush and purge the feedline and the chamber. A schematic of the feedlines and measurement points is given in Fig. A.2.

A.2.1 GN2 Cooling Feed-line

The new specimens are cooled with gaseous nitrogen. Due to the low heat capacity of this fluid, a significant amount of gas must be introduced to the channels to properly cool the segment and prevent excessively high temperatures. To achieve this objective, a new feeding line has been installed and is presented in Fig. A.1, thereby extending the test bench's capabilities.

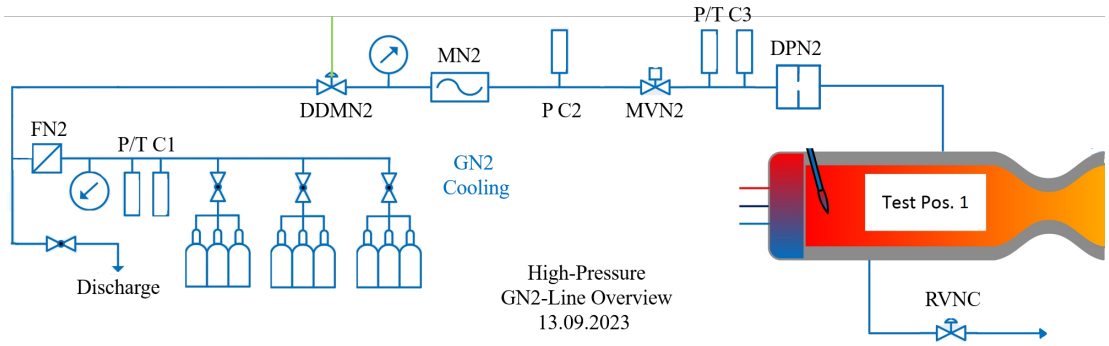

Figure A.1: High-pressure gaseous nitrogen feeding line.

Three gaseous nitrogen reservoirs with a maximum pressure of 300 *bar* are needed to run the system. A dome-loaded pressure regulator is used to set the desired flow pressure, while a Coriolis mass flow meter measures the mass flow. Additionally, a differential pressure measurement is used to retrieve the mass flow with a much faster response time compared to the Coriolis mass flow meter. Downstream the cooling channels, a controlled valve is used to set the coolant mass flow.

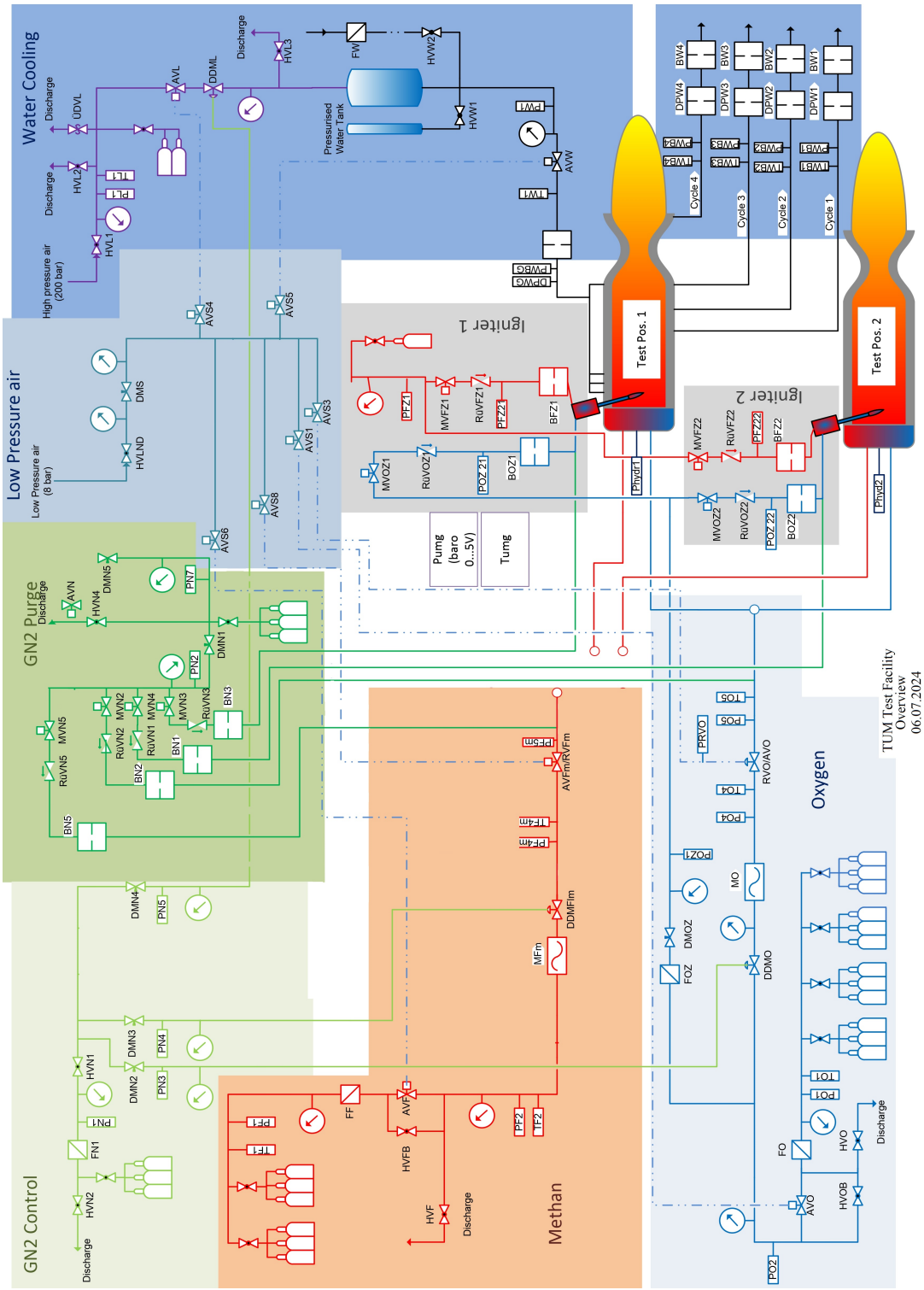

Figure A.2: TUM Space Propulsion facility.

A.3 Validation of the DNS Solver

Although the same code was widely used in our research group, [27][30][31][32], the application case was different, and some features of the code have been modified. Thus, a new validation was considered necessary. Simulations of smooth square ducts have been considered as a reference test for our solver, and the case of a square duct with friction Reynolds number $Re_\tau = hu_\tau/\nu = 110$ was selected as a test case since this relatively low Reynolds number allows us to compare and validate a fully turbulent simulation without the high computational cost of higher Reynolds numbers. Consequently, a smooth square duct is simulated as a validation case for the results of the ribbed duct presented in section 5.2.
Available data on this test case can be found in previous work from Pinelli et al., [120], who created a database of simulations for square ducts spanning a range of low bulk Reynolds numbers. The same dataset has been widely used for validation in many other works, as in [103] [24] [114]
For this simulation, a control box of dimensions $2h$ x $2h$ x $20h$ is used, and periodic boundary conditions are enforced on the inlet and outlet as it has been done for the ribbed case. For all the simulations in this work, the same solver and boundary conditions have been used.
When simulating wall-bounded ducts, accurately predicting secondary flow is the most challenging phenomenon to capture, and statistical errors due to poor convergence may dominate over the numerical error itself. For this reason, the in-plane velocity components are often used for validating purposes. If the simulation is well-converged, both the exact position of secondary flows and their magnitude are accurately captured.
In Fig. A.3a we present the spanwise velocity component, along with the results from the work of Pinelli et al..

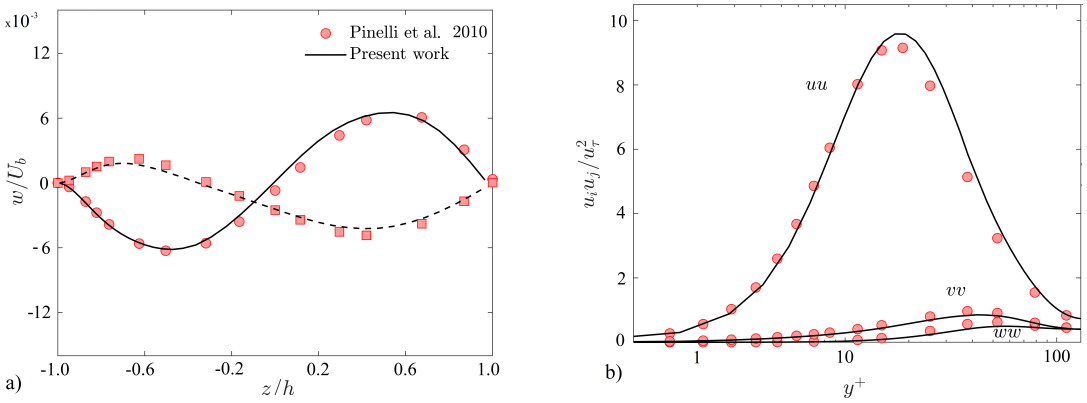

Figure A.3: a) The spanwise velocity component is presented on the planes located at $y = -0.75h$ (continuous line) and at $y = -0.25h$ (dashed line); b) The turbulent normal stresses are shown after scaling with respect to inner-wall characteristic length and velocity. The results are compared with those from [120].

In addition, the more difficult-to-catch second-order statistics are presented on the right of Fig. A.3b. The comparison of the turbulent normal stresses with [120], is carried out in inner-wall scaling.
The good agreement with previous simulations from other authors is clear and assures us that the computation technique is correct. The slight deviation that is visible between the present simulation and data from Pinelli has to be attributed to the different averaging time: Pinelli averaged his simulation for 4160 convective times (h/u_b), while in this work, we averaged the validation case over an interval of ≈ 1900 convective times, saving more than half of the computational time but still getting very satisfactory results.

A.4 Pressure Loss Calculation

The sharp turn of the flow, when flowing from the manifold to the cooling channels and then again from the channels to the outlet manifolds, is presented in Fig. A.4.

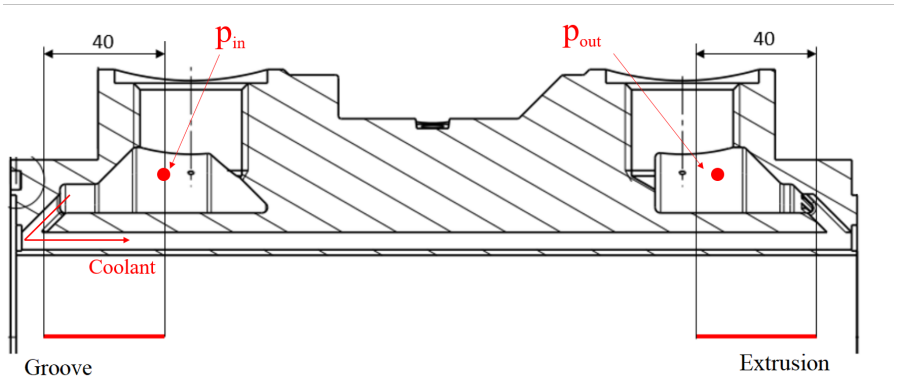

Figure A.4: Cut of the test article: measurement point and actual flow path of the coolant.

This turn causes a pressure loss. Thus, the measured values of the pressure, P_{in} and P_{out} (measured at the locations indicated by a red circle in Fig. A.4) need to be corrected. To account for the loss caused by the sharp turn, the model presented in [161] is used. The model is reported in Eq. (A.1).

$$p_{loss} = \frac{1}{2}\rho V^2 \left(\frac{\pi r_0}{d_H} \frac{\theta}{180} + k_b \right) \tag{A.1}$$

where k_b is a loss coefficient taken from the graph presented in Fig. A.5 and readapted from [161]. This coefficient is only based on the channel dimensions, d and b, and the curvature radii r_0 and r_i.

Fig. A.5 presents the pressure loss coefficient for a 90-degree turn. Losses for bends other than 90 degrees are commonly considered to be proportional to the bend angle, θ, which is introduced in Eq. (A.1). In the equation, the first contribution is due to the friction losses in the bend, and the second is due to the flow distortion. Due to the very small curvature radius, r_0, the first contribution is neglected, while only the distortion loss is considered, being the dominant factor in this case.

Lastly, the corrected pressures are calculated as:

$$p_{in,corrected} = p_{in} - p_{loss}$$

$$p_{out,corrected} = p_{out} + p_{loss}$$

.

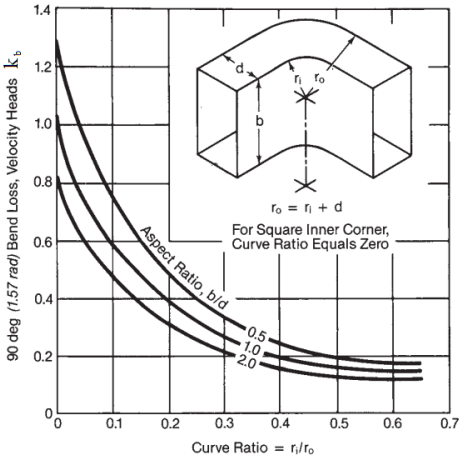

Figure A.5: Pressure loss coefficient for the 90-degree bend in a rectangular duct.

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